STAT0003 (former STAT1005)

Further Probability and Statistics

Lecture Notes

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Contents

Part 1: *Statistics for populations*

Chapter 1

Probabilities

1.1 The axioms of probability

We are already acquainted with the classical definition of probability as a relative frequency (limit of the number of 'successes' over the number of trials, as the number of trials tends to infinity). For example, if we want to define the probability that a coin (it might not be fair) gives 'Head', we toss the coin n times and count the number of times s that the outcome was 'Head'. Then the ratio s/n tends to the wanted probability, as $n \to \infty$.

Intuitively, we can understand why this cannot be a negative number, as the number of 'successes' can be either zero or a positive number. Also the number of 'successes' cannot be greater than the number of trials. The sample space Ω should cover all possible outcomes and it should be related to the 100% probability, the probability of 'the whole'. For two different events A and B, we would expect the probability of their union $A \cup B$ to be related to the sum of the two separate probabilities. All these intuitive ideas are summarized in the next definition.

Definition (*The Three Axioms of Probability*): Let Ω be a sample space. A *probability* assigns a real number $P(A)$ to each event $A \subseteq \Omega$ in such a way that

1. $P(A) \geq 0$ for every A.
2. If $A_1, A_2, \ldots$ are pairwise disjoint events ($A_i \cap A_j = \emptyset,\ i \neq j,\ i, j = 1, 2, \ldots$), then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$. (This property is called *countable additivity*.)
3. $P(\Omega) = 1$.

The fact that (1),(2) and (3) are axioms means that they are not to be proved and we take them as they are. Some more results might be derived as their consequences.

In fact,

a) $P(A) \leq 1$ for all A.

b) $P(\emptyset) = 0$

c) $P(A^c) = 1 - P(A)$

d) Suppose A and B are disjoint. Then $P(A \cup B) = P(A) + P(B)$

Can you prove these remarks?

It is often useful to gain practical intuition of many probabilistic properties by visualising events through their corresponding Venn diagrams. A Venn diagram shows all possible logical relations between a finite group of sets. For example, in the diagrams below (Figure 1.1) the sample space Ω (represented by the whole rectangle) contains events A, B, as well as some other events (represented by the area outside A, B). These events may not be mutually disjoint, as they can intersect.

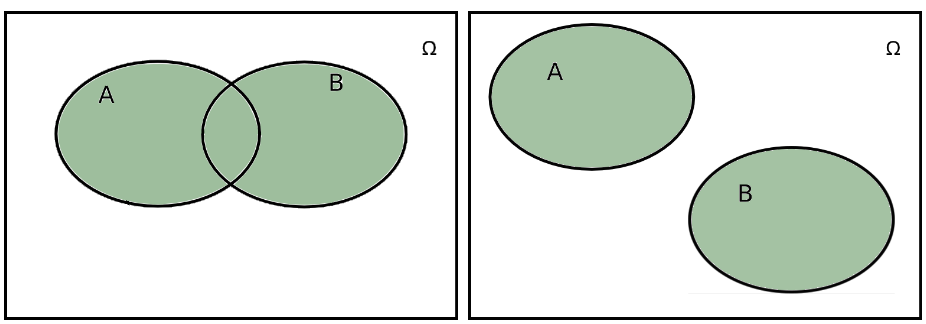

Figure 1.1: A Venn diagram where the sample space Ω (whole reactangle) contains events A, B, as well as some other events (the rest of the rectangle). In the left-hand diagram, events A and B are not mutually exclusive (i.e., disjoint), whereas in the right-hand diagram they are.

If probabilities are represented by regions, one can immediately see that in the case of the left-hand diagram of Figure 1.1, $P(A \cup B) \neq P(A) + P(B)$ since we would be double-counting the middle (intersection) region. In the right-hand diagram, where A and B are mutually exclusive, indeed $P(A \cup B) = P(A) + P(B)$.

CAREFUL: Venn diagrams cannot replace a rigorous mathematical proof. Additionally, Venn diagrams can only be used for a finite collection of sets, whereas the axioms of probability are defined for an infinite (but countable) collection of sets.

While the axioms of probability refer to the properties of probabilities, we have still not discovered any ways to assign probabilities to events of the sample space. For a sample space with a FINITE number of elements, we may, for example, consider all the outcomes to be equally likely.

Example 1.1: A fair die is thrown. Take $\Omega = \{1, 2, 3, 4, 5, 6\}$. Each outcome is equally likely, so we assign probability p to each outcome. What is this p equal to?

By axiom (3), it holds that

$$P(\{1\}) + P(\{2\}) + \ldots + P(\{6\}) = 1 \text{ or } 6p = 1 \text{ or } p = \frac{1}{6}.$$

Now, we can find the probability of any event, e.g.

$$\begin{aligned} P(\text{even number}) &= P(\{2\} \cup \{4\} \cup \{6\}) \\ &= P(\{2\}) + P(\{4\}) + P(\{6\}) \qquad \text{(countable additivity)} \\ &= 3\,\frac{1}{6} = \frac{1}{2} \end{aligned}$$

Note that

$$\frac{3}{6} = \frac{\#\{2, 4, 6\}}{\#\Omega}.$$

In general, we may define

$$P(A) = \frac{\#A}{\#\Omega}$$

for any event $A \subseteq \Omega$. Does this probability P, as it has just been defined, satisfy (1), (2) and (3)?

Example 1.2: Two fair dice are thrown. Take $\Omega = \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\} = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}$. Let A be the event 'total is 10', then

$$A = \{(4, 6), (5, 5), (6, 4)\}$$

and

$$P(A) = \frac{\#A}{\#\Omega} = \frac{3}{36} = \frac{1}{12}.$$

What is the probability that $(3, 4)$ occurs? What is the probability that one die shows a 3 and another die shows a 4? What is the probability that $(2, 2)$ occurs? What is the probability that one die shows a 2 and another die shows a 2?

Example 1.3: Promotional packs of cornflakes contain vouchers for 10p, £1 or £10. Each promotional pack contains exactly one voucher. Assume that to every pack containing a £10 voucher, there correspond 10 packs with a £1 voucher and 100 packs with a 10p voucher. Assume I buy a promotional pack at random.

Can you calculate the probability that it contains a £10 voucher? Can you calculate the probability that it contains a voucher for at least £1?

1.2 Conditional probability

Definition (*Conditional Probability*): Let Ω be a sample space and the two events $A, B \subseteq \Omega$ with $P(B) > 0$. A *conditional probability* of A given B, is defined to be equal to

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Toy example: Consider the setup shown in Figure 1.2, and suppose we are interested in the probability $P(A \mid B)$. The statement 'given B' means that we know that only the shaded region is possible. The conditional probability is then the proportion of probability of the darker region (representing $P(B)$) which lies also within A, i.e. is within the dark shaded region representing $P(A \cap B)$, which gives us the desired result.

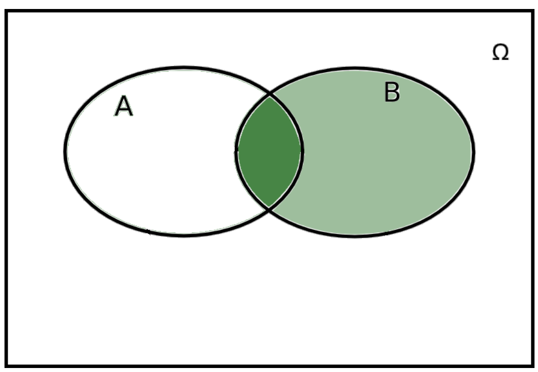

Figure 1.2: A Venn diagram where events A and B intersect and we are interested in the conditional probability $P(A \mid B)$

Definition (*Partition*): Let Ω be a sample space and the k events $A_1, \ldots, A_k \subseteq \Omega$. Then $A_1, \ldots, A_k$ are called a *partition* of Ω, if it holds that (i) $A_i \cap A_j = \emptyset$, $i \neq j$, $i, j = 1, \ldots, k$ (or mutually exclusive events), (ii) $A_1 \cup A_2 \cup \ldots \cup A_k = \Omega$ (or exhaustive events).

Toy example An example of a partition of a sample space Ω is shown in Figure 1.3 as a Venn diagram.

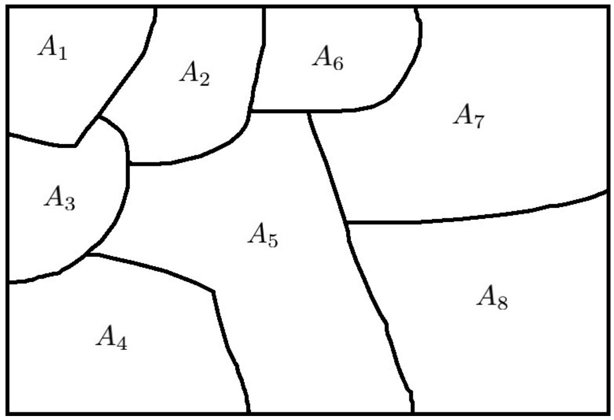

Figure 1.3: An example of the Venn diagram of a partition of a sample space Ω into eight events $A_1, \ldots, A_8$.

> **Definition (*Total Probability Law*):** Let Ω be a sample space, let $A_1, \ldots, A_k$, be a partition of Ω and let $B \subseteq \Omega$. Then it holds that
>
> $$\begin{aligned} P(B) &= P(B \cap (A_1 \cup A_2 \ldots \cup A_k)) = P(B \cap A_1) + P(B \cap A_2) + \ldots + P(B \cap A_k) \\ &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \ldots + P(A_k)P(B|A_k) = \sum_{i=1}^{k} P(A_i)P(B|A_i). \end{aligned}$$

This is also known as the **Partition Theorem**.

Toy example: An example of the partition theorem is shown in Figure 1.4, where the sample space Ω is split into 8 disjoint events $A_1, \ldots, A_8$. The probability of an event B is then the sum of the intersections of B with each individual event of the partition, in this case $P(B) = P(B \cap A_1) + \ldots + P(B \cap A_8) = P(B \cap A_5) + P(B \cap A_7) + P(B \cap A_8)$, since B has no intersection with any except 3 of the A events. This can be further expanded using the corresponding conditional probabilities to give us that $P(B) = P(B \mid A_5)P(A_5) + P(B \mid A_7)P(A_7) + P(B \mid A_8)P(A_8)$

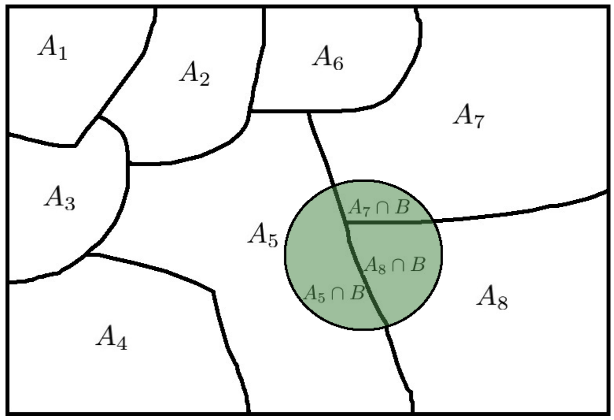

Figure 1.4: An example of a the law of total probability of an event B (shown in green) given a partition of a sample space Ω into eight events $A_1, \ldots, A_8$.

Bayes Theorem: Let Ω be a sample space, let $A_1, \ldots, A_k$, be a partition of Ω and let $B \subseteq \Omega$, such that $P(B) > 0$. Then it holds that

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^{k} P(A_i)P(B|A_i)}, \ j = 1, \ldots, k.$$

Example 1.4: A woman has two children; we write $B_i, \ i = 1, 2$, for the event that 'the i-th child is a boy' and $G_i, \ i = 1, 2$, for the event that 'the i-th child is a girl' (thus, we agree to use the index 1 for the oldest child). Are B_1, G_1 a partition of the sample space? Are B_2, G_2 another partition of the sample space? Can you think of a partition of the sample space, made of four sets then?

Note: ***Unions and intersections of more than two sets.*** We know that, for two events $B_1, \ B_2 \subseteq \Omega$, it holds for the probability of their union

$$P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 \cap B_2).$$

If there are three events B_1, B_2, B_3 of the sample space, we can write

$$P(B_1 \cup B_2 \cup B_3) = P(B_1) + P(B_2) + P(B_3) - P(B_1 \cap B_2) - P(B_1 \cap B_3) - P(B_2 \cap B_3) + P(B_1 \cap B_2 \cap B_3).$$

This is also known as the ***Inclusion-Exclusion formula***, because this is exactly what is does; once a probability has been included more times than it should, it has to be excluded, and if it is excluded more times than it should, it will be included again.

The general form for events $B_1, \ldots, B_n$ of the sample space is

$$P(\cup_{i=1}^n B_i) \ = \ \sum_{i=1}^n P(B_i) \quad - \sum_{i,j=1,\ldots,n,\ i<j} P(B_i \cap B_j) \ + \ \ldots \ + \ (-1)^{n+1} P(\cap_{k=1}^n B_k).$$

Also for the probability of intersection of three events, we may write

$$P(B_1 \cap B_2 \cap B_3) = P(B_1)P(B_2 \cap B_3|B_1) = P(B_1)P(B_2|B_1)P(B_3|B_1, B_2).$$

This comes straight from the fact that

$$P(B_3|B_1, B_2) = \frac{P(B_1 \cap B_2 \cap B_3)}{P(B_1 \cap B_2)} = \frac{P(B_2 \cap B_3|B_1)P(B_1)}{P(B_2|B_1)P(B_1)}$$

and, so

$$P(B_3|B_1, B_2) = \frac{P(B_2 \cap B_3|B_1)}{P(B_2|B_1)} \quad \text{or} \quad P(B_2 \cap B_3|B_1) = P(B_3|B_1, B_2)P(B_2|B_1).$$

For the general case of the events $B_1, \ldots, B_n$ of the sample space, we write for the probability of their intersection

$$P(\cap_{i=1}^n B_i) = P(B_1)P(B_2|B_1) \ldots P(B_n|B_1, \ldots, B_{n-1}).$$

The above formula can be used for any ordering of the n events. However, when there is a **natural order** implied by the indices of the sets, then it can be extremely useful. A similar comment might be found in the next section for tree diagrams.

Example 1.5: An urn contains 3 black and 4 red balls. We draw three balls at random without replacement. If we know that the third ball was black, what is the probability that the first one was red?

1.3 Combinatorics

This section concerns the cases when the sample space is comprised of a FINITE number of EQUALLY LIKELY outcomes. If this number is not small (as it was in our example with the fair die), then the rules presented next could be useful.

Example 1.6: A coin and a die are thrown. There are 2 possible outcomes for the coin, and 6 possible outcomes for the die. Hence the number of possible outcomes of the experiment is $2 \times 6 = 12$.

In general, if an experiment consists of two INDEPENDENT tasks and there are n_1 possible outcomes for the first task and n_2 possible outcomes for the second task in an experiment, then there are $n_1 \times n_2$ possible outcomes for the experiment consisting of the two tasks together. Next, we write an example of not independent tasks.

Example 1.7: A coin is thrown. If it is 'Head', a die is thrown next. Otherwise, nothing is done. How many possible outcomes are there?

The multiplication rule can be extended to the case where the experiment consists of k independent tasks with n_i possible outcomes for the i-th task. In that case, there are $\prod_{i=1}^{k} n_i$ possible outcomes for the experiment consisting of the k tasks together.

Example 1.8: Can you find the number of possible outcomes for each of the following experiments?

(i) A die is thrown twice

(ii) A die is thrown n times

(iii) Three MPs (one Labour, one Conservative and one Other) are chosen from the British House of Commons. (At the time of writing - in December 2017 and after the 8 June 2017 general election - the House of Commons consists of 317 Conservative, 262 Labour and 71 Other MPs.)

What would change if two instead of one Labours were chosen instead?

Permutations

When we have n different units (suppose each one has a different name) and we want to place them at n different positions (suppose each one is numbered $1, \ldots, n$), then we are dealing with permutations.

Example 1.9: In how many ways can 3 children be lined up?

Answer: For the first position we can choose any of the three children. Two children will be left to fill the second position. The child not selected for either the first or the second position, will have to fill the third position. Therefore

$$\#\text{ways} = 3 \times 2 \times 1 = 6.$$

Notice that a sample space for the experiment of lining up three children, say C_1, C_2 and C_3, is

$$\Omega = \{(C_1, C_2, C_3), (C_1, C_3, C_2), (C_2, C_1, C_3), (C_2, C_3, C_1), (C_3, C_1, C_2), (C_3, C_2, C_1)\}.$$

If we assume that each way of lining up the children is equally likely then each has a probability of $1/6$ of occurring.

Each of the *orders* in which n distinct objects can be placed in a line is called a *permutation*. There are

$$n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1 = n! \qquad (n \text{ factorial})$$

permutations of n distinct objects.
Note $0! = 1$.

Special case: It might be that instead of lining all the n distinct objects, we only want to use $r \leq n$ of them. In that case, we will obviously stop at

$$n \times (n-1) \times \ldots \times (n-r+1) = \frac{n!}{(n-r)!},$$

when we have filled r instead of n positions.

Example 1.10: Two children chosen from five are to be lined up. In how many ways can this be done?

$$\#\text{ways} = 5 \times 4 = 20.$$

Combinations

When we have n different units (suppose each one has a different name) and we want to select some of them to be in a group (and the rest of them to be in its complement group), then we are dealing with combinations. The order does not matter.

Example 1.11: In how many ways can we choose two letters from A, B, C, D, if the order of choice is irrelevant?

Answer: First suppose that we want to line the two selected letters. Then the number of permutations is

$$4 \times 3 = \frac{4!}{2!} = 12,$$

according to everything we have said so far. More specifically, these groups are the AB, BA, AC, CA, AD, DA, BC, CB, BD, DB, CD, DC. However, we are not interested in the order and the number we are looking for should be smaller than this (AB and BA both say the same thing). The number we are looking for, multiplied by 2! should give us the number 12. Thus, the correct number is

$$\frac{4!}{2!2!} = 6.$$

Why 2!?

The number of groups of r distinct objects (complement groups consist of the remaining $(n-r)$ objects left out) out of n distinct objects that we can create is

$$\frac{n!}{r!(n-r)!} = \binom{n}{r}.$$

This is called the number of *combinations* of r from n. We have that

$$\#\{\text{combinations of } r \text{ from } n\} \times r! = \#\{\text{permutations of } r \text{ from } n\}.$$

Example 1.12: A department consists of 6 men and 4 women. We wish to choose 3 men and 2 women to form a committee. In how many ways can this be done? Write down a sample space for this event.

Answer: We can think of this experiment of being in two stages. In the first stage we choose two men; in the second stage we choose two women. Use the combination rule to see that there are $\binom{6}{3} = 20$ ways of choosing 3 men out of the 6 in the department, and $\binom{4}{2} = 6$ ways of choosing 2 women out of the 4. Now use the multiplication rule to conclude that there are $20 \times 6 = 120$ ways of performing the experiment.

Permutations and combinations

We may generalize the concept of COMBINATIONS, for the case when we have n distinct units and we want to find in how many ways these can be put in k different groups, each one consisting a fixed number of $n_i,\ i = 1, \ldots, k$, units and $n_1 + \ldots + n_k = n$. In the previous section, it was that $k = 2$ (the main group and its complement). It holds then that the number of groups are

$$\binom{n}{n_1} \times \binom{n - n_1}{n_2} \cdots \binom{n - n_1 - \ldots n_{k-2}}{n_{k-1}} = \frac{n!}{n_1!n_2! \ldots n_k!}.$$

Example 1.13: How many *distinguishable* permutations are there of the letters of the word BABBLE?

Answer: Yes, you have read well, the question is asking you to find the permutations (not combinations!). However, we will turn this into a 'combinations' question. The word to be written will have 6 letters, the first letter 1 up to the last letter 6 (the order matters, of course, since it is changing the meaning of the word). Now, we want to make a group of 3 B-s, a group of 1 A, 1 L and 1 E. Thus, deciding how these 6 ordered letters will be put into the 4 groups (or matching the six ordered letters with these 4 groups) will be done in

$$\frac{6!}{3!1!1!1!}$$

ways.

This can be extended for the case of PERMUTATIONS: for n objects consisting of n_1 of type 1, n_2 of type 2, ..., n_k of type k, there are

$$\frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$$

distinguishable permutations.

Chapter 2

Discrete random variables

2.1 Definition of discrete r.v., *pmf* and *cdf*

We have met examples where we have been interested in some function of the outcome of an experiment, rather than in the outcome itself. For example, if we throw two dice we may be interested in the total score, and if we throw a coin 10 times we may be interested in the number of heads obtained rather than the precise sequence of H and T which is observed. These numerical quantities of interest are called *random variables*. The precise definition of a random variable (r.v.) is that it is a real-valued function on the sample space Ω. Remember that a function is just a mapping: a random variable is just a rule for allocating numbers to each outcome of an experiment.

Example 2.1: Suppose a coin is thrown 3 times. Let X be the number of heads obtained. The correspondence between the outcomes of the experiment and the values taken by X is:

ω	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	3	2	2	1	2	1	1	0

X maps each outcome to a real number (e.g. if $\omega = \text{HHT}$ then $X(\omega) = 2$). In mathematical notation, $X : \Omega \to I\!R$. X is a random variable. The set of possible values for X is $\{0, 1, 2, 3\}$.

A r.v. that takes its values in a finite or countable set is called a *discrete random variable*. X in Example 2.1 is a discrete r.v.

Another **example**: a coin is thrown repeatedly until the first head is obtained. If X is the number of throws required then X is a random variable taking values in $\{1, 2, 3, \ldots\}$, which is countable so X is a discrete random variable.
An **example** of a *non-discrete* random variable is the time between 2 phone calls.

If Ω is itself finite or countable, then any r.v. on Ω must be discrete. For the remainder of this chapter, assume Ω is countable.

It is important to distinguish the random variable (which is a rule for allocating numbers to the outcome of an experiment) from the values it takes (which are numbers). **We write capital letters for r.v.s**, and **lower case letters for the values they take**. E.g. for X in Example 2.1, x could be 0,1, 2 or 3.

Example 2.1 continued: Suppose the coin in Example 2.1 is fair, so $P(H) = 1/2$, and that successive throws are independent. Then the eight possible outcomes (HHH, HHT etc.) are equally likely with $P(HHH) = \ldots = P(TTT) = 1/8$. We can find the probability that X takes a particular value. For example, $P(X = 2) = P\{\omega : X(\omega) = 2\} = P\{HHT, HTH, THH\} = 3/8$. Repeating this for all possible values x that X can take, we get the following table:

x	0	1	2	3
$P(X = x)$	1/8	3/8	3/8	1/8

Note that $P(X = x) \geq 0$ for all values of x, and that the probabilities add up to 1. Note again the distinction between X and x.

Definition: Let X be a discrete r.v. The *probability mass function* (pmf) of X is

$$p(x) = P(X = x).$$

Example 2.1 continued: For X in Example 2.1, $p(0) = 1/8,\ p(1) = 3/8,\ p(2) = 3/8$ and $p(3) = 1/8$.

Each r.v. has its own pmf. We write $p_X(x)$ if we wish to emphasize that the pmf belongs to the r.v. X.

Properties of pmfs: Let X take the values x. Then

1. $p(x) \geq 0$ for all x
2. $\displaystyle\sum_{\text{all } x} p(x) = 1.$

We saw these properties in Example 2.1, where $x = 0, 1, 2, 3$.

When we know the pmf for a r.v. X, we can calculate probabilities like $P(X > 2)$, $P(X \leq 0)$, $P(-0.5 < X \leq 3.9)$ etc.

Example 2.1 continued:

- $P(X > 2) = P(X = 3) = 1/8$
- $P(X \leq 2) = 1 - P(X > 2) = 7/8$
- $P(-1 < X \leq 3/2) = P(0 \leq X \leq 1) = P(X = 0) + P(X = 1) = 1/8 + 3/8 = 1/2$.

Example 2.2: Suppose that a coin is not fair, it has $P(H) = 1/3$ and it is thrown 3 times, independently. For each H you win £1 and for each T you lose £1. Let X be your winnings in £(if X is negative, you have a loss). Find the pmf of X and the probability that you make a loss.

There are 8 possible outcomes as in Example 2.1, but now they are not all equally likely. E.g. $P(HHT) = (1/3) \times (1/3) \times (2/3) = 2/27$. The following table gives the outcomes, their probabilities and the corresponding X- values:

ω	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
Probability	1/27	2/27	2/27	4/27	2/27	4/27	4/27	8/27
X	3	1	1	-1	1	-1	-1	-3

X is a discrete random variable taking values in $\{-3, -1, 1, 3\}$.
Its pmf is

x	-3	-1	1	3
$p(x)$				

(*Check*: sum of probabilities is 1).
The probability that you make a loss is $P(X < 0) = P(X = -3) + P(X = -1) = 20/27$.

Example 2.3: A fair coin is thrown twice. Let X be the number of heads. X is a discrete r.v. taking values in $\{0, 1, 2\}$, with pmf

x	0	1	2
$P(X = x)$	1/4	1/2	1/4

Then we can calculate $P(X \leq x)$ for any value of x we choose. For example:

$$\begin{aligned} P(X \leq 1) &= P(X = 0) + P(X = 1) = 3/4 \\ P(X \leq 0) &= P(X = 0) = 1/4 \end{aligned}$$

$$\begin{aligned}
P(X \leq 1/2) &= P(X=0) = 1/4 \\
P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) = P(X \leq 2) = 1 \\
P(X \leq 2.3) &= P(X \leq 2) = 1 \\
P(X \leq -1) &= 0
\end{aligned}$$

We can plot $P(X \leq x)$ against x in Figure 2.1.

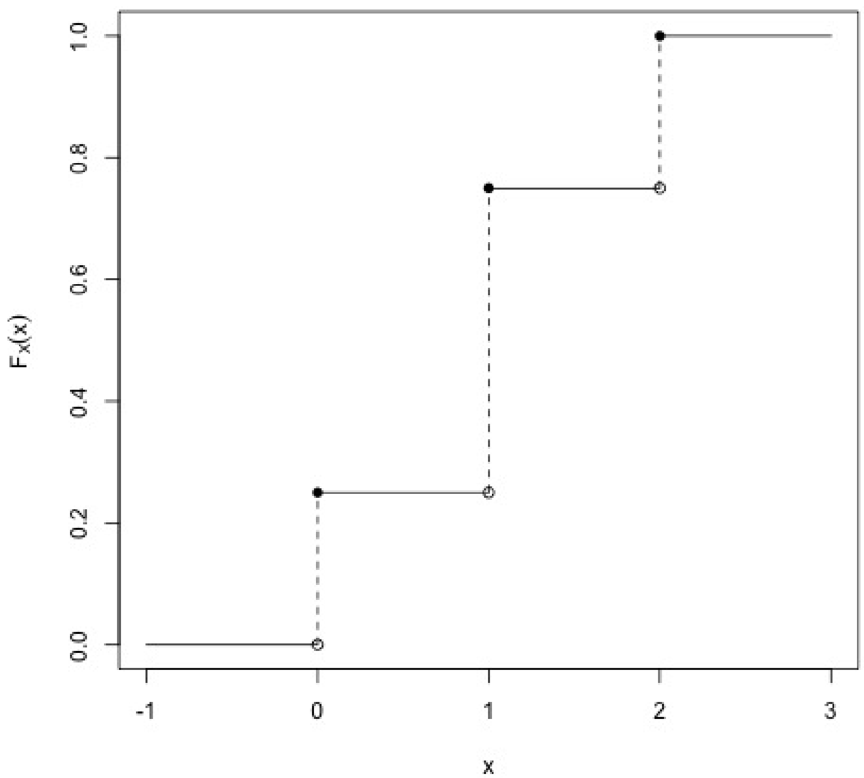

Figure 2.1: The probability $P(X \leq x)$ (i.e., the cumulative distribution function of Example 2.3.

Definition: The *(cumulative) distribution function* of a r.v. X is

$$F(x) = P(X \leq x) \text{ for all } x \text{ in } \mathbb{R}.$$

The distribution function may be referred to as a cdf.
Note: (for those who did Stats at school!) **when we say 'distribution function', we *always* mean the cdf**.

In Example 2.3 we have plotted $F(x)$ against x for the r.v. X. Note that $F(x)$ is defined for *all* x in $I\!R$, not just for the values taken by X, and that it has the following properties:

Properties of cdf

1. F is non-decreasing.

2. $0 \leq F(x) \leq 1$ (because $F(x) = P(X \leq x)$ is a probability), $F(x) \to 1$ as $x \to \infty$, $F(x) \to 0$ as $x \to -\infty$.

3. For a discrete r.v. taking values x, F is a step function with steps of size $p(x)$ at x. If X is discrete then F is not continuous but it is **right-continuous**.

4. For a discrete r.v. X, $F(x) = \displaystyle\sum_{\text{all } y \leq x} p(y)$.

Proofs: We will not prove all of these properties, but we will look at the proof of the first. To show that F is non decreasing we need to show

$$F(x) \leq F(y) \text{ if } x < y.$$

To show this note that if $x < y$

$$\{X \leq x\} \subseteq \{X \leq y\}.$$

Then, we know that it holds that $P(X \leq x) \leq P(X \leq y)$. Hence $F(x) \leq F(y)$. This is what was required and we have shown that F is non-decreasing.

The reason the distribution function is such an important function is that if we know $F(x)$ we can find various probabilities, e.g.

$$\begin{aligned} P(X > a) &= 1 - P(X \leq a) = 1 - F(a) \\ P(a < X \leq b) &= P(X \leq b) - P(X \leq a) = F(b) - F(a) \quad (a < b) \end{aligned}$$

In fact if we know the distribution function, we know everything we need to know about the random variable. Similarly for the probability mass function. One might be obtained if the other is known.

Exercise: By way of revision, use the axioms of probability to justify the first step in the second example here. Start by writing:

$$\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\}$$

Now continue. Justify every step in your working.

2.2 Expectation of a discrete random variable

If a fair die is thrown a large number of times, then each outcome $1, 2, \ldots, 6$, occurs on approximately 1/6 of trials. The average score (averaged over a large number of trials) is approximately

$$\left(1 \times \frac{1}{6}\right) + \left(2 \times \frac{1}{6}\right) + \left(3 \times \frac{1}{6}\right) + \left(4 \times \frac{1}{6}\right) + \left(5 \times \frac{1}{6}\right) + \left(6 \times \frac{1}{6}\right) = 3.5 \; . \tag{2.1}$$

The value 3.5 is an idealized long-term average: if you play a game where, each time the die is thrown and shows x, you receive £x, then in the long term you will have received about £3.50 per play, on average. Note that (2.1) is equal to

$$1 \times P(X = 1) + 2 \times P(X = 2) + \ldots + 6 \times P(X = 6).$$

Definition: The *expectation* (or *expected value* or *mean*) of a discrete r.v. X taking values x is

$$E(X) = \sum_{\text{all } x} x \; P(X = x) = \sum_{\text{all } x} x \; p(x) \; ,$$

provided that the sum is well-defined.

The sum above will always be well-defined if the number of values x that the random variable X is allowed to take is finite. If the set of values of X is infinite, difficulties arise, since infinite sums are not guaranteed to be well-defined. In this case, we shall require absolute convergence of the sum, so as to get the same answer irrespective of the order of summation.

We will require: $\sum_{\text{all } x} |x| \; p(x) < \infty$ in order for $E(X)$ to be defined.

If X is the score on a fair die, then $E(X) = 3.5$.

Note that $E(X)$ is a *number*, not a random variable (X is not a number, it's a rule for allocating numbers to outcomes of an experiment). $E(X)$ need not be a value that X can take (e.g. $E(X) = 3.5$ in the example of the die). We often use the Greek letter μ to denote $E(X)$. Sometimes we may use μ_X to emphasize 'the mean of X'.

Example 2.4: For X in Example 2.2, we have

$$E(X) = (-3 \times 8/27) + (-1 \times 12/27) + (1 \times 6/27) + (3 \times 1/27) = -1 \; .$$

So in the long run we would expect to lose £1 per game (where each game consists of 3 throws - see example 2.2).

A game in which the winnings X are such that $E(X) = 0$ is said to be a *fair game* — in example 2.4, the game is not fair.

Expectation of a function of a r.v.

If $g : I\!R \to I\!R$ and X is a r.v., then $g(X)$ is also a r.v. If we write $Y = g(X)$, we mean that whenever X takes the value x, Y takes the value $g(x)$. For example, X^2, $\sqrt{X}$, $5X + 10$ and 2^X are all r.v.'s.

Proposition: Let X be a discrete r.v. taking values x, and let $g : I\!R \to I\!R$. Put $Y = g(X)$. Then

$$E(Y) = E(g(X)) = \sum_x g(x)\ P(X = x)\ .$$

Proof: Note that $Y = g(X)$ is a r.v. Its pmf is given by

$$P(Y = y) = P(g(X) = y) = \sum_{x:g(x)=y} P(X = x)\ ,$$

so that

$$\begin{aligned} E(Y) &= \sum_y y\ P(Y = y) \\ &= y_1\ P(Y = y_1) + y_2\ P(Y = y_2) + \dots \\ &= y_1 \sum_{x:g(x)=y_1} P(X = x) + y_2 \sum_{x:g(x)=y_2} P(X = x) + \dots \\ &= \sum_{x:g(x)=y_1} g(x)\ P(X = x) + \sum_{x:g(x)=y_2} g(x) P(X = x) + \dots \\ &= \sum_x g(x) P(X = x)\ . \end{aligned}$$

Example 2.5: X is the number of heads obtained when a fair coin is thrown twice. From Example 2.3, its pmf is

x	0	1	2
$P(X = x)$	1/4	1/2	1/4

Find $E(X)$, $E(X^2)$ and $E(2X + 5)$.

Solution:

$$E(X) = 0 \times 1/4 + 1 \times 1/2 + 2 \times 1/4 = 1.$$

$E(X^2)$ is of the form $E(g(X))$, where $g(x) = x^2$. So,

$$E(X^2) = \sum_x x^2\ P(X = x) = 0^2 \times 1/4 + 1^2 \times 1/2 + 2^2 \times 1/4 = 3/2.$$

$E(2X+5)$ is of the form $E(g(X))$, where $g(x) = 2x+5$. So,

$$E(2X+5) = (2\times 0+5)\times(1/4) + (2\times 1+5)\times(1/2) + (2\times 2+5)\times(1/4) = 7.$$

Note in this example that $E(2X+5) = 2E(X)+5$, but that $E(X^2)$ is **NOT** equal to $(E(X))^2$. The $2X+5$ example is a special case of one of the properties of expectation (i.e. that it is LINEAR operator).

Jensen's inequality: If $g(x)$ is a twice-differentiable function and it holds that $g''(x) \geq 0$ or $g''(x) \leq 0$ for all $x \in \mathbb{R}$, then it holds that

$$E(g(X)) \geq g(E(X)) \qquad \text{or} \qquad E(g(X)) \leq g(E(X)),$$

respectively.

Properties of Expectation

These should be learned.

1. If a and b are constants then $E(aX+b) = aE(X)+b$. To see this, note that

   $$E(aX+b) = \sum_x (ax+b)P(X=x) = a\sum_x x\ P(X=x) + b\sum_x P(X=x)\ .$$

   The first sum is $E(X)$ by definition, and the second sum is 1 — the result follows.

2. If X does not take negative values then $E(X) \geq 0$ (because $E(X) = \sum_x x\ p(x)$, and none of the terms in the sum can be negative).

3. If X is a constant c (by which we mean that $P(X=c)=1$), then $E(X) = c$.

These properties are very useful.
E.g. if Y has $E(Y) = 2$ then $E((Y+6)/3) = E(Y/3+2) = E(Y)/3+2 = 8/3$.

The expectation of X gives a *measure of location* (or *central tendency*) of X. We can think of $E(X)$ as the center of gravity (or balance point) of mass distributed on $\mathbb{R}$ according to the pmf of X.

Examples:

1. X is the score on a fair die: We see immediately that $E(X) = 3.5$ by symmetry (no need to calculate).

2. X as in Example 2.3: Again, the pmf is symmetric so $E(X) = 1$.

Remark: The symmetry argument is a useful labor-saving one. It cannot generally be applied if X takes values in an infinite set though. **Why not?**

Example 2.6: Let Y be the total score when two fair dice are thrown. We may write the following table:

		Die 1					
		1	2	3	4	5	6
Die 2	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Thus, the random variable Y takes values in $\{2, 3, \ldots, 12\}$. Its pmf (count points in the sample space) is

y	2	3	4	5	6	7	8	9	10	11	12
$P(Y = y)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

By symmetry, $E(Y) = 7$.
Now observe that Y can be thought of as $X_1 + X_2$, where X_1 is the score on the first die and X_2 the score on the second. We already know $E(X_1) = E(X_2) = 3.5$.
Note that $E(Y) = E(X_1) + E(X_2)$.
Indeed, we may write

$$\begin{aligned}
E(X_1 + X_2) &= \sum_{x_1=1}^{6} \sum_{x_2=1}^{6} (x_1 + x_2)\ P(X_1 = x_1, X_2 = x_2) \\
&= \sum_{x_1=1}^{6} \sum_{x_2=1}^{6} x_1\ P(X_1 = x_1, X_2 = x_2) + \sum_{x_2=1}^{6} \sum_{x_1=1}^{6} x_2\ P(X_1 = x_1, X_2 = x_2) \\
&= \sum_{x_1=1}^{6} x_1 \sum_{x_2=1}^{6} P(X_1 = x_1, X_2 = x_2) + \sum_{x_2=1}^{6} x_2 \sum_{x_1=1}^{6} P(X_1 = x_1, X_2 = x_2) \\
&= \sum_{x_1=1}^{6} x_1\ P(X_1 = x_1) + \sum_{x_2=1}^{6} x_2\ P(X_2 = x_2) = E(X_1) + E(X_2).
\end{aligned}$$

This example is a special case of the following result (we may think of 'induction', in order to prove it):

Proposition: For random variables $X_1, \ldots, X_n$, we have

$$E(X_1 + \ldots + X_n) = E(X_1) + \ldots + E(X_n)\ .$$

2.3 Variance of a discrete random variable

Example 2.7: A fair coin is thrown. In Game A you win £1 if a head is obtained, and lose £1 otherwise. In Game B you win £100 for a head, and lose £100 for a tail. Let X be your winnings in Game A and Y the winnings in Game B (both in pounds). The random variable X takes the values +1 and -1, each with probability 1/2: $E(X) = 0$. The r.v. Y takes the values +100 and -100, each with probability 1/2: $E(Y) = 0 = E(X)$. So both games are fair. However, in Game B you can win more or lose more than in Game A; your winnings vary more.

This example illustrates the need to measure the variability of a r.v. We introduce the following quantity:

Definition: Let X be a r.v. with $E(X) = \mu$. The *variance* of X is

$$\mathrm{Var}(X) = E\Big[(X - E(X))^2\Big] = E\Big[(X - \mu)^2\Big] \ ,$$

providing the expectation is defined (i.e. the appropriate sum is absolutely convergent).

The *standard deviation* of X is $+\sqrt{\mathrm{Var}(X)}$.

Notes:

1. From the definition of expectation of a function of a r.v.,

$$\mathrm{Var}(X) = \sum_x (x - \mu)^2 P(X = x).$$

2. $\mathrm{Var}(X)$ is often denoted by σ^2 (or σ_X^2 if we wish to emphasize that we mean the variance of X rather than of another r.v.). The standard deviation of X is often denoted by σ or σ_X.

3. $\mathrm{Var}(X) \geq 0$, with equality only when all the probability is at a single point (i.e. $\mathrm{Var}(X) = 0$ only if $P(X = c) = 1$, for some $c \in I\!R$; in that case $c = E(X)$). The variance gets bigger as points with positive probability are spread out more.

4. The standard deviation of X has the same units of measurement as X itself. E.g. if X is a length in m, the standard deviation is also in m; the variance would be in m^2.

Example 2.7 (*continued*): For X in Example 2.7, the pmf is

x	-1	1
$P(X = x)$	1/2	1/2

$\mu_X = E(X) = 0$, and $\text{Var}(X) = E\Big[(X - \mu)^2\Big] = E(X^2) = ((-1)^2 \times 1/2) + (1^2 \times 1/2) = 1$.

For Y in Example 2.7, the pmf is

y	-100	100
$P(Y = y)$	1/2	1/2

$\mu_Y = E(Y) = 0$, and $\text{Var}(Y) = (-100 - 0)^2 \times 0.5 + (100 - 0)^2 \times 0.5 = 10,000$,

which is much greater than $\text{Var}(X)$, reflecting the increased variability in your winnings for this game.

There are two useful alternative formulae for $\text{Var}(X)$:

(a) $\text{Var}(X) = E(X^2) - \mu^2$

(b) $\text{Var}(X) = E(X(X-1)) - \mu(\mu - 1)$

where $\mu = E(X)$ in both cases.

The first of these formulae is derived from the definition of variance as follows:

$$\begin{aligned}
\text{Var}(X) &= E\Big[(X-\mu)^2\Big] = E\Big[X^2 - 2\mu X + \mu^2\Big] \\
&= E(X^2) - 2\mu E(X) + \mu^2 \\
&= E(X^2) - 2\mu^2 + \mu^2 \qquad \text{(since } E(X) = \mu\text{)} \\
&= E(X^2) - \mu^2 \,.
\end{aligned}$$

The second formula can be derived very quickly from the first. Can you do that?

Example 2.8: For X as in Example 2.2 (3 throws of a biased coin with $P(H) = 1/3$), we know from Example 2.4 that $E(X) = -1$. To find $\text{Var}(X)$, formula (a) above is easier to use than the definition:

$$E(X^2) = \left((-3)^2 \times \frac{8}{27}\right) + \left((-1)^2 \times \frac{12}{27}\right) + \left(1^2 \times \frac{6}{27}\right) + \left(3^2 \times \frac{1}{27}\right) = 11/3 \,.$$

So $\text{Var}(X) = E(X^2) - (E(X))^2 = 11/3 - (-1)^2 = 8/3$.

Properties of variance

Again, these should be learned.

1. If a and b are constants then

$$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X).$$

To see this, let $Y = aX + b$. Then $E(Y) = aE(X) + b$ so that $Y - E(Y) = a(X - E(X))$. So

$$\begin{aligned}\mathrm{Var}(aX+b) &= \mathrm{Var}(Y) = E\Big[(Y - E(Y))^2\Big] \\ &= E\Big[a^2\,(X - E(X))^2\Big] = a^2 E\Big[(X - E(X))^2\Big] = a^2\mathrm{Var}(X)\ .\end{aligned}$$

If $a = 1$ we get $\mathrm{Var}(X+b) = \mathrm{Var}(X)$: translation does not affect spread.

2. If X is constant i.e. $P(X = c) = 1$, then $\mathrm{Var}(X) = 0$.
To see this, note that in that case $E(X) = c$, so that

$$\mathrm{Var}(X) = E\Big[(X-c)^2\Big] = (c-c)^2 = 0.$$

Compare these properties with those of expectation — and be aware of differences!

Note: Variance of a function of a random variable:
If $g : I\!\!R \to I\!\!R$ and X is a r.v., then $\mathrm{Var}(g(X)) = E\Big[(g(X) - E(g(X)))^2\Big]$.

We can find it from

$$\begin{aligned}\mathrm{Var}(g(X)) &= E\Big[(g(X))^2\Big] - [E(g(X))]^2 \\ &= \sum_x \left(g(x)^2\right) P(X=x) - \left(\sum_x g(x)P(X=x)\right)^2\ .\end{aligned}$$

Higher moments

The mean $E(X)$ and variance $\mathrm{Var}(X)$ of a random variable quantify the location and spread of its distribution. However, further quantities of interest might be the skewness $E(X^3)$ and kurtosis $E(X^4)$ to characterise the random variable. In general, we can define any such quantity providing the relevant sums are absolutely convergent:

1. The k - *th moment of* X is defined as $E(X^k)$, for $k = 1, 2, \ldots$
2. If α is a real number, then the k - *th moment of* X *about* α is defined as $E((X-\alpha)^k)$, for $k = 1, 2, \ldots$

So $E(X)$ is the first moment of X, $E(X^2)$ is the second moment of X and $\mathrm{Var}(X)$ is the second moment of X about μ.

Unfortunately, calculating $E(X^k)$ is not always straightforward. A common 'trick' is to instead calculate $E(X \times (X-1)\ldots(X-k+1)$ or $E(X(X+1)\ldots\ (X+k-1)$ and then back-compute $E(X^k)$.

For **example:**

$$\begin{aligned} E(X^2) &= E(X(X-1)+X) = E(X(X-1)) + E(X) \\ E(X^2) &= E(X(X+1)-X) = E(X(X+1)) - E(X). \end{aligned}$$

Similar expressions can be derived for $E(X^3)$ and so on.

2.4 Discrete random variables and independence

Earlier in the course, we said that two events A and B were (stochastically) independent if $P(A \cap B) = P(A)P(B)$. We can extend this concept to the case of random variables.

Definition: Discrete random variables X and Y are *independent* if, for all x and y, the events $\{X = x\}$ and $\{Y = y\}$ are independent i.e.

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

The following important property can be shown to hold for **independent** random variables (note that it is **not** generally true otherwise):

Property: If X and Y are independent, then $E(XY) = E(X)E(Y)$.

Note: This condition $E(XY) = E(X)E(Y)$ on its own does **not** imply that X and Y are independent. To see this consider the example where X can equal 0 or 1 and Y can equal 0, 1 or 2. Suppose the probabilities of the events $\{X = x, Y = y\}$ are given in the following table:

		Y	
	0	1	2
X 0	1/4	0	1/4
1	0	1/2	0

Then

$$\begin{aligned}
E(X) &= 0 \times P(X = 0) + 1 \times P(X = 1) = 0 \times 1/2 + 1 \times 1/2 = 1/2 \\
E(Y) &= 0 \times P(Y = 0) + 1 \times P(Y = 1) + 2 \times P(Y = 2) = 0 \times 1/4 + 1 \times 1/2 + 2 \times 1/4 = 1 \\
E(XY) &= 0 \times 0 \times 1/4 + 0 \times 1 \times 0 + 0 \times 2 \times 1/4 + 1 \times 0 \times 0 + 1 \times 1 \times 1/2 + 1 \times 2 \times 0 = 1/2.
\end{aligned}$$

Though $E(XY) = E(X)E(Y)$, the variables X and Y are **not** independent!

For example,
$P(X = 0) = 1/2$ and $P(Y = 0) = 1/4$ but $P(X = 0, Y = 0) = 1/4 \neq P(X = 0)P(Y = 0)$.

Proposition: If X and Y are independent, then

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Proof (**exercise**): Let X and Y be independent random variables. Then, for example,

$$\begin{aligned}
\mathrm{Var}(X-Y) &= E\Big[(X-Y)^2\Big] - [E(X-Y)]^2 \\
&= E\Big(X^2+Y^2-2XY\Big) - (\mu_X - \mu_Y)^2 \\
&= E(X^2) + E(Y^2) - 2E(XY) - \mu_X^2 + 2\mu_X\mu_Y - \mu_Y^2 \\
&= [E(X^2) - \mu_X^2] + [E(Y^2) - \mu_Y^2] - 2[E(XY) - \mu_X\mu_Y] \\
&= \mathrm{Var}X + \mathrm{Var}(Y).
\end{aligned}$$

For INDEPENDENT random variables $X_1, \ldots, X_n$, it holds that

$$\mathrm{Var}(X_1 + \ldots + X_n) = \mathrm{Var}(X_1) + \ldots + \mathrm{Var}(X_n).$$

2.5 Further results on expectation and variance

When we introduced the idea of expectation, it was intended to represent some kind of long-run average behavior of a random variable. However, we have not yet proved that it really does have this interpretation. This is what we will now do.

What do we mean, in mathematical terms, by 'long-run average behavior' of a random variable? The idea is something like this: you repeat an experiment over and over again. At each repetition you observe the outcome of the experiment, and the associated value of the r.v. You then calculate the average of all the values you've observed so far, and see what happens as you continue indefinitely. Let's try and make that precise:

Let X_n be the random variable associated with the n-th *independent* repetition of the experiment (e.g. in a sequence of coin-tossing experiments, X_n might take the value 1 if the n-th toss is a head, and 0 for a tail). All the X's have the same *identical* probability distribution, so they are called *Independent and Identically Distributed* or *IID* random variables. As they are identically distributed they have the same mean and variance which we will call μ and σ^2. On the n-th repetition, the outcome you observe is ω_n say, which is in the sample space Ω for the experiment. Once you know ω_n, you can translate that to a value for X_n, which we will call x_n (formally, we would write $X_n(\omega_n) = x_n$). The average of the first n values is itself a random variable, say

$$\overline{X}_n = \frac{1}{n}\left(X_1 + \ldots + X_n\right) \ .$$

This means that when X_1 takes the value x_1, X_2 takes the value $x_2, \ldots$, and X_n takes the value x_n, then $\overline{X}_n$ takes the value

$$\overline{x}_n = \frac{1}{n}\left(x_1 + \ldots + x_n\right) \ .$$

In Part 2 of these notes, we will see how we count on quantities such as $\overline{X}_n$ from a random sample, in order to make some formal conclusions on the relevant quantities of the population (eg. μ).

So when we talk about the 'long-run average', we are talking about the random variable $\overline{X}_n$ when n is large. What are the properties of this random variable? Its expectation is

$$\begin{aligned} E\Big(\overline{X}_n\Big) &= E\Big[\frac{1}{n}\left(X_1 + \ldots + X_n\right)\Big] \\ &= \frac{1}{n}E\left(X_1 + \ldots + X_n\right) \\ &= \frac{1}{n}\left(E(X_1) + \ldots + E(X_n)\right) \\ &= \frac{1}{n}\left(\mu + \ldots + \mu\right) = \mu \ . \end{aligned}$$

In a similar way, we can show that the variance of $\overline{X}_n$ is σ^2/n. This is left as an **exercise.**

So as $n \to \infty$, $E(\overline{X}_n)$ remains at μ and the variance tends to zero. This looks like the kind of result we're looking for — because we know that if a random variable has a very small variance then it's very unlikely to take values which are a long way from its mean. But do we really know that? We haven't properly proved *that* yet, either. If we can prove it, then we will have succeeded in showing that expectation represents long-run average behavior. The result we need is

Chebyshev's inequality: If X is a random variable with mean μ and variance σ^2 then, for any value $k > 0$ or $m = k/\sigma$, it holds that

$$P\left(|X-\mu| \geq k\right) \leq \frac{\sigma^2}{k^2} \quad \text{or} \quad P\left(|X-\mu| \geq m\sigma\right) \leq \frac{1}{m^2}, \quad \text{for} \quad m = \frac{k}{\sigma}.$$

Note that the probability on the left hand side is $P\left(X \leq \mu - m\sigma\right) + P\left(X \geq \mu + m\sigma\right)$, which is the probability that X is more than m standard deviations away from its mean.

Let us now put Chebyshev's inequality together with the previous results. We have seen that $E(\overline{X}_n) = \mu$ and $\text{var}(\overline{X}_n) = \sigma^2/n$. Hence

$$P\left(|\overline{X}_n - \mu| \geq k\right) \leq \frac{\sigma^2/n}{k^2} = \frac{\sigma^2}{nk^2} \to 0 \text{ as } n \to \infty.$$

This is called the **Weak Law of Large Numbers**. It demonstrates that the long-run average of a random variable (obtained by repeating an experiment over and over again) is very unlikely to be far from its expectation, since the variance of the long-run average gets smaller and smaller.

2.6 Basic and further probability distributions for discrete random variables

The Bernoulli distribution - *Revision*

An experiment takes place and there are 2 possible outcomes (if there are more, we can turn them into 2), say ‘S’ for ‘success’ and ‘F’ for ‘failure’. ‘Success’ and ‘failure’ are conventional names, meaning what we are or we are not interested in, respectively. The probability of success is $P(S) = p$ with $p \in (0, 1)$, and the probability of failure is $P(F) = 1 - P(S) = 1 - p$.

We consider the Bernoulli random variable X to be ‘the number of successes we have if we perform the experiment’. We write

$$X \sim \text{Bin}(1, p).$$

We can either have no success at all, or one success. Thus,

$$P(X = x) = p, \text{ if } x = 1, \quad P(X = x) = 1 - p, \text{ if } x = 0.$$

Notice that this is well-defined, since $p + (1 - p) = 1$. It holds that

$$E(X) = p, \quad \text{Var}(X) = p(1 - p).$$

Can you show that?

The Binomial distribution - *Revision*

We repeat a number of n Bernoulli trials INDEPENDENTLY ($n \in \mathbb{N}$). In each trial, the probability of success is the same $P(S) = p \in (0, 1)$.

The Binomial random variable X is ‘the number of successes we have counted, after the n independent Bernoulli trials have taken place’. We write

$$X \sim \text{Bin}(n, p).$$

The probability mass function is

$$P(X = x) = \binom{n}{x} p^x \, (1 - p)^{n-x}, \; x = 0, \ldots, n.$$

Can you use everything you know to explain why the probabilities are equal to that?

DO NOT FORGET to write $x = 0, \ldots, n$, in order to make sure that the probability is greater than zero (and equal to this number) only for these values. In the contrary, the cumulative distribution function is defined for all the real numbers.

Can you write the cumulative distribution function?

It holds that

$$\sum_{x=0}^{n} P(X = x) = \sum_{x=0}^{n} \binom{n}{x} p^x \, (1-p)^{n-x} = 1.$$

Can you prove that? If not, do not forget to TAKE IT FOR GRANTED FOR YOUR PROOFS. Given this, show that

$$E(X) = np, \quad \mathrm{Var}(X) = np(1-p).$$

Exercise: For any positive integer k, what is $E(X(X-1)\ldots(X-k+1))$ equal to (if it exists)?

Solution:

$$\begin{aligned}
E(X(X-1) & \cdots (X-k+1)) \\
&= \sum_{x=0}^{n} x \cdot (x-1) \cdots (x-k+1) P(X=x) \\
&= \sum_{x=k}^{n} x \cdot (x-1) \cdots (x-k+1) P(X=x) \\
&= \sum_{x=k}^{n} \frac{x!}{(x-k)!} \binom{n}{x} p^x (1-p)^{n-x} \\
&= \sum_{x=k}^{n} \frac{x!}{(x-k)!} \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\
&= \sum_{x=k}^{n} \frac{n!}{(x-k)!(n-x)!} p^x (1-p)^{n-x} \\
x^* = x - k : & \\
&= \sum_{x^*=0}^{n-k} \frac{n!}{x^*!(n-k-x^*)!} p^{x^*+k} (1-p)^{n-k-x^*} \\
n^* = n - k : & \\
&= p^k \sum_{x^*=0}^{n^*} \frac{(n^*+k)!}{x^*!(n^*-x^*)!} p^{x^*} (1-p)^{n^*-x^*}
\end{aligned}$$

$$
\begin{aligned}
&= p^k \frac{(n^*+k)!}{n^*!} \sum_{x^*=0}^{n^*} \frac{n^*!}{x^*!(n^*-x^*)!} p^{x^*}(1-p)^{n^*-x^*} \\
&= p^k \frac{n!}{(n-k)!} \sum_{x^*=0}^{n^*} \left(\begin{array}{c} n^* \\ x^* \end{array} \right) p^{x^*}(1-p)^{n^*-x^*} = \frac{n! p^k}{(n-k)!}
\end{aligned}
$$

Alternatively, remember that the number of successes in the n independent trials is equal to 'the number of successes in the first trial', say X_1, + 'the number of successes in the second trial', say X_2, $+ \ldots +$ 'the number of successes in the n-th trial', say X_n. Then

$$X = X_1 + \ldots + X_n.$$

The random variables $X_i,\ i = 1, \ldots, n$, are Independent and Identically Distributed (all are Bernoulli random variables with same parameter-probability of success p). Can you use that to prove what $E(X)$ and $\mathrm{Var}(X)$ are equal to now?

Example 2.9: We toss a fair coin three times. Every time it shows 'Head' we gain £1. Otherwise, we get nothing. Next, we toss a fair die four times. Every time it shows an even number, we gain £0.5. Otherwise, we get nothing. After that, we toss the die another three times and we gain £1, every time it shows $\{1\}$. What is the expected number of money we are going to gain? What is the variance of the same random variable?

The Geometric distribution - *Revision*

We repeat a number of independent Bernoulli trials with probability of success $p \in (0, 1)$, until we record the first success. Thus, the number of trials (or the number of failures) is a random variable.

Let the r.v. X be 'the number of independent Bernoulli trials with same probability of success, until we record one success'. We may also write the r.v. X^* for 'the number of failures until we record one success'. See that

$$X^* = X - 1.$$

We write

$$X \sim \mathrm{NB}(1, p)$$

The probability mass function is

$$P(X = x) = (1-p)^{x-1} p,\ x = 1, 2, \ldots.$$

Can you tell why we do not have any combinations-permutations factor now?

It is not hard to tell that $P(X > x) = (1-p)^x$. Can you write down the cumulative distribution function?

It holds that
$$\sum_{x=1}^{\infty} P(X=x) = \sum_{x=1}^{\infty} (1-p)^{x-1} p = 1.$$
Then
$$E(X) = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}$$

Exercise (not examinable): For any positive integer k, what is $E(X(X+1)\ldots(X+k-1))$ equal to (if it exists)? Can you also compute $E(X \times (X-1) \times \ldots \times (X-k+1))$?

Solution (not examinable): See lecture slides.

Example 2.10: John likes to play 'Minesweeper' with his laptop. As he is really good, he finishes successfully approximately 9 out of 10 games. His younger brother James wants to play too. 'When are you going to let me play?' he asks John. 'When I lose my first game', John replies. Around how many games will James have to wait until he is allowed to play? (BE CAREFUL WHAT YOU CONSIDER AS 'SUCCESS'!)

The Poisson distribution - *Revision*

This distribution is named after Siméon Poisson (1781–1840). We consider an interval or surface (over time or space), on which an event takes place rarely with rate $\mu > 0$.

> Let the r.v. X be 'the number of events we recorded on one such surface (or length) we had available'. We write
> $$X \sim \mathrm{Poi}(\mu).$$
> The probability mass function is
> $$P(X=x) = \frac{e^{-\mu}\mu^x}{x!}, \; x = 0, 1, \ldots.$$

So far, we have been able to understand why mass functions take the forms we give them. For the case of the Poisson distribution, we will see later how it arises as an approximation to another distribution.

It holds that
$$\sum_{x=0}^{\infty} P(X=x) = \sum_{x=0}^{\infty} \frac{e^{-\mu}\mu^x}{x!} = 1.$$
Then

$$E(X) = \mu, \quad \mathrm{Var}(X) = \mu$$

The fact that the expected value of a Poisson random variable is equal to its variance is a rare property (rarely verified in practice).

Exercise: For any positive integer k, what is $E(X(X-1)\dots(X-k+1))$ equal to (if it exists)?

Solution:

$$\begin{aligned}
E(X(X-1)\dots(X-k+1)) &= \sum_{x=0}^{\infty} x(x-1)\dots(x-k+1)p(x) \\
&= \sum_{x=k}^{\infty} x(x-1)\dots(x-k+1)p(x) \\
&= \sum_{x=k}^{\infty} \frac{x!}{(x-k)!} p(x) = \sum_{x=k}^{\infty} \frac{x!}{(x-k)!} \frac{e^{-\mu}\mu^x}{x!} \\
&= \sum_{x=k}^{\infty} \frac{e^{-\mu}\mu^x}{(x-k)!} \\
(\text{ substitute: } x^* = x-k) &= \sum_{x^*=0}^{\infty} \frac{e^{-\mu}\mu^{x^*+k}}{x^*!} = \mu^k \sum_{x^*=0}^{\infty} \frac{e^{-\mu}\mu^{x^*}}{x^*!} = \mu^k \cdot 1 \\
&= \mu^k
\end{aligned}$$

Example 2.11: Springer-Verlag claim that it is very unlikely to discover a typo in their books and they give a rate of 1 typo for every 10,000 fully printed pages. They are willing to pay the amount of £100 to anyone, who will discover one (at least) typo in one of their books of 500 pages. If you search for a typo in the book, what is the probability that you will be paid by Springer-Verlag?

The Poisson distribution as an approximation to the binomial distribution

One of the uses of the Poisson distribution is as an approximation to the binomial distribution when n is large, p is small and np is moderate.

To show this, we need to know that for any $x \in \mathbb{R}$:

$$\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x.$$

Suppose X is a discrete random variable distributed as $\text{Bin}(n, p)$. We will set $\mu = np$ (so that $p = \mu/n$), and examine the behavior of the distribution of X as $n \to \infty$ with μ fixed. Now

$$\begin{aligned} P(X = x) = \left(\begin{array}{c} n \\ x \end{array} \right) p^x (1-p)^{n-x} &= \frac{n(n-1) \times \ldots \times (n-x+1)}{x!} \left(\frac{\mu}{n}\right)^x \left(1 - \frac{\mu}{n}\right)^{n-x} \\ &= \frac{n(n-1) \times \ldots \times (n-x+1)}{n^x x!} \mu^x \left(1 - \frac{\mu}{n}\right)^n \left(1 - \frac{\mu}{n}\right)^{-x} \end{aligned}$$

for any (fixed) $x \in \{0, 1, \ldots, n\}$. As $n \to \infty$, the following things happen:

- $\frac{n(n-1)\times\ldots\times(n-x+1)}{n^x} \to 1$
- $\left(1 - \frac{\mu}{n}\right)^n \to e^{-\mu}$ (by the result quoted above)
- $\left(1 - \frac{\mu}{n}\right)^x \to 1^x = 1$

(note the difference between the last two lines — in one, the exponent itself is tending to ∞ whereas in the other the exponent is fixed). Thus, letting $n \to \infty$ with $\mu = np$ fixed, we find that

$$\lim_{n \to \infty} P(X = x) = \frac{\mu^x e^{-\mu}}{x!}$$

which is equal to $P(Y = x)$ where Y is a Poisson random variable with mean μ. Note that this is true for every x. The conclusion therefore is that, provided that n is large enough and p is small enough, the Poisson distribution with mean $\mu = np$ provides a good approximation to the Binomial(n, p) distribution.

As a check, we can examine the behavior of the mean and the variance of the binomial distribution as $n \to \infty$ with $np = \mu$. If $X \sim \text{Bin}(n, p)$ then $E(X) = np = \mu$, so the Poisson approximation is exact for $E(X)$. Also, we have $\text{Var}(X) = np(1-p) = \mu(1-p) = \mu(1-\mu/n)$, which tends to μ as $n \to \infty$.

Example 2.12 Suppose that the probability of any particular character being misprinted in a book is 0.01, independently of other characters. Let X be the number of misprints in a block of text 600 characters long. Then $X \sim \text{Bin}(600, 0.01)$. Find $P(X = 0)$, $P(X = 1)$, $E(X)$ and $\text{Var}(X)$: (i) exactly, and (ii) using the Poisson approximation.

Solution: We first derive the exact results:

$$\begin{aligned} P(X = 0) &= \left(\begin{array}{c} 600 \\ 0 \end{array} \right) (0.01)^0 (0.99)^{600} = 0.00245 \\ P(X = 1) &= \left(\begin{array}{c} 600 \\ 1 \end{array} \right) (0.01)^1 (0.99)^{599} = 0.01458 \\ E(X) &= np = 600 \times 0.01 = 6 \\ \text{Var}(X) &= npq = 600 \times 0.01 \times 0.99 = 5.94\ . \end{aligned}$$

And under the Poisson approximation, with $\mu = 600 \times 0.01 = 6$:

$$\begin{aligned} P(X=0) &\approx \frac{6^0 e^{-6}}{0!} = 0.00248 \\ P(X=1) &\approx \frac{6^1 e^{-6}}{1!} = 0.01487 \\ E(X) &= \mu = 6 \\ \mathrm{Var}(X) &\approx \mu = 6\ . \end{aligned}$$

We see that the Poisson distribution gives very reasonable approximations to the exact results in this example.

The Poisson approximation can be a very useful one in practice, because Poisson probabilities are far easier to calculate than Binomial ones. In particular, Poisson probabilities can be computed very quickly and easily on a hand calculator, for note that if $X \sim \mathrm{Poi}(\mu)$ then for $x = 1, 2, \ldots$:

$$P(X=x) = \frac{\mu^x e^{-\mu}}{x!} = \frac{\mu}{x} \frac{\mu^{x-1} e^{-\mu}}{(x-1)!} = \frac{\mu}{x} P(X=x-1)$$

and $P(X=0) = e^{-\mu}$. So for hand calculation of a sequence of Poisson probabilities, we first calculate $P(X=0)$; then $P(X=1) = \mu P(X=0)/1$; then $P(X=2) = \mu P(X=1)/2$ and so on.

Example 2.13 Consider a gene consisting of $10,000$ nucleotides (i.e. 'letters' A,C,G,T). Assume that when this gene is passed on from parent to child, each nucleotide site mutates (i.e. changes letter) with probability 10^{-12}, independently of all other nucleotide sites. Find the probability that when the gene is passed on from parent to child, at least one nucleotide site mutates.

The Poisson Process — *general case*:

The Poisson process may be 'defined' as follows:

Assume that:

- Events ('accidents') occur randomly in time, in such a way that with each accident is associated a unique time instant. We will talk about 'accidents' from now on, to avoid confusion with 'events' in the context of sample spaces.
- The probability of 1 accident in a small time interval of length h time units is approximately λh (where λ is a constant).

- The probability of no accidents in the same time interval is approximately $1 - \lambda h$.
- The number of accidents in any time interval is independent of the number of accidents in any other time interval, providing the two intervals do not overlap.

If, for any $t > 0$, we denote by $N(t)$ the number of accidents occurring in the time interval $(0, t]$, then $N(t)$ is a random variable which has a Poisson distribution with mean λt. The *collection* of all these random variables, $\{N_t : t \in \mathbb{R}^+\}$, forms a *Poisson Process.* The parameter λ is called the *rate* of the process, and has the interpretation that it is the mean number of accidents occurring during one unit of time. We say that 'accidents occur according to a Poisson Process of rate λ'.

It can also be shown that the number of accidents occurring in *any* time interval $(a, b]$, with $b > a$, has a Poisson distribution with mean $\lambda(b - a)$. Note in particular that the number of accidents occurring during a given time interval will depend on the length of the interval but not on its position.

Example 2.14 Patients arrive at a doctor's surgery according to a Poisson process of rate 0.5 per minute. What is the probability that no patients arrive in a given 10 minute interval?

Example 2.15 Suppose that earthquakes occur on a particular island according to a Poisson process of rate 0.7 per year. (a) What is the probability that during a particular year at least two earthquakes occur? (b) What is the expected number of earthquakes during a time period of 20 years? (c) Calculate the probability that less than 5 earthquakes occur during a time period of 20 years.

The Poisson process model is often used in situations where accidents occur randomly in time *or space*, and where accidents occur

- Singly (i.e. $P(2 \text{ accidents at same point}) = 0$);
- Uniformly (Expected number of accidents in an interval is proportional to the length of the interval, but does not depend on its location);
- Independently (the number of accidents in any interval is independent of the number of accidents in any other non-overlapping interval).

In a two-dimensional spatial Poisson process of rate λ, the number of accidents in any region of area A is Poisson distributed with mean λA. The rate λ is measured in units of $(\text{area})^{-1}$, e.g. 'accidents per square meter'. References to 'interval' in the preceding discussion are replaced by references to 'region'. Similar extensions apply to processes in higher dimensions.

Example 2.16 Suppose that in a certain area, plants are randomly dispersed with an average density of 1.3 per square meter. A biologist randomly locates 100 sampling *quadrats* (rectangles), each of size 2m^2, in such a way that none of them overlap. Assuming a Poisson Process model for plant locations, find the expected number of sampling quadrats that contain no plants.

The Poisson Process model is often used as a 'first-guess' model for real-world phenomena, because of its simplicity. In practice it may often be unrealistic because of the independence between the number of accidents on two different intervals that do not overlap. However, it does provide a quick way to see whether dependence is present between intervals that do not overlap: if the Poisson Process model is correct, then the mean and the variance of counts should be equal. If, from counts of numbers of accidents in non-overlapping intervals of equal size, we find that the mean and the variance differ substantially, then we may conclude that there is interaction in the process being studied.

The Negative Binomial distribution

We repeat a number of independent Bernoulli trials with probability of success $p \in (0, 1)$, until we record the r-th success, where $r = 1, 2, \ldots$. Thus, the number of trials (or the number of failures) is a random variable.

Let the r.v. X be 'the number of independent Bernoulli trials with same probability of success, until we have r successes'. We may also write the r.v. X^* for 'the number of failures until we get r successes'. See that

$$X^* = X - r.$$

We write

$$X \sim \mathrm{NB}(r, p)$$

The probability mass function is

$$P(X = x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r,\ x = r, r+1, \ldots.$$

We can clearly see that the Geometric distribution (introduced before) is a special case of the Negative Binomial distribution.

Can you tell why we have the factor $\binom{x-1}{r-1} = \binom{x-1}{x-r}$ now?

Can you write down the cumulative distribution function for this probability mass function?

It holds that

$$\sum_{x=r}^{\infty} P(X=x) = \sum_{x=r}^{\infty} \left(\begin{array}{c} x-1 \\ r-1 \end{array} \right) (1-p)^{x-r} p^r = 1.$$

If you cannot prove it, TAKE IT FOR GRANTED. Then

$$\boxed{E(X) = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}}$$

Exercise: For any positive integer k, what is $E(X(X+1)\dots(X+k-1))$ equal to (if it exists)?

Solution:

$$\begin{array}{rl}
& E(X(X+1)\dots(X+k-1)) \\
= & \dfrac{p^r}{(1-p)^r} \displaystyle\sum_{x=r}^{\infty} x(x+1)\dots(x+k-1) \left(\begin{array}{c} x-1 \\ r-1 \end{array} \right) (1-p)^x \\
= & \dfrac{p^r}{(1-p)^r} \displaystyle\sum_{x=r}^{\infty} \dfrac{(x+k-1)!}{(x-1)!} \dfrac{(x-1)!}{(r-1)!(x-r)!} (1-p)^x \\
= & \dfrac{p^r}{(1-p)^r} \displaystyle\sum_{x=r}^{\infty} \dfrac{(x+k-1)!}{(r-1)!(x-r)!} (1-p)^x \\
\text{Call} & x^* = x+k \text{ (or } x = x^* - k) \text{ and rewrite:} \\
= & \dfrac{p^r}{(1-p)^r} \displaystyle\sum_{x^*-k=r}^{\infty} \dfrac{(x^*-1)!}{(r-1)!(x^*-k-r)!} (1-p)^{x^*-k} \\
= & \dfrac{p^r}{(1-p)^{r+k}} \displaystyle\sum_{x^*=r+k}^{\infty} \dfrac{(x^*-1)!}{(r-1)!(x^*-k-r)!} (1-p)^{x^*} \\
\text{Call} & r^* = r+k \text{ (or } r = r^* - k) \text{ and rewrite:} \\
= & \dfrac{p^r}{(1-p)^{r+k}} \displaystyle\sum_{x^*=r^*}^{\infty} \dfrac{(x^*-1)!}{(r^*-k-1)!(x^*-r^*)!} (1-p)^{x^*} \\
= & \dfrac{p^r (r^*-1)!}{(1-p)^{r+k} (r^*-k-1)!} \displaystyle\sum_{x^*=r^*}^{\infty} \dfrac{(x^*-1)!}{(r^*-1)!(x^*-r^*)!} (1-p)^{x^*} \\
= & \dfrac{p^r (r^*-1)!}{(1-p)^{r+k} (r^*-k-1)!} \displaystyle\sum_{x^*=r^*}^{\infty} \left(\begin{array}{c} x^*-1 \\ r^*-1 \end{array} \right) (1-p)^{x^*} \\
= & \dfrac{p^r (r^*-1)!}{(1-p)^{r+k} (r^*-k-1)}! \dfrac{(1-p)^{r^*}}{p^{r^*}} \\
= & \dfrac{p^r (r+k-1)!}{(1-p)^{r+k} (r-1)!} \dfrac{(1-p)^{r+k}}{p^{r+k}} = \dfrac{(r+k-1)!}{(r-1)!} \dfrac{1}{p^k}
\end{array}$$

Alternatively, remember that the number of trials until the r-th success is equal to 'the number of trials until the first success', say X_1, + 'the number of trials until the second success', say X_2,$+ \ldots +$ 'the number of trials until the r-th success', say X_r. Then

$$X = X_1 + \ldots + X_r.$$

The random variables $X_i,\ i = 1, \ldots, r$, are Independent and Identically Distributed (all are Geometric random variables, with same parameter, probability of success p). Can you use that to prove what $E(X)$ and $\mathrm{Var}(X)$ are equal to now?

Example 2.17 A large stockpile of used pumps contains 20% that are unusable and need repair. There are 3 repair kits. Pumps are selected for testing 1 at a time, at random. If one needs repairing, then one of the repair kits is used to repair it. It takes 10 minutes to test a pump if it works; 30 minutes to test and repair a pump that does not work. Find the expected value and variance of the total time it takes to use up the three kits.

Solution: Let X be the number of pumps tested until the repair kits are all used up. Then $X \sim \mathrm{NB}(3, 0.2)$, so $E(X) = 3/0.2 = 15$ and $\mathrm{Var}(X) = 3 \times 0.8/0.2^2 = 60$. Now let T denote the total time it takes to use up the repair kits. We note that
$T = (10\,(X - 3) + 90)$ minutes.
So $E(T) = 10(E(X) - 3) + 90 = 210$ minutes, and $\mathrm{Var}(T) = 100\ \mathrm{Var}(X) = 6000$.

Example 2.18 It is a well-attested fact that cats have 9 lives. Montgomery is a cat who lives in a house on a busy main road. On the other side of the road is a fish shop. Every Friday, Montgomery sprints across the road during the rush hour to steal a haddock, and then sprints back. The probability of his being hit by a car in any week is 1/20, independently from week to week; if he is hit, he loses one of his lives. Find his life expectancy (in weeks); the standard deviation of his lifespan; and the probability that he will survive for another 2 years (104 weeks), if (i) he still has all 9 lives intact (ii) he has 5 lives intact (iii) he has only 1 life intact.

Solution: Let X_r denote Montgomery's lifespan if he currently has r lives left. Then $X_r \sim NB(r, 1/20)$. His life expectancy is

$$E(X_r) = 20r.$$

The variance of his lifespan is

$$\mathrm{Var}(X_r) = r \times (19/20)/(1/20)^2 = 380r,$$

so the standard deviation is $\sqrt{380r}$. We have $E(X_9) = 180$ weeks and $\mathrm{Sd}(X_9) = 58.48$ weeks; $E(X_5) = 100$ weeks and $\mathrm{Sd}(X_5) = 43.6$ weeks; $E(X_1) = 20$ weeks and $\mathrm{Sd}(X_1) = 19.49$ weeks.

Note firstly, that the standard deviation is on the same scale as the random variable itself, so we can talk about a sd of '19 weeks'; and secondly, that the standard deviation increases with the number of lives remaining, reflecting an increased uncertainty about Montgomery's exact lifespan.

To find the probability of survival for another 2 years, we could try to evaluate

$$P(X_r > 104) = \sum_{x=105}^{\infty} \binom{x-1}{r-1} \left(\frac{1}{20}\right)^r \left(\frac{19}{20}\right)^{x-r}.$$

But this is not feasible. Recall when we found the probability of a geometric random variable exceeding some value x by noting that this was equivalent to the first x trials failing.

A similar approach holds for the negative binomial distribution. In this case we have

$$P(X_r > 104) = P(Y < r)$$

say, where Y is the number of lives lost by an IMMORTAL cat in 104 weeks. We have $Y \sim \text{Bin}(104, 1/20)$, so that

$$P(Y = y) = \binom{104}{y} \left(\frac{1}{20}\right)^y \left(\frac{19}{20}\right)^{104-y}.$$

We require $P(X_r > 104) = \sum_{y=0}^{r-1} P(Y = y)$ for $r = 1, 5$ and 9. For $r = 1$ we have

$$P(X_1 > 104) = P(Y = 0) = \left(\frac{19}{20}\right)^{104} = 0.005.$$

For $r = 5$ and $r = 9$, the calculations are still going to be quite lengthy (although feasible).

The Hypergeometric distribution

We explain the standard experiment, such that the hypergeometric distribution arises. Suppose that we have N units (N is a positive integer and $N \geq 2$), from which M (M is a positive integer too and $M < N$) share the same characteristic (and $N - M$ do not have that characteristic). We select n units (n is a positive integer and $n \leq N$) from the population of N, WITHOUT REPLACEMENT.

Let the r.v. X be 'the number of units in the sample of n (that we selected without replacement) that have the characteristic'. We write

$$X \sim \text{Hyper}(n, M, N).$$

The probability mass function is

$$P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \ldots, \min(n, M).$$

Using everything you know from combinatorics, could you explain why

$$\sum_{x=0}^{\min(n,M)} P(X=x) = \sum_{x=0}^{\min(n,M)} \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} = 1?$$

Can you show that

$$E(X) = n\,\frac{M}{N}, \qquad \mathrm{Var}(X) = \frac{nM(N-M)(N-n)}{N^2(N-1)}?$$

Exercise: For any positive integer k, what is $E(X(X-1)\dots(X-k+1))$ equal to (if it exists)?

Solution: $E(X(X-1)\dots(X-k+1))$

$$= \sum_{x=0}^{\min(n,M)} x(x-1)\cdots(x-k+1)\binom{M}{x}\binom{N-M}{n-x}\Big/\binom{N}{n}$$

$$= \sum_{x=k}^{\min(n,M)} \frac{x!}{(x-k)!}\,\frac{M!}{(M-x)!x!}\binom{N-M}{n-x}\Big/\binom{N}{n}$$

$$= \sum_{x=k}^{\min(n,M)} \frac{M!}{(M-x)!(x-k)!}\binom{N-M}{n-x}\Big/\binom{N}{n}$$

Call $x^* = x-k$ (or $x = x^*+k$) and rewrite:

$$= \frac{1}{\binom{N}{n}} \sum_{x^*=0}^{\min(n-k,M-k)} \frac{M!}{(M-x^*-k)!x^*!}\binom{N-M}{n-x^*-k}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x^*=0}^{\min(n-k,M-k)} \frac{M!}{(M-x^*-k)!x^*!}\binom{N-M}{n-x^*-k}$$

Call $n^* = n-k, M^* = M-k, N^* = N-k$ and rewrite:

$$= \frac{1}{\binom{N}{n}} \sum_{x^*=0}^{\min(n^*,M^*)} \frac{(M^*+k)!}{(M^*-x^*)!x^*!}\binom{N^*-M^*}{n^*-x^*}$$

$$= \frac{1}{\binom{N}{n}}\,\frac{(M^*+k)!}{M^*!} \sum_{x^*=0}^{\min(n^*,M^*)} \frac{M^*!}{(M^*-x^*)!x^*!}\binom{N^*-M^*}{n^*-x^*}$$

$$= \frac{1}{\binom{N}{n}} \frac{(M^*+k)!}{M^*!} \sum_{x^*=0}^{\min(n^*,M^*)} \binom{M^*}{x^*}\binom{N^*-M^*}{n^*-x^*}$$

$$= \frac{1}{\binom{N}{n}} \frac{(M^*+k)!}{M^*!} \sum_{x^*=0}^{\min(n^*,M^*)} \binom{M^*}{x^*}\binom{N^*-M^*}{n^*-x^*}$$

$$\text{Using } \sum_{x=0}^{\min(n,M)} \binom{M}{x}\binom{N-M}{n-x} = \binom{N}{n}:$$

$$= \frac{(M^*+k)!}{M^*!}\binom{N^*}{n^*}\Big/\binom{N}{n} = \frac{M!}{(M-k)!}\binom{N-k}{n-k}\Big/\binom{N}{n}$$

$$= \frac{M!}{(M-k)!}\frac{(N-k)!}{(N-n)!(n-k)!}\Big/\frac{N!}{(N-n)!n!}$$

$$= \frac{M!}{(M-k)!}\frac{(N-k)!}{N!}\frac{n!}{(n-k)!}$$

Alternatively, you should think as follows. First, write X_i, $i = 1, \dots, n$, for the result of the i-th trial, in the sense that X_i takes the value 1, if a unit with the characteristic of interest is drawn in the i-th trial. Of course, X_i takes the value 0, if the selected unit does not have the characteristic. What is the distribution of each random variable X_i?

Now, think of the probability of 'success' (finding a unit with the characteristic), say p_1, of the first Bernoulli random variable X_1. It holds that

$$p_1 = P(X_1 = 1) = M/N$$

However, we can compute

$$P(X_2 = 1|X_1 = 1) = (M-1)/(N-1), \qquad P(X_2 = 1|X_1 = 0) = M/(N-1).$$

What does this mean? Are the Bernoulli random variables X_1 and X_2 independent?

The fact that the random variables are not Independent should not be confused with the fact they might be Identically Distributed (are they?). We know that they are all Bernoulli random variables. If the probabilities of 'success' $p_1, \dots, p_n$, of all trials are equal, then the random variables are identically distributed. Can you compute $p_2 = P(X_2 = 1)$ now (use the total probability law)? Is that equal to $P(X_1 = 1)$?

Actually, it should not be a surprise that the Bernoulli random variables are identically distributed. All trials are taking place on the same population of N units with M units sharing the characteristic. The probability of 'success' is M/N. For example, if we have no information at all about what happened in the first trial, and we want to 'guess in the best way possible' the probability of success of the second trial, what can we say?

Since $X = X_1 + \dots + X_n$, can you compute again $E(X)$? What happens with $\mathrm{Var}(X)$?

Example 2.19 A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components at random from a lot and to accept the lot only if none of them are defective. If 30% of the lots have 4 defective components and 70% have only 1, what proportion of lots does the purchaser reject?

Approximating the Hypergeometric with the binomial

Intuitively, it seems reasonable to think that if N (the size of the population) is very large compared to n (the number of items we draw from it), then the probability of drawing an item with the characteristic of interest at each stage will hardly vary, and hence the experiment will be very similar to a sequence of independent Bernoulli trials with success probability $p = M/N$ at each trial. This in turn suggests that,

> when the ratio n/N is small, the hypergeometric (n, M, N) distribution may be approximated by the binomial (n, p) distribution with $p = M/N$.

It can be shown (but not here) that this is indeed the case.

As a quick check we can examine the behavior of $E(X)$ and $\text{Var}(X)$ when $X \sim \text{Hyper}(n, M, N)$ as $N \to \infty$ with (for convenience) n fixed and M/N fixed at a value of p (so $M = Np$). We have

$$E(X) = \frac{nM}{N} = np$$

and

$$\begin{aligned}
\lim_{N\to\infty} \text{Var}(X) &= \lim_{N\to\infty} \frac{nM(N-M)(N-n)}{N^2(N-1)} = \lim_{N\to\infty} n\frac{M}{N}\,\frac{N-M}{N}\,\frac{N-n}{N-1} \\
&= \lim_{N\to\infty} n\frac{M}{N}\left(1-\frac{M}{N}\right)\frac{1-n/N}{1-1/N} = np(1-p)
\end{aligned}$$

— both these expressions correspond to the appropriate formulae for the binomial distribution with parameters (n, p).

Chapter 3

Continuous random variables

In the previous chapter, we considered discrete r.v.'s where X takes only a finite or countable number of values. However, there are many r.v.'s that have more than countably many possible values, e.g. time until the train arrives (of course, we should already know which distribution might be used for DISCRETE time until an event happens - WHICH ONE IS IT?).

Example 3.1 - *The 'uniform' example*: When a pointer is spun, suppose that it is equally likely to stop anywhere in $[0, 4)$.

Let X be the value obtained. Then X is a random variable which can take any value in $[0, 4)$: *uncountably* many values, so X is *not* discrete.

> **(a)** In the discrete case, we assigned a probability to each possible value of X. Unfortunately this will not work here; a continuous random variable CAN NEVER BE EQUAL EXACTLY TO A NUMBER - the corresponding probability is equal to 0. However, we can use INTERVALS and check the probability that the random variable takes values in the interval of interest. We have an infinite number of intervals, to which we should assign probabilities (also the intervals do not correspond to mutually exclusive events like in the discrete case of a countable number of values).
>
> For example, we may be interested in computing the probabilities $P(X \leq 2)$, $P(X \leq 1)$, $P(2 < X \leq 3)$ or $P(1 < X \leq 1.1)$. Thus, we can find the probability that X is in a certain interval.
>
> **(b)** We may create intervals that do not cover each other (mutually exclusive events) but cover the whole of the set $[0, 4)$. Then, we may remember the CLASSICAL DEFINITION OF PROBABILITY. Imagine spinning the pointer many times, and recording the outcomes in a frequency table, grouping the results into intervals with boundaries $0, 1, 2, 3, 4$ (4 intervals). For a large number of spins we would expect approximately

equal numbers in each interval. We could represent the distribution of X like this:

$$P(X \in \mathcal{A}) = \frac{1}{4}, \ \mathcal{A} = [0,1), \ [1,2), \ [2,3), \ [3,4).$$

We see that the total area $= 4 \times \frac{1}{4} = 1$. The choice of interval length is arbitrary, so we can remove the interval boundaries.

(c) The distribution of the X is represented by a function

$$f(x) = \begin{cases} \frac{1}{4} & \text{if } 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}.$$

This is called the probability density function of X. The probability that X takes a value in $(a, b]$ is the area under $f(x)$ between a and b:

$$P(a < X \leq b) = \textbf{ Area under } f(x) \textbf{ between } a \textbf{ and } b \ = \int_a^b f(x)dx = \frac{b-a}{4},$$

where the last equality holds if $0 \leq a \leq b \leq 4$.

3.1 Continuous r.v., *pdf* and *cdf*

Definition: A *probability density function* (pdf) is a function f such that

(i) $f(x) \geq 0$ for every $x \in \mathbb{R}$

(ii) $\int_{-\infty}^{\infty} f(x)dx = 1$ (i.e. the integral over the whole real line).

Thus, pdf's are always non-negative and integrate to 1.

In Example 3.1(c), the total area under the graph of f is equal to 1.

Definition: A random variable X is called a *continuous random variable* if there exists a pdf $f(x)$ such that

$$P(a < X \leq b) = \int_a^b f(x)dx \qquad \text{for all } a, b \in \mathbb{R} \text{ with } a < b.$$

(i.e. if the distribution of X can be given by a pdf).

Examples:

- A toy example of a probability density function of a continuous random variable is given in Figure 3.1.
- X in Example 3.1 is a continuous r.v.

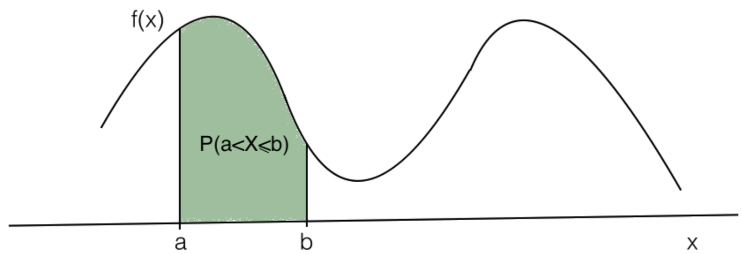

Figure 3.1: The probability density function of a random variable X. The area of the shaded region represents $P(a < X \leq b)$.

Example 3.2: Let X be a continuous r.v. with pdf

$$f(x) = \begin{cases} c(2-x), & \text{if } 0 \leq x \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

We can also plot this probability density function (see Figure 3.2).
Find c and find $P(X > 1.5)$, shown shaded in the plot.
Hint: the plot can often help you simplify your calculations.

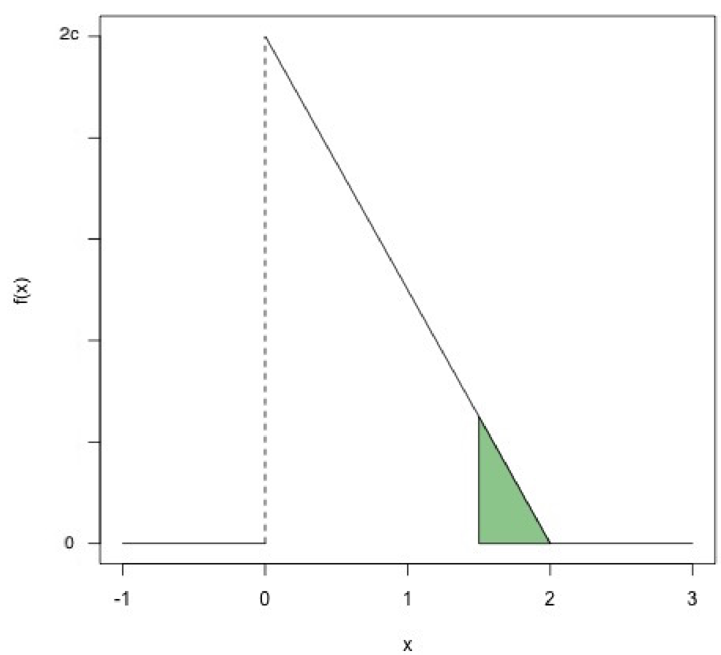

Figure 3.2: The probability density function of Example 3.2. The shaded region represents $P(X \geq 1.5)$.

BE CAREFUL! The number c is a CONSTANT - it must be equal to a specific value, so that f is a probability density function. Later, we will know the meaning of (and difference from) a PARAMETER.

Notes on pdf's:

1. For a continuous r.v. X,

$$P(X = a) = \int_a^a f(x)dx = 0.$$

So

$$\begin{aligned} P(a < X \leq b) &= P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) \\ &= \int_a^b f(x)dx. \end{aligned}$$

2. Heuristic interpretation of $f(x)\delta x$:

$$P(x \leq X \leq x + \delta x) = \int_x^{x+\delta x} f(t)dt \approx f(x)\delta x$$

i.e. for very small δx, the covered area is approximately equal to the area of the rectangle of height $f(x)$ and width δx. We think of $f(x)\delta x$ as being the (approximate) probability that X is in $[x, x + \delta x]$ (for small δx).

3. Recall from Chapter 2 that the *(cumulative) distribution function* F of a r.v. X is given by

$$F(x) = P(X \leq x), \text{ for every } x \in \mathbb{R}.$$

If the r.v. X is continuous with density f then the **(cumulative) distribution function** F will be a continuous function and

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt.$$

(Note: this could as well be written $\int_{-\infty}^x f(u)du$ or $\int_{-\infty}^x f(v)dv$ or $\int_{-\infty}^x f(y)dy$, as all these expressions give the area under the graph of f to the left of x.)

So given the pdf f, we find the cdf F by INTEGRATING. For example, see Figure 3.3 for a visual example of how we calculate the cumulative distribution at a point x_0 given a probability density function $f(x)$.

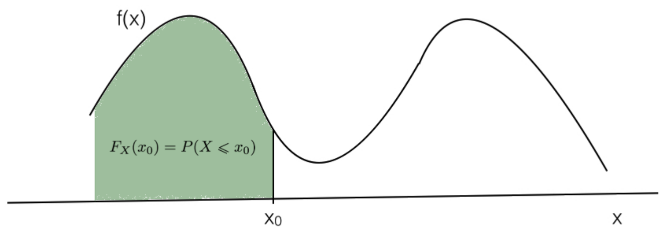

Figure 3.3: The probability density function of a random variable X. The area of the shaded region represents the cumulative distribution function at point x_0, $P(X \leq x_0) = F_X(x_0)$.

4. Given the distribution function F, we find f by DIFFERENTIATING:

$$f(x) = F'(x) = dF/dx.$$

5. We sometimes write $f_X(x)$ for the pdf of X, and $F_X(x)$ for the cdf of X.

Example 3.3 (*pointer example continued*):
Let X be the r.v. in Example 3.1, with pdf

$$f(t) = \begin{cases} \frac{1}{4} & \text{if } 0 \leq t \leq 4 \\ 0 & \text{otherwise.} \end{cases}$$

We can also plot this probability density function, shown in Figure 3.4.

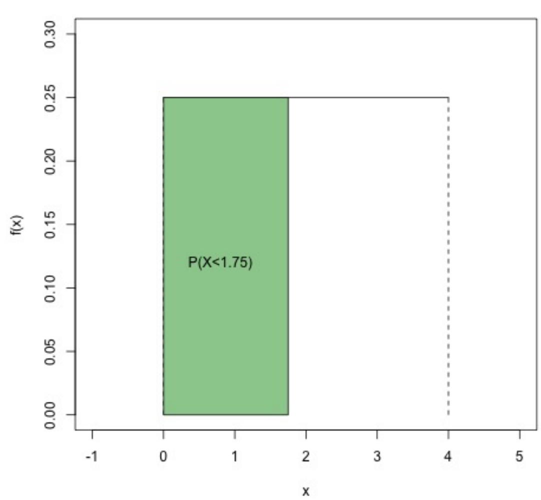

Figure 3.4: The probability density function of Example 3.3. As an example, the shaded region represents $P(X \leq 1.75)$.

Write down and plot the cumulative distribution function for all real numbers x.

Note that the distribution function F is non-decreasing and between 0 and 1. Note also that F is a continuous function. This is IN CONTRAST to the discrete case, where the distribution functions are step functions.

Summarizing, we have the following properties for the cdf (compare with the case of discrete r.v.):

Properties of the distribution function F:

(i) F is non-decreasing.

(ii) $0 \leq F(x) \leq 1$, $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to \infty$.

(iii) If X is a continuous random variable then its distribution function F is a continuous function.

(iv) If X is a continuous random variable then $F(x) = \int_{-\infty}^{x} f(t)dt$.

Recall also that from the distribution function F we can find various other probabilities (actually we can find ANY probability we want), e.g. for $a, b \in \mathbb{R}$ with $a < b$:

$$P(X > a) = 1 - P(X \leq a) = 1 - F(a)$$

$$P(a < X < b) = P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a), \ a < b.$$

3.2 Expectation and variance

Recall that for a discrete r.v. Y taking values $\{y\}$, the expectation of Y is

$$E(Y) = \sum_y y \ P(Y = y).$$

We aim to find an analogue of this for continuous r.v.'s.

Some cases where the pdf is symmetric are easy. E.g. in example 3.1 (the example of the pointer) we should have that $E(X) = 2$ by symmetry.

For a general continuous r.v. X, suppose that we divide the range of possible values into small intervals of length δx:

We have seen previously that we may write, for any value x,

$$P(x < X \leq x + \delta x) \approx f(x)\delta x.$$

We can think of the distribution of X being approximated by a discrete distribution with probability mass $f(x)\delta x$ at x. The expectation of this (approximating) discrete distribution is $\sum_x xp(x) = \sum_x xf(x)\delta x$. As we let $\delta x \to 0$, this quantity approaches $\int_{-\infty}^{\infty} xf(x)dx$. We take this as our definition of $E(X)$.

Definition: Let X be a continuous r.v. with pdf $f(x)$. The *expected value* (or *expectation* or *mean*) of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

(provided that $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$, which makes sure that $E(X) \leq E|X| < \infty$).

Just as in the discrete case,

if we have a function $g : I\!R \to I\!R$, then we can consider the r.v. $Y = g(X)$. If X is a continuous r.v. then we can calculate the expected value of $g(X)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

(provided that $\int_{-\infty}^{\infty} |g(x)|f(x)dx < \infty$).

This leads to the following definition for the variance of a continuous r.v.:

Definition: Let X be as above, with $E(X) = \mu$. The *variance* of X is

$$\begin{aligned} \text{Var}(X) &= E[(X-\mu)^2] && \text{(same as in discrete case)} \\ &= \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx && \text{(formula for } E[g(X)]) \end{aligned}$$

(provided this expectation is defined).

The *standard deviation* of X is $+\sqrt{\text{Var}(X)}$.

As in the discrete case,

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

which is often easier for calculations.

Example 3.4: Let X be as in Example 3.1. We have already seen that $E(X) = 2$ by symmetry. Verify this result using the formula in the definition of $E(X)$, and find $\text{Var}(X)$.

Properties of expectation and variance:

(a) If X does not take negative values then $E(X) \geq 0$.

(b) $E(aX + b) = aE(X) + b$ for all a, b.

(c) $\text{Var}(X) \geq 0$.

(d) $\text{Var}(aX + b) = a^2\text{Var}(X)$.

These are the same properties as we had for discrete r.v.'s. The proofs are similar but now involve integrals and pdf's instead of sums and pmf's. Check the proofs yourself as an **exercise**.

Further measures of location/central tendency

> **(a)** A *median* of a continuous r.v. X with pdf f is a value M such that $\int_{-\infty}^{M} f(x)dx = 0.5$, i.e. $M \in \mathbb{R}$ such that $P(X \leq M) = 0.5$.

Note that for a discrete random variable, defining the median is not that straightforward, as the probability that $P(X = M)$ risks to be a positive value (rather than 0 like in the continuous case). For example, for the random variable X with probability mass function

$$P_X(x) = \begin{cases} 1/8, & x = 1, 2, 15, 20 \\ 1/4, & x = 5, 7 \end{cases},$$

it holds that $P(X \leq 5) = 0.5$ and $P(X > 5) = 0.5$. This is the same as $P(X < 7) = 0.5$ and $P(X \geq 7) = 0.5$. Thus, both values 5 and 7 could be taken as the median.

Note that this is different from the case that we have a SAMPLE of $n = 8$ values, which would be ordered as

$$1 \quad 2 \quad 5 \quad 5 \quad 7 \quad 7 \quad 15 \quad 20.$$

Then the median is defined as the arithmetic mean of the two 'middle' observations and it is unique (equal to $(5+7)/2 = 6$). However, if these were the values of a discrete random variable in a population of $N = 8$ units, then the previous method for finding the median is the correct one.

> **(b)** A *mode* of a continuous r.v. X is a value of x for which the pdf $f(x)$ is maximum.

Note that a r.v. X can have more than one mode.

3.3 Independence of random variables

We have already seen in the previous chapter that two discrete random variables X and Y are independent, if for all x and y: $P(X = x, Y = y) = P(X = x)P(Y = y)$.

Clearly, this definition is not appropriate for the case of two continuous random variables (**exercise: why not?**).

The following general definition of independence of two random variables can be given:

> **Definition:** Two random variables X and Y (discrete, continuous or neither) are *independent* if for all $x, y \in \mathbb{R}$: $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$.

If X and Y are discrete random variables then this definition is equivalent to the definition of independence given in Section 2.3. For the case of two continuous random variables, an equivalent definition in terms of pdf's can also be provided.

The following property still holds:

> **Property:** If X and Y are independent, then $E(XY) = E(X)E(Y)$.

3.4 The Moment-Generating Function

Recall that for a continuous random variable X and a function $g : \mathbb{R} \to \mathbb{R}$, the expectation of $g(X)$ is given by

$$E[g(X)] = \int g(x)f(x)dx$$

where $f(.)$ is the pdf of X.

Now suppose we put $g(X) = e^{tX}$ for some real number t. It is often quite straightforward to evaluate the required integral $\int e^{tx}f(x)dx$, if it exists (which in mathematical terms is related to the *Laplace transform* of the function $f(.)$). We call this the **moment generating function**, $M(t)$.

> The moment generating function, $M(t)$, is calculated from the formula
>
> $$M(t) = E(e^{tX}) = \begin{cases} \sum e^{xt}p(x) & \text{if } X \text{ is discrete} \\ \int e^{xt}f(x) & \text{if } X \text{ is continuous} \end{cases}$$

We know that

$$e^m = 1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots$$

for any real number m, so

$$M(t) = E(e^{tX}) = E\left(1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots\right)$$

(for those values of t for which the above expectations are defined).

If we put $t = 0$ we get $M(0) = E(1 + 0 + 0 + \dots) = 1$.

Suppose we differentiate $M(t)$ with respect to t. We shall use the result that we can change the order of differentiating and taking expectations (in a more advanced course we would need to prove this) and conclude that

$$\frac{dM(t)}{dt} = \frac{d}{dt}E\left(e^{tX}\right) = E\left(\frac{d}{dt}e^{tX}\right) = E\left(Xe^{tX}\right).$$

If we put $t = 0$ we get $\frac{dM}{dt}_{|t=0} = E(X)$.

By differentiating again:

$$\frac{d^2M}{dt^2} = E\left(X^2 e^{tX}\right).$$

If we put $t = 0$ we get $\frac{d^2M}{dt^2}_{|t=0} = E(X^2)$.

In general, $\frac{d^kM}{dt^k}_{|t=0} = E(X^k)$, is the k-th moment of X.

For this reason $M(.)$ is called the *moment-generating function* (mgf) of X.
Mgf's often provide a very quick way of obtaining means and variances. They are defined for all random variables (discrete, continuous or mixed).

Moment generating functions are especially useful when dealing with sums of independent random variables. This is because of the following very important result.

Proposition Suppose X and Y are independent random variables with moment generating functions M_X and M_Y respectively. Let $Z = X + Y$. Then the mgf of Z is $M_X M_Y$.

Proof $M_Z(t) = E(e^{Zt}) = E(e^{Xt}e^{Yt})$. We can now use independence to see that this equals $E(e^{Xt})E(e^{Yt})$. We have shown $M_Z(t) = M_X(t)M_Y(t)$ or "the moment generating function of the sum equals the product of the moment generating functions." Notice that the independence condition is crucial. The result is not in general true for dependent random variables.

Another reason the moment generating function is so important, is because of following the uniqueness result. It should be clear that if two random variables have the same probability mass function (in the discrete case) or probability density function (in the continuous case) then they also have the same distribution function. (i.e If f_X equals f_Y then F_X must equal F_Y.) In this sense the probability mass (or density) function uniquely defines the distribution of the random variable. In a similar way it can be shown that the moment generating function uniquely defines the distribution of the random variable *provided it (the mgf) is continuous at the origin.*

Proposition Suppose X and Y are random variables (discrete, continuous or neither) with the same moment generating function, i.e. $M_X = M_Y = M$. Suppose also that $M(t)$ is continuous at $t = 0$. Then $F_X = F_Y$ (i.e. if they have the same moment generating function they have the same distribution function).

We shall use this result in later sections. For now we only note that the condition that $M(t)$ must be continuous at $t = 0$ holds for many of the most common random variables.

Proposition Suppose X has Moment Generating Function $M_X(t)$. Then the random variable $Y = aX + b$ has MGF $M_Y(t) = e^{bt}M_X(at)$.

Proof

$$
\begin{aligned}
M_Y(t) &= E\left(e^{Yt}\right) \\
&= E\left(e^{(aX+b)t}\right) \\
&= E\left(e^{bt}e^{aXt}\right) \\
&= e^{bt}E\left(e^{X(at)}\right) \\
&= e^{bt}M_X(at)
\end{aligned}
$$

Example 3.5: Calculate the moment generating functions for each of the following variables and in each case use the mgf to calculate the expected value and variance of the random variable:

1. A Bernoulli random variable with parameter 1/2.
2. A continuous random variable with the following probability density function.

$$f(x) = e^{-x} \ (x > 0)$$

   (is this a density indeed?)

3.5 Basic and further probability distributions for continuous random variables

The continuous Uniform distribution - *Revision*

A continuous random variable X has a *uniform distribution over the interval* (a, b) (both a, b are real numbers and $a < b$), if it has pdf

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise.} \end{cases}$$

We write $X \sim U(a, b)$.

For instance in example 3.1, $X \sim U(0, 4)$.

For $f(x)$ to be a valid pdf we must check:
(i) $f(x) \geq 0$ everywhere (trivial)
(ii) $\int_{\mathbb{R}} f(x)dx = 1$:

$$\int_{-\infty}^{\infty} f(x)dx = \int_a^b \frac{1}{b-a}dx = \frac{1}{b-a}[x]_a^b = \frac{1}{b-a}(b-a) = 1 \ ,$$

so $f(x)$ is a valid pdf.

For any interval $(c, c+d) \subseteq (a, b)$, we have

$$P(c < X < c+d) = \int_c^{c+d} \frac{1}{b-a}dx = \frac{1}{b-a}[x]_c^{c+d} = \frac{d}{b-a},$$

so that $P(X \in (c, c+d))$ depends only on the length of the interval, and not on its location.

Can you write the cdf $F(x)$ for any real x?

For the mean, we have $E(X) = (a+b)/2$ by symmetry.

Exercise: Can you derive this from the definition? Can you find $E(X^2)$ and $E(X-E(X))^2$?

Example 3.6: Buses arrive at a specified stop exactly every 15 minutes, i.e. at 7.15, 7.30, 7.45 etc. A passenger arrives at the stop at a time that is uniformly distributed between 7am and 7:30am. Find the probability that the bus will arrive within 5 minutes(note that the reverse problem is very common too, i.e. that we know the time the passenger arrived at the stop, but we are uncertain about the time that the bus will arrive and, thus, it is a random variable).

Use of the uniform distribution: The uniform distribution is not so much of interest in its own right. Its most important use is in the simulation of random numbers (on a computer or hand calculator, for example): it turns out that if we can generate a random number from the distribution $U(0,1)$ then we can quite easily transform this into a random number from almost any other distribution.

Also remember that there is a discrete uniform distribution too (Bin(1,0.5)/ experiment with a fair die e.t.c.)

The Exponential distribution - ***Revision***

We have a rate $\lambda > 0$ of 'rare' events (per unit of time / space), as we have explained before for the Poisson distribution.

> We consider the r.v. X to be the ' continuous time (or interval in space e.t.c.) until the first event takes place'.
> We write $X \sim \mathrm{Exp}(\lambda)$.

Note that this is the continuous analogue of a Geometric random variable.

We will derive the cumulative distribution function (first) in the same way as for the geometric distribution. For convenience, we define the Poisson r.v. N_x to be 'the number of events, which happen with rate λ per unit time, after x time units have gone by'.

For $x > 0$

$$\begin{aligned} P(X > x) &= P(\text{first event occurs after time } x) \\ &= P(\text{no events in } (0, x]) \\ &= P(N_x = 0) = \frac{(\lambda x)^0 e^{-\lambda x}}{0!} = e^{-\lambda x}. \end{aligned}$$

But that gives us the cdf of X.

The cdf of $X \sim \mathrm{Exp}(\lambda)$ is:

$$F(x) = P(X \leq x) = 1 - P(X > x) = 1 - e^{-\lambda x}, \; x > 0.$$

Differentiating we obtain the pdf of X:

$$f(x) = \frac{dF}{dx} = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

We first check that $f(.)$ defines a valid pdf: $f(x) \geq 0$ everywhere, and

$$\int_{\mathbb{R}} f(x)dx = \int_0^{\infty} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_0^{\infty} = 0 - (-1) = 1 \;,$$

as required.

Can you find $E(X)$ and $\mathrm{Var}(X)$ now?

$$E(X) = \frac{1}{\lambda}, \qquad \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

We can also derive all moments of the exponential distribution, via its moment-generating function, very easily. Can you find $M(t)$?

From the moment generating function (if it exists), can you find what $E(X^r)$ is equal to for any $r = 1, 2, \ldots$?

Example 3.7: The amount of sugar that is needed in one day in a sugar refinery has an exponential distribution with mean 4 tons. How much raw sugar should be stocked each day to ensure that the chance of running out is only 0.05?

Example 3.8 - *Loss of memory property*: It is said that the time (in years) until a bone cancer patient needs to be hospitalized, is described by Exp(0.4) (starting from the time the sickness has been diagnosed). Is that realistic? To answer that, first compute the probability that for a patient, it will take longer than one year to go to hospital. For another patient, who we know has not been hospitalized for one year since his diagnosis, find the probability that he will not be hospitalized for another full year. What can you observe?

The Gaussian distribution - *Revision*

A continuous random variable X has a *normal distribution with parameters* μ *and* σ^2, (μ is any real number and σ^2 is a positive real number), if its probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \text{ , for all } x \in \mathbb{R}.$$

We write $X \sim N(\mu, \sigma^2)$.

It is straightforward to verify that $f(x) \geq 0$ everywhere. Also, since this is a density function, we can take for granted that

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] dx = 1.$$

The density is symmetric about $x = \mu$, and it is unimodal; it is often described as 'bell-shaped':

Can you show that $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$?

Definition: The $N(0,1)$ distribution is called the *standard normal* distribution. We write $\phi(z)$ for its pdf and $\Phi(z)$ for its cdf, so that

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \qquad \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}t^2} dt$$

for any real z.

$\Phi(z)$ does not have a simple functional form, but it has been tabulated.

It also holds that if $X \sim N(\mu, \sigma^2)$ and $Z = (X - \mu)/\sigma$, then $Z \sim N(0, 1)$.

This allows us to convert probability statements about any normal random variable into statements about a standard normal random variable, whence we can use tables to find any required probabilities. For example, if $X \sim N(\mu, \sigma^2)$ then

$$P(a < X < b) = P\left(\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right) = P\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) \text{ ,}$$

where $Z \sim N(0,1)$. The cdf of Z is $\Phi(.)$ (by definition) and so

$$P\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \; .$$

These probabilities can be read from tables.

It is useful to remember that $\Phi(0) = 0.5$ and that, by symmetry, $\Phi(-z) = 1 - \Phi(z)$ for all $z \in \mathbb{R}$.

Example 3.9: X is a random variable distributed as $N(20, 25)$. Find $P(X < 18)$.

Solution: We have $\mu = 20, \sigma^2 = 25$ so that $\sigma = 5$.

$$P(X < 18) = P\left(\frac{X-20}{5} < \frac{18-20}{5}\right) = P(Z < -0.4)$$

where $Z \sim N(0,1)$. Consulting the tables of the cdf of Z, we see that they are only tabulated for $z > 0$. We therefore need to use the symmetry results and note that

$$P(Z < -0.4) = \Phi(-0.4) = 1 - \Phi(0.4) \; .$$

$\Phi(0.4)$ is read from the tables as 0.6554. Hence $P(X < 18) = 1 - 0.6554 = 0.3446$.

It is not difficult to remember (from tables) that:

if $Z \sim N(0,1)$, then approximately $P(|Z| \leq 1) = 0.68$ and $P(|Z| \leq 2) = 0.95$ and $P(|Z| \leq 3) = 0.99$.

What does this imply about the corresponding probabilities of a r.v. $X \sim N(\mu, \sigma^2)$? How is that related to...CHEBYSHEV'S INEQUALITY? Does this verify it and in what way?

Real uses of the normal distribution

The normal distribution is perhaps the most important model for continuous random variables, and much statistical theory is based upon it. The reasons for its importance are:

1. Empirically, many 'real-world' phenomena such as heights, weights etc. often have a distribution which is approximately normal. Even if not, a transformation can often be found (e.g. by taking logs or cube roots) that produces an approximately normal distribution.

2. The distribution often provides a good model for random measurement errors in experiments.

3. The sum of a large number of Independent and Identically Distributed random variables from *any* distribution is approximately normally distributed. This result is known as the *Central Limit Theorem*, and underlies many statistical techniques in common use. For example, the sample mean of a data set is obtained by summing all the values and then dividing by the number of points; so the mean of a large sample has an approximately normal distribution.

4. There are useful normal approximations to other distributions, such as the binomial, Poisson and Gamma distributions (this is because Poisson is sum of Poisson, Binomial is ..., Negative Binomial is ..., Gamma is... which one is Gamma???? - see next). Note that, if the normal distribution is used to approximate a *discrete* distribution, then some correction needs to be made for the fact that the normal distribution is continuous. This correction is called a *continuity correction*, and an example is used next to demonstrate that.

5. If X is a normal random variable, and $Y = aX + b$ for some constants a and b, then Y is also a normal random variable. Moreover, sums of normal random variables (not necessarily independent) are themselves normally distributed.

Example 3.10: A machine automatically fills bottles with juice. The amount of juice dispensed into a bottle has a normal distribution with standard deviation 2cm^3. The expected value μ can be altered by a dial on the machine. Find μ such that a 1 litre bottle will overflow only 5% of the time.

Example 3.11 - *Continuity Correction*: A railway company aims to ensure that at most 11% of its services arrive at their destination more than 5 minutes late. In one year, the company runs 10,000 services. If the probability of any one service being more than 5 minutes late is 0.1, independently of all other services, find the probability that the company fails to meet its target.

Solution: Let X be the number of services which are more than 5 minutes late; then $X \sim \text{Bin}(10000, 0.1)$. Note that $E(X) = 1000$ and $\text{Var}(X) = 900$.

Here, the parameters of the binomial distribution are such that n is large but np is not small, so that the Poisson approximation is not appropriate. Under these circumstances, it is appropriate to use a normal approximation to the binomial distribution. We therefore define a new random variable $Y \sim N(1000, 900)$, which is normally distributed with the same mean and variance as X.

Note that we require $P(X > 1100)$; because X is a discrete random variable, we could equally write

$$P(X > 1100) = P(X \geq 1100 + c), \;\; c \in (0, 1].$$

However, if Y is a continuous random variable, it holds that

$$P(Y > 1100) \neq P(Y \geq 1100 + c), \ \ c \in (0, 1].$$

To work around this, we introduce a *continuity correction*, and we write

$$P(X > 1100) = \ldots = P(X \geq 1101) \approx P(Y > 1100.5).$$

Now

$$P(Y > 1100.5) = P\left(\frac{Y - 1000}{30} > \frac{1100.5 - 1000}{30}\right) = P(Z > 3.35) = 0.0004.$$

Example 3.12: When we stretch a rope of length 2m, we have no idea at which point (which length from its left end), it is going to break; we consider that it is 'equally likely' that it will break anywhere. We have broken now 100 ropes in that way, and we have recorded the lengths $X_1, \ldots, X_{100}$. What is the probability that the mean length of 100 is less than 0.5m?

The Gamma distribution

We have a rate $\lambda > 0$ of 'rare' events (per unit of time / space), as we have explained before for the Poisson distribution.

> For positive integer α, we consider the r.v. X to be the 'continuous time (or interval in space e.t.c.) until the α-th event takes place'.
> We write $X \sim \text{Gamma}(\alpha, \lambda)$.

Note that this is the continuous analogue of a Negative Binomial random variable.

What is $\text{Gamma}(1, \lambda)$?

We shall derive the cumulative distribution function, like in the case of an exponential random variable.
For $x > 0$, it holds that

$$\begin{aligned} P(X > x) &= P(\alpha\text{-th event occurs after time } x) \\ &= P(N_x = 0) + P(N_x = 1) + \ldots + P(N_x = \alpha - 1) \\ &= \sum_{v=0}^{\alpha-1} \frac{(\lambda x)^v e^{-\lambda x}}{v!} = \sum_{v=0}^{\alpha-1} \frac{\lambda^v}{v!} \ x^v e^{-\lambda x} \end{aligned}$$

and

$$F(x) = P(X \leq x) = 1 - \sum_{v=0}^{\alpha-1} \frac{\lambda^v}{v!} \ x^v e^{-\lambda x}.$$

We will need to differentiate, as follows for $x > 0$

$$\begin{aligned}
f(x) = \frac{d}{dx}F(x) &= -\frac{d}{dx}\sum_{v=0}^{\alpha-1}\frac{\lambda^v}{v!}\,x^v e^{-\lambda x} = -\sum_{v=0}^{\alpha-1}\frac{\lambda^v}{v!}\,\frac{d}{dx}\,x^v e^{-\lambda x} \\
&= \sum_{v=0}^{\alpha-1}\frac{\lambda^v}{v!}(\lambda\,x^v e^{-\lambda x} - v\,x^{v-1}e^{-\lambda x}) \\
&= \lambda\sum_{v=0}^{\alpha-1}\frac{(\lambda x)^v}{v!}\,e^{-\lambda x} - \sum_{v=0}^{\alpha-1}\frac{\lambda^v}{v!}\,v\,x^{v-1}e^{-\lambda x} \\
&= \lambda\sum_{v=0}^{\alpha-1}\frac{(\lambda x)^v}{v!}\,e^{-\lambda x} - \lambda\sum_{v=1}^{\alpha-1}\frac{(\lambda x)^{v-1}}{(v-1)!}\,e^{-\lambda x} \\
&= \lambda\sum_{v=0}^{\alpha-1}\frac{(\lambda x)^v}{v!}\,e^{-\lambda x} - \lambda\sum_{v-1=0}^{\alpha-2}\frac{(\lambda x)^{v-1}}{(v-1)!}\,e^{-\lambda x} \\
&= \lambda\,P(N_x \le \alpha-1) - \lambda\,P(N_x \le \alpha-2) = \lambda\,P(N_x = \alpha-1) \\
&= \frac{\lambda^\alpha}{(\alpha-1)!}\,x^{\alpha-1}\,e^{-\lambda x},\;\; x > 0.
\end{aligned}$$

Now, there is the more general case, where we consider:

$X \sim$ Gamma(α, λ) where $\lambda > 0$, but also $\alpha > 0$ is not necessarily a positive integer. Then, the density function of the random variable is

$$f(x) = \begin{cases} \dfrac{\lambda^\alpha x^{\alpha-1}e^{-\lambda x}}{\Gamma(\alpha)} & \text{for } x \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\Gamma(.)$ denotes the *gamma function*, defined as

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}dx$$

for any $\alpha \in I\!R^+$.

From the definition of the gamma function we see that $\Gamma(1) = \int_0^\infty e^{-x}dx = 1$. Hence, we verify that the pdf of the $\Gamma(1, \lambda)$ distribution is

$$f(x) = \lambda e^{-\lambda x},\;\; x > 0$$

as we expected.

Can you find a relationship between $\Gamma(a)$ and $\Gamma(a-1)$? How is that related to the special cases that a is an integer and to $a!$?

We can check that $f(x)$ is a valid pdf: it is non-negative everywhere, and

$$\begin{aligned}\int_{I\!R} f(x)dx &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^\infty x^{\alpha-1}e^{-\lambda x}dx \\ &= \text{(making the substitution } u=\lambda x \text{ so that } dx = du/\lambda) \\ & \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_{u=0}^\infty \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-u}\frac{du}{\lambda} = \frac{1}{\Gamma(\alpha)}\int_0^\infty u^{\alpha-1}e^{-u}du = \frac{1}{\Gamma(\alpha)}\Gamma(\alpha) = 1\ ,\end{aligned}$$

the last step following from the definition of the Gamma function.

We call α the *index* or *shape parameter*, and λ the *scale parameter* (or sometimes, confusingly, just the *parameter*) of the distribution.

Given the fact that $f(x)$ is a pdf indeed, can you compute the mean and variance of the random variables? Can you compute $E(X^r)$ for any positive integer r (if it exists)?

It holds that

$$E(X) = \frac{\alpha}{\lambda}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2}.$$

Finally, can you compute the moment generating function of X?

Suppose that you have the Independent and Identically Distributed random variables $X_1, \ldots, X_\alpha$ (α is an integer), which are all exponential with parameter λ. Can you find the moment generating function of their sum? What conclusions can you make immediately about the sum of independent exponential (or gamma) random variables?

Example 3.13: The response times (in tenths of a second) of an online terminal are found to have a gamma distribution with expectation 4 and variance 8. Find the parameters of this gamma distribution and write down the pdf for these response times.

Example 3.14: A mean rate of two trains arriving every five minutes is true for the Marble Arch station (central line) and the Holborn station (Piccadilly line). Christine's every day route to work includes taking the train from Marble Arch to Holborn on the central line, and then changing for the Piccadilly line. Today, Christine will spend exactly 9 minutes to walk to the Marble Arch station, 10 minutes into the central line train, another 1.5 minutes to move from the central line platform to the Piccadilly line platform, 5.2 minutes in the Piccadilly line train, and finally another 7.8 minutes to walk to work. What is the expected time she will take from home to work? What is the variance of this random variable?

Relationships between some distributions

Look at the following table carefully:

	Distribution	
Random variable	**Discrete time**	**Continuous time**
No. of 'events'	Binomial	Poisson
Time till first 'event'	Geometric	Exponential
Time till r-th 'event'	Negative binomial	Gamma

We have already said that Exponential and Gamma are the continuous analogues of Geometric and Negative Binomial. Is Poisson the continuous analogue of Binomial? Why?

Chapter 4

Functions of variables

4.1 Sums of independent random variables

An important problem in probability theory concerns investigating the distribution of sums of random variables (convolution). Suppose that $X_1, X_2, X_3, \ldots$ are a sequence of random variables. We define their n-th partial sum as

$$S_n = X_1 + X_2 + \ldots + X_n.$$

If the random variables are independent and all have the same distribution function, we say that they are I.I.D. (independent and identically distributed). In some special cases the distribution of the n-th partial sum of certain I.I.D. random variables is easily obtained. For example:

Example 4.1:

- If $X_1, X_2, \ldots X_n$ are I.I.D. Bernoulli(p) random variables, then S_n has a Bin(n, p) distribution.
- If $X_1, X_2, \ldots X_n$ are I.I.D. Exp(λ) random variables, then S_n has a Gamma(n, λ) distribution.
- If $X_1, X_2, \ldots X_n$ are I.I.D. Poi(λ) random variables, then S_n has a Poi($n\lambda$) distribution.
- If $X_1, X_2, \ldots X_n$ are I.I.D. $\mathrm{N}(\mu, \sigma^2)$ random variables, then S_n has a $\mathrm{N}(n\mu, n\sigma^2)$ distribution.

Proof of Poisson case The random variables $X_1, X_2, \ldots X_n$ are all identically distributed so they all have the same moment generating function. We have seen that the mgf of a Poi(λ) random variable is $\exp\{\lambda(e^t - 1)\}$. Because the X_i-s are independent, we can use the result that the moment generating function of the sum, S_n, is the product of the moment generating functions to conclude that

$$M_{S(n)}(t) = \prod_{i=1}^{n} e^{\lambda(e^t-1)} = e^{\lambda(e^t-1)} \cdot \ldots \cdot e^{\lambda(e^t-1)} = e^{n\lambda(e^t-1)}.$$

Because $M_{S(n)}(t)$ is continuous at the point $t = 0$ the uniqueness result tells us that it uniquely determines the distribution of S_n.
However, it is easy to see that $M_{S(n)}(t)$ is the mgf of a Poi$(n\lambda)$ random variable. Thus, $S_n \sim$ Poi$(n\lambda)$.

In the specific cases above we have been able to work out the distribution of S_n but, in general, it is usually not possible to calculate its distribution and we will usually only be able to find an approximation to it (e.g. see the Central Limit theorem below).

However, calculating the expected value and variance of S_n is easier. We use the following results:

Expected value of S_n: Suppose $X_1, X_2, \ldots X_n$ are random variables and $E(X_i) = \mu_i$ (they do not need to be independent or identically distributed). Then $E(S_n) = \sum_1^n \mu_i$.
Variance of S_n: Suppose $X_1, X_2, \ldots X_n$ are independent random variables and $\text{Var}(X_i) = \sigma_i^2$ (they must be INDEPENDENT but do not need to be identically distributed). Then $\text{Var}(S_n) = \sum_1^n \sigma_i^2$.

In addition, we can get information about linear combinations of the X_i-s by combining the results above with the results we have seen in Chapter 3.

Proposition: Let X be a random variables and a and b be constants. Then

$$E(aX + b) = aE(X) + b; \qquad \text{Var}(aX + b) = a^2\text{Var}(X).$$

Example 4.2: Suppose $X_1 \sim$ Bin(4, 0.5) and $X_2 \sim$ Poi(3), and suppose that X_1 and X_2 are independent. Find the expected value and variance of $X_1 + X_2$ and $3X_1 + 2X_2 + 7$.

Answer: From our previous results $E(X_1) = 2$, $\text{Var}(X_1) = 1$, $E(X_2) = 3$ and $\text{Var}(X_2) = 3$. Thus, $E(X_1 + X_2) = 2 + 3 = 5$ and $\text{Var}(X_1 + X_2) = 1 + 3 = 4$.

For the second part note that $E(3X_1) = 3 \cdot 2 = 6$ and $\text{Var}(3X_1) = 9 \cdot 1 = 9$ and $E(2X_2) = 2 \cdot 3 = 6$ and $\text{Var}(2X_2) = 4 \cdot 3 = 12$.

This implies that

$$E(3X_1 + 2X_2 + 7) = 6 + 6 + 7 = 19; \text{Var}(3X_1 + 2X_2 + 7) = 9 + 12 = 21.$$

Proposition (I.I.D. random variables): Suppose $X_1, X_2, \ldots X_n$ are I.I.D. random variables with expected value equal to μ and variance equal to σ^2.
Then $E(S_n) = n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$.

We have already seen the Weak Law of Large Numbers. This told us that if we have n I.I.D. random variables, then the sample mean S_n/n, will converge to the population mean μ as n goes to infinity. (Standard notation uses $\overline{X}_n$ for the sample mean S_n/n.) In other words, if n is large the sample mean will be approximately equal to μ. The natural next step is to examine the difference

$$S_n/n - \mu = \overline{X}_n - \mu.$$

We know that (with a high probability) this quantity will be small, provided that n is large. The Central Limit Theorem examines this difference further and gives us some information on exactly how small this is. It tells us the difference will be of order $n^{1/2}$. It also provides the distribution of the sample mean.

Central Limit Theorem Let $X_1, X_2, X_3, \ldots$ be I.I.D. random variables with expected value equal to μ and variance equal to σ^2. Then, the distribution of $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$ approaches the standard normal distribution as $n \to \infty$ in the sense that for any real number z

$$P(Z_n \le z) \to \Phi(z) \text{ as } n \to \infty.$$

We can also write this as

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{S_n/n - \mu}{\sigma/\sqrt{n}} = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Hence, the Central Limit Theorem tells us that for large n, $\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}$ is approximately $N(0,1)$. This is sometimes stated as $\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$ (the previous statement is right, as it is DISTRIBUTION FREE).

The importance of the Central Limit Theorem is that it tells us that we can approximate the distribution of S_n by a normal distribution provided n is large and this is true, *regardless of the distribution of the X_i-s.*

Examples 4.3:

1. Electric light bulbs have lifetimes X in hours where $X \sim \mathrm{Exp}(1/1000)$. Find approximately the probability that the sum of the lifetimes of 100 bulbs is less than 110,000 hours.

2. An astronomer wishes to measure the distance d to a particular star (in light years). It is known that repeated independent measurements will vary owing to instrument error, atmospheric conditions etc., but will have expectation d and standard deviation 5 light years. How many observations should be made to be 95% sure that the sample mean is within 1/2 light year from d?

3. *Normal approximation to the binomial distribution.* Silicon wafers coming to a microchip plant are inspected for conformance to specifications. From a large lot of wafers a sample of 100 are inspected. The lot is accepted only if the number of non-conformances in the sample is no more than 12. Find the approximate probability of acceptance if the proportion of non-conformances is $p = 0.2$.

4.2 Transformation of random variables

This section deals with the following important problem:

Suppose X is a continuous random variable with pdf f_X and $g : I\!R \to I\!R$. Consider the random variable $Y = g(X)$. Can we find f_Y, the pdf of Y?

First note that Y might not be a continuous variable. (For example, if $g(.) \equiv int(.)$, then Y will be discrete.) However, Y will be continuous if g is a "nice" function. There are several methods of finding its pdf.

Method of distribution functions

Example 4.4: Suppose $X \sim U(0,1)$ and $Y = X^3$. Find f_Y.

Solution: We know

$$f_X(x) = 1,\ 0 < x < 1 \ \text{ and } F_X(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x \geq 1 \end{cases} .$$

Now find the distribution function of Y. We know that $X \in (0,1)$ so $Y = X^3 \in (0,1)$. For $0 < y < 1$ only

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X^3 \leq y) = P(X \leq y^{1/3}) \\ &= F_X(y^{1/3}) = y^{1/3}. \end{aligned}$$

Now differentiate to find $f_Y(y)$:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = 1/3\ y^{-2/3},\ 0 < y < 1.$$

What does $f_Y(y)$ equal for $y \leq 0$ and $y \geq 1$?

Example 4.5: Let X be a continuous r.v. with pdf f_X and let $Y = X^2$. Find f_Y in terms of f_X.

Solution: Clearly $Y \geq 0$, so $F_Y(y) = 0$ for $y < 0$. For $y \geq 0$ only

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \end{aligned}$$

Differentiate with respect to y (using the chain rule) to get

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy}[F_X(\sqrt{y}) - F_X(-\sqrt{y})] \\ &= \frac{d\sqrt{y}}{dy} \frac{d}{d\sqrt{y}}[F_X(\sqrt{y}) - F_X(-\sqrt{y})] = \frac{1}{2\sqrt{y}}\,[f_X(\sqrt{y}) + f_X(-\sqrt{y})]. \end{aligned}$$

Example 4.6: Suppose $X \sim U(0,1)$ and $Y = -log_e(X)$. Find f_Y.

Solution: As $X \in (0,1)$ we see that $Y > 0$, so $F_Y(y) = 0$ for $y \leq 0$. For $y > 0$

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(-log_e(X) \leq y) = P(X \geq e^{-y}) \\ &= 1 - F_X(e^{-y}) \\ &= 1 - e^{-y}. \end{aligned}$$

Differentiate with respect to y to get

$$f_Y(y) = e^{-y}, \; y > 0$$

and you will notice that Y has an exponential distribution.

Remember the rule that 'if we can generate realizations from a continuous uniform distribution (if not between $(0,1)$ make it such), then we can always turn them into realizations from ANY other distribution'!

Transformation formula

> This can be used when g is either strictly increasing or decreasing (one-to-one transformations) and it is differentiable.

Suppose that g is strictly increasing over the set of all possible X-values. Let $h = g^{-1}$ be the inverse of g. Because g is increasing the following sets are equal

$$\{x : g(x) \leq y\} = \{x : x \leq h(y)\}.$$

So

$$\begin{aligned} P(g(X) \leq y) &= P(X \leq h(y)) \text{ by the above} \\ P(Y \leq y) &= P(X \leq h(y)) \\ F_Y(y) &= F_X(h(y)) \end{aligned}$$

Differentiating with respect to y (using the chain rule) we find that

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(h(y)) = f_X(h(y))h'(y).$$

Now consider the case where g is strictly decreasing over the set of all possible X-values. Denoting $h = g^{-1}$ again we now have

$$\begin{aligned} P(g(X) \leq y) &= P(X \geq h(y)) \\ P(Y \leq y) &= P(X \geq h(y)) \\ F_Y(y) &= 1 - F_X(h(y)) \end{aligned}$$

Differentiating with respect to y we find

$$f_Y(y) = -f_X(h(y))h'(y).$$

When g is decreasing $h'(y)$ is negative so $-h'(y) =\mid h'(y) \mid$.

Putting this altogether, for **both** cases

$$f_Y(y) = f_X(h(y)) \mid h'(y) \mid$$

This is called the *density transformation formula* or *change of variables formula.*

When using it, consider carefully the range of Y-values and check that your density f_Y integrates to 1.

This formula can also be written as

$$f_Y(y) = f_X(g^{-1}(y)) \mid \frac{d}{dy} \left(g^{-1}(y)\right) \mid .$$

Example 4.6 (*again*): Here $X \sim U(0,1)$ and $Y = -log_e(X)$, so $y = g(x) = -log_e(x)$ is strictly decreasing in the interval $(0,1)$ and the inverse of g is

$$x = h(y) = e^{-y}, \text{defined for } y > 0.$$

Using the change of variable formula we obtain that for $0 < x < 1$ or $y > 0$

$$f_Y(y) = f_X(h(y)) \mid h'(y) \mid=\mid h'(y) \mid=\mid -e^{-y} \mid= e^{-y}.$$

Examples 4.7:

1. Let $X \sim \text{Exp}(\lambda)$ and let $Y = \sqrt{X}$. Find f_Y.

2. We have solved Example 4.6 using the distribution function method and the method of transformations. Now solve it using the method based on moment generating functions.

Part 2: *Statistics for samples*

Chapter 5

From populations to samples

In the first part of these notes, we were concerned with statistics in populations. In practice, we only have a random sample available (what is a RANDOM sample?), in order to make some conclusions about what is really going on in the unknown population. The population might be infinite or it might be finite but huge. Even if the population is finite, it is often shown that selecting a sample and making conclusions about the quantities of the population, is preferred over measuring all the units of the population (it is time saving but it also reduces the number of mistakes!). The 'random' ways in which we select our samples guarantee that there are certain probabilities that our sample quantities cannot be too 'far away' from reality.

For a better understanding, we give the following example.

Example 5.1: We have the population of $N = 6$ people, three men and three women. Alex weighs 85kg, George weighs 79kg and Jonathan weighs 85kg too. Amanda weighs 58kg, Susan weighs 62kg and, finally, Christine weighs 52kg. Suppose that we write the r.v. X to be 'the weight in kg' of the people in the population.

(i) Can you write down the probability mass function $p(x)$ of the r.v. X?

(ii) What is the mean of weights $\mu = E(X)$ in the population consisted of 6 units? What is the variance of weights $\sigma^2 = \mathrm{Var}(X)$?

(iii) Suppose now that we select a sample of $n = 2$ units, say X_1 and X_2, from the population. The two units 'happen' to be Jonathan first ($x_1 = 85$) and Amanda second ($x_2 = 58$). Can you compute the following quantities

$$\begin{aligned}
\overline{x} &= \frac{x_1 + x_2}{2} \\
s^{*2} &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2}{2} \\
s^2 &= \frac{(x_1 - \overline{x})^2 + (x_2 - \overline{x})^2}{2 - 1} \qquad ?
\end{aligned}$$

(iv) Consider that we call random sample the Simple Random Sample WITH REPLACEMENT. In other words, for the first selection X_1, we give equal probability to all our N units. What is the probability mass function of X_1? Whatever our selection x_1, we replace the selected unit back in. For the second selection X_2, we AGAIN give equal probability to all our N units. What is the probability mass function of X_2? Whatever the selection x_2 may be, we replace the unit in the population...

Is it fair to say that the random sample $X_1, \ldots, X_n$ from the population described by the random variable X is such that $X_1, \ldots, X_n$ are Independent and Identically Distributed? What is the distribution of $X_1, \ldots, X_n$?

(v) Can you find (combinatorics) how many random samples of size n we can select from a population of size N? For our example, could you write down these possible samples? For each one of these samples, could you compute $\overline{x}, s^2$ and s^{*2}?

(vi) Are the probabilities that each one of these samples is selected in practice equal? Why? Can you write down the probability mass function of $\overline{X}, S^2$ and S^{*2} then? What are $E(\overline{X}), E(S^2)$ and $E(S^{*2})$ equal to? What is $\text{Var}(\overline{X})$ equal to?

(vii) Suppose that we were dealing with a Simple Random Sample WITHOUT REPLACEMENT (leave out a unit, once it has been selected in the sample). How much of all this would change? Would $X_1, \ldots, X_n$ be independent r.v.? Would they be identically distributed? Which distribution would they follow? What would be $E(\overline{X})$ be equal to? What could we say about $\text{Var}(\overline{X})$? Which two distributions can you recall that highlighted these differences? When would one tend to the other? Thus, does it really matter whether we say SRS (simple random sample) with or without replacement when N is too large (or infinite)? Call it RANDOM SAMPLE anyway...

5.1 A summary

- The population (of finite size N or infinite size) is driven by a r.v. X which has a specific distribution. Even though we might know the distribution (eg. Gaussian), we may not know the PARAMETERS involved (eg μ, σ^2).

- We select a random sample $X_1, \ldots, X_n$. These are n I.I.D. random variables from the same distribution as X (and also giving information on the unknown parameters). The joint probability mass function $p_{X_1,\ldots,X_n}(x_1, \ldots, x_n)$ or probability density function $f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)$ is equal to the product of the individual probability (mass or density) functions.

- From the sample available $x_1, \ldots, x_n$ we compute similar quantities to the unknown parameters (eg $\overline{x}, s^2$). We keep in mind that these are the realizations of the corresponding r.v. $(\overline{X}, S^2)$ for the specific sample that we 'happened' to select. Thus, all functions of the random sample, say $T(X_1, \ldots, X_n)$, are RANDOM VARIABLES (therefore, they have a distribution/expected value/variance).

- FROM NOW ON, we do not forget that in CLASSICAL STATISTICS (Bayesian statistics thinks the other way around), the population quantities (often called parameters) are just...NUMBERS. The sample quantities (later we will call them 'estimators' or 'statistics') are RANDOM VARIABLES. Of course, for a specific sample, we manage to see only a realization of these random variables (the estimate).

5.2 Some useful distributions

The χ^2–Distribution

Definition: If $Z \sim N(0,1)$ and $Y = Z^2$, then the distribution of Y is called χ_1^2, i.e. Chi–squared with 1 degree of freedom.

Properties of χ_1^2:

1. The Moment Generating Function (mgf) given by $M(t) = E(e^{tY})$ of $Y \sim \chi_1^2$ is

$$M(t) = (1-2t)^{-\frac{1}{2}}.$$

Proof:

$$\begin{aligned}
M(t) &= E(e^{tY}) \quad \text{by definition} \\
&= E(e^{tZ^2}) \\
&= \textstyle\int_{-\infty}^{\infty} e^{tz^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\
&= \textstyle\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(1-2t)z^2} dz \\
&= \textstyle\frac{1}{\sqrt{1-2t}} \int_{-\infty}^{\infty} \underbrace{\frac{1}{\sqrt{2\pi}}(1-2t)^{\frac{1}{2}} e^{-\frac{1}{2}(1-2t)z^2} dz}_{density\ of\ N(0,\frac{1}{1-2t})} \\
&= \textstyle\frac{1}{\sqrt{1-2t}} \times 1 \\
&= (1-2t)^{-1/2}
\end{aligned}$$

We can use the mgf to check the expectation and variance of Y as follows:

$$\begin{aligned}
E(Y) &= M'(0) \\
\text{here } M'(t) &= -\frac{1}{2}(1-2t)^{-\frac{3}{2}}(-2) \\
&\overset{t=0}{=} -\frac{1}{2}(-2) = 1
\end{aligned}$$

and use $M''(0) = E(Y^2)$ to find the variance.

2. The χ_1^2 distribution is the same as Gamma$(\frac{1}{2}, \frac{1}{2})$.

Proof:

$$\begin{aligned}
\text{Gamma}(\alpha, \lambda) &\quad \text{has m.g.f} \quad (1-\tfrac{t}{\lambda})^{-\alpha} \\
\Longrightarrow \text{Gamma}(\tfrac{1}{2}, \tfrac{1}{2}) &\quad \text{has m.g.f} \quad (1-2t)^{-\frac{1}{2}}
\end{aligned}$$

3. P.d.f of Y is found from p.d.f of Gamma(α, λ).

$$f(y) \quad = \quad \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}, \quad \text{for} \quad y > 0$$

$E(Y) = \frac{\alpha}{\lambda} = 1$ and $Var(Y) = \frac{\alpha}{\lambda^2} = 2$.

Definition: If $Z_1, Z_2, ..., Z_\nu$ are independent $N(0,1)$ and $Y = Z_1^2 + Z_2^2 + ... + Z_\nu^2$ then:

$$Y \sim \chi^2_\nu$$

i.e. the sum of ν squared *independent* standard normal variables has a χ^2–distribution with ν degrees of freedom (*or* the sum of independent χ^2 random variables is also a χ^2 random variable with degrees of freedom the sum of the separate degrees of freedom - see 5. next).

Properties of χ^2_ν:

1. Moment Generating Function of Y: $M(t) = (1 - 2t)^{-\frac{1}{2}\nu}$.
2. χ^2_ν distribution is the same as Gamma $\left(\frac{\nu}{2}, \frac{1}{2}\right)$.
3. χ^2_2 is the same as Gamma$(1, \frac{1}{2})$, same as Exponential$\left(\frac{1}{2}\right)$.
4. If $Y \sim \chi^2_\nu$ then $E(Y) = \frac{\alpha}{\lambda} = \nu, \quad Var(Y) = \frac{\alpha}{\lambda^2} = 2\nu$.
5. $Y_i \sim \chi^2_{\nu_i}$ independently for $i = 1, 2, ..., k$, and $U = Y_1 + Y_2 + ... + Y_k$, then
$$U \sim \chi^2_{\nu_1 + \nu_2 + ... + \nu_k}.$$
6. If $Y_1 \sim \chi^2_{\nu_1}$ and Y_1, Y_2 independent and $Y_1 + Y_2 \sim \chi^2_{\nu_1 + \nu_2}$, then
$$Y_2 \sim \chi^2_{\nu_2}.$$
7. By the Central Limit Theorem, we have that if $Y \sim \chi^2_\nu$
$$\frac{Y - \nu}{\sqrt{2\nu}} \sim N(0,1), \text{ for } \nu \to \infty.$$

Student's t–Distribution

Definition If $Z \sim N(0,1)$, $Y \sim \chi^2_\nu$, Y and Z are independent, then
$$T = \frac{Z}{\sqrt{Y/\nu}} \sim t_\nu.$$
We say that T has a Student- or t-distribution with ν degrees of freedom.

The name *Student* comes from the mathematician William Sealy Gosset who 'discovered' this distribution.

Properties of t–Distribution:

1. The p.d.f of T is symmetric around 0 and has thicker tails than a $N(0,1)$.
2. If $\nu > 2$, then $E(T) = 0$ and $Var(T) = \frac{\nu}{\nu-2}$.
3. As $\nu \to \infty$, $t_\nu \to N(0,1)$. Intuitively, $E\left(\frac{Y}{\nu}\right) = 1$, $Var\left(\frac{Y}{\nu}\right) = \frac{2}{\nu} \to 0$, as $\nu \to \infty$ (remember that if $Y \sim \chi^2_\nu$ then $Var(Y) = 2\nu$).
   So, $\sqrt{\frac{Y}{\nu}} \to$ constant $1 \Rightarrow$ distribution of $T \to$ distribution of Z.
4. In the special case $\nu = 1$, the p.d.f is:
$$f(t) = \frac{1}{\pi} \times \frac{1}{1+t^2}, \quad -\infty < t < \infty.$$
   For this, $E(T)$ and $Var(T)$ do not exist (the corresponding integrals give ∞).
   For the special case $\nu = 1$, it is called the *Cauchy distribution.*
5. The c.d.f and various quantities are tabulated for various ν values.

F–Distribution

Definition If $Y_1 \sim \chi^2_{\nu_1}$ and $Y_2 \sim \chi^2_{\nu_2}$ are two independent random variables, then
$$F = \frac{Y_1/\nu_1}{Y_2/\nu_2} \sim F_{\nu_1,\nu_2}.$$
We say that the random variable F has an F distribution with ν_1 and ν_2 degrees of freedom.

Properties of F–Distribution:

1. The pdf $f_F(x)$ of the r.v. F takes nonzero values only for $x > 0$.
2. If $F \sim F_{\nu_1,\nu_2}$, then $\frac{1}{F} \sim F_{\nu_2,\nu_1}$.
3. $P(F \leq x) = 1 - P(1/F \leq 1/x),\ x > 0$.
4. If $T \sim t_n$ then $T^2 \sim F_{1,n}$.
5. $E(F) = \frac{\nu_2}{\nu_2 - 2}$, for $\nu_2 > 2$.

5.3 Quantiles of distributions

In this course we use the same definition of quantiles as in STAT0002 (former STAT1004):

> **Definition** The 100p% quantile of X is defined to be the value x_p such that
>
> $$F_X(x_p) = P(X \leq x_p) = p.$$

We will be using the quantiles of the standard normal, the χ^2, t and F distributions a lot. Therefore we introduce specific symbols for quantiles of these distributions.

> A quantile for the standard normal distribution is defined in the following way
>
> $$z_p : P(Z \leq z_p) = p \text{ where } Z \sim N(0,1),$$

i.e. the standard normal quantile z_p is that number for which the probability *that a standard normal distributed variable is less than (or equal to) that number* is exactly equal to p.

> The quantiles for the χ^2, t and F distributions are defined in a similar way
>
> $$\chi^2_{\nu;p} \ :\ P(T \leq \chi^2_{\nu;p}) = p \text{ where } T \sim \chi^2_\nu$$
>
> $$t_{\nu;p} \ :\ P(T \leq t_{\nu;p}) = p \text{ where } T \sim t_\nu$$
>
> $$F_{\nu_1,\nu_2;p} \ :\ P(T \leq F_{\nu_1,\nu_2;p}) = p \text{ where } T \sim F_{\nu_1,\nu_2}$$

Example 5.2 What are the following probabilities?

$$P(Z \leq z_{0.963}) = \qquad , \quad P(Z < z_{0.963}) = \qquad , \quad P(T > t_{3;0.45}) = \qquad \text{where } T \sim t_3$$

As we will be using the aforementioned quantiles for confidence intervals and testing, it is important that you are able to find specific values using statistical tables. The statistical tables often are constructed in different ways and as such should be read in different ways, so practice with the next exercise.

Example 5.3

$z_{0.995} =$	$z_{0.950} =$	$z_{0.600} =$	$z_{0.400} =$
$\chi^2_{9;0.995} =$	$\chi^2_{30;0.950} =$	$\chi^2_{4;0.600} =$	$\chi^2_{4;0.400} =$
$t_{6;0.995} =$	$t_{6;0.950} =$	$t_{50;0.600} =$	$t_{50;0.400} =$
$F_{3,4;0.995} =$	$F_{8,7;0.950} =$	$F_{24,10;0.100} =$	$F_{3,10;0.025}$
$F_{4,3;0.995} =$	$F_{7,8;0.950} =$	$F_{10,24;0.100} =$	$F_{10,3;0.025} =$

Chapter 6

Point estimation

Suppose that we do know the distribution that is taking place in the population (which drives a r.v. X) and the only thing that remains unknown are the exact values of the parameters (maybe the values of some parameters are known, or maybe we are not really interested in the values of some other parameters/ nuisance parameters). We are now going to find out different ways to use the sample available $X_1, \ldots, X_n$, in order to find the sample 'analogues' of the parameters of interest. These will, of course, be functions of the sample $T(X_1, \ldots, X_n)$ and, thus, RANDOM VARIABLES that we will call ESTIMATORS. The specific realization of the estimator for the specific sample available, will be called ESTIMATE.

Definition: ***Point Estimator*** Let θ be an unknown population parameter. A *point estimator* for θ is a real valued function $T = T(X_1, \ldots, X_n)$ of the sample variables. It uses the available data to estimate the unknown value of a parameter when some statistical model has been proposed.

Often an estimator for θ is also denoted by $\hat{\theta}$ (or $\tilde{\theta}$ or θ^*), in order to make clear which parameter is being estimated.

6.1 Properties of Estimators

From above, it seems that any function of the sample can be used as an estimator of the unknown parameter. We will learn ways of finding 'good' estimators later on. But what is a 'good' estimator? Once an estimator has been defined, what are the nice properties that we would like it to have (and that will make us single it out as a 'better' estimator, against other estimators)?

For a *good* estimator we require that it is indeed 'informative' for the unknown parameter. There are different ways to measure how 'informative' an estimator is.

Unbiasedness: An estimator T is called unbiased for θ if $E(T) = \theta$. Unbiasedness means that, if we apply T to several samples, then we get on average the true value of θ.

Asymptotic unbiasedness: If the estimator T of θ has an expected value that is a function of the sample size n, and which tends to θ, as $n \to \infty$, then T is called an asymptotically unbiased estimator of θ.

Bias: The bias of T is given as $b(T, \theta) = E(T) - \theta$.
Obviously, an unbiased estimator has zero bias.

Consistency: An unbiased estimator T is called consistent, if $Var(T) \to 0$ for $n \to \infty$. Consistency means that the precision of the estimator increases, if the sample size increases.

Efficiency: Consider only unbiased estimators. Then T is an efficient estimator for θ if its $Var(T)$ is smaller than the variance of any other unbiased estimator for θ.

Relative efficiency (*between two unbiased estimators*): When the variance of one estimator T_1 is smaller than the variance of another estimator T_2, we call T_1 relatively more efficient than T_2.

Standard error: As it can be seen from above, the variance or standard deviation of an estimator is itself of interest. Typically $Var(T)$ depends on the unknown parameters that are estimated by T. A 'trick' in order to still get an idea of the standard deviation of T is to plug in estimates for the unknown parameters in $\sqrt{Var(T)}$ — this is then called the standard error of T.

Example 6.1: Suppose that we have a random sample $X_1, \ldots, X_n$, from a population with unknown mean μ. We propose four different estimators for the unknown parameter:

$$T_1 = \frac{X_1 + X_2 + X_3}{2}, \quad T_2 = \frac{X_1 + 2X_2 + 3\ X_3}{6}, \quad T_3 = \frac{\sum_{i=1}^n X_i}{n-1}, \quad T_4 = \frac{\sum_{i=1}^n X_i}{n} = \overline{X}.$$

Which one is the 'best', according to the properties described and why?

Example 6.2: Consider a normal sample $X_1, \ldots, X_n$, from $N(\mu, \sigma^2)$. $\overline{X}$ is an unbiased estimator of μ and it is consistent. (WHY??) We also know its standard error. For the estimators

$$S^{*2} = \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{n}, \qquad S^2 = \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{n-1},$$

find their expected value, and check whether they are unbiased for σ^2. Then, check whether they are consistent too (you might use the result that $\sum_{i=1}^n (X_i - \overline{X})^2/\sigma^2 \sim \chi^2_{n-1}$).

6.2 Classical Methods of Estimation

Least Squares Estimators (LSE)

We create a quantity, which is a function of the observations $X_1, \ldots, X_n$ (for which we have values available) and the unknown parameter θ. The quantity is equal to a sum of squared differences of each observation from its expected value; this expected value is, hopefully, a convenient (linear?) function of θ. We select our estimator $\hat{\theta}_{LS}$, such that the quantity is minimized.

Example 6.3: $X_1, ..., X_n \sim Poi(\mu)$, then we write the quantity

$$Q(\mu) = \sum_{i=1}^n (X_i - E(X_i))^2 = \sum_{i=1}^n (X_i - \mu)^2$$

and its derivative with respect to the unknown parameter μ

$$\begin{aligned} \frac{d}{d\mu} Q(\mu) &= \frac{d}{d\mu} \sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n \frac{d}{d\mu} (X_i - \mu)^2 = \sum_{i=1}^n \sum_{i=1}^n (-2)(X_i - \mu) \\ &= (-2) \sum_{i=1}^n (X_i - \mu). \end{aligned}$$

Then we obtain the minimum Q for, say $\hat{\mu}_{LS}$, when

$$\left. \frac{d}{d\mu} Q(\mu) \right|_{\mu = \hat{\mu}_{LS}} = 0$$

or

$$\sum_{i=1}^n (X_i - \hat{\mu}_{LS}) = 0 \qquad \text{or} \qquad \sum_{i=1}^n X_i - n\, \hat{\mu}_{LS} = 0.$$

This results in $\hat{\mu}_{LS} = \overline{X}$ (we may check that this is a minimum with the second derivative).

The method is mostly known for linear regression where the least squares method has a nice geometric interpretation. It is non-parametric; i.e. in the previous example, if we did not know this is Poisson, the method would still provide the same result.

Method of Moments Estimators (MME)

Population moments: The k–th (noncentral) moment of a random variable X is given by $E(X^k)$. The mgf is helpful in deriving these moments. The k–th central moment of a random variable X is given by $E((X - E(X))^k)$.

Sample moments: The k–th (noncentral) sample moment of $X_1, \ldots, X_n$ is given by

$$k\text{–th sample moment} = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

Method of Moments: If r parameters $\theta_1, \ldots, \theta_r$ are unknown, equate the first r sample moments (in terms of the data) with the corresponding population moments (in terms of the unknown parameters) and determine $\hat{\theta}_1, \ldots, \hat{\theta}_r$ as solutions of these equations, i.e. express $E(X^k)$ as function of $\theta_1, \ldots, \theta_r$ and solve

$$\frac{1}{n}\sum_{i=1}^{n} X_i^k = \hat{E}(X^k), \quad k = 1, \ldots, r,$$

(r unknowns and r equations). $\hat{E}(X^k)$, $k = 1, \ldots, r$, has been expressed as a function of $\hat{\theta}_1, \ldots, \hat{\theta}_r$ (like $E(X^k)$ is a function of $\theta_1, \ldots, \theta_r$).

Example 6.4 : $X_1, ..., X_n \sim Exp(\lambda)$. Find the moment estimator for λ.

Maximum Likelihood Estimators (MLE)

Let $X_1, ..., X_n$ be an I.I.D. sample with individual p.m.f $p(x;\theta)$ (if discrete) or p.d.f $f(x;\theta)$ (if continuous) — note that we make clear that the p.m.f or p.d.f depend on the parameter θ.

Likelihood function: The Likelihood function of the unknown parameter(s) θ based on a sample $x_1, ..., x_n$ is given by

$$L(\theta) = \begin{cases} \prod_{i=1}^n p(x_i;\theta), & \text{if discrete} \\ \prod_{i=1}^n f(x_i;\theta), & \text{if continuous} \end{cases}$$

Note that the likelihood is a function of θ, only! The sample observations $x_1, ..., x_n$ are considered as fixed and given. Heuristically, the likelihood is then the 'probability' that the actual sample is observed under θ. Note that it is not accurate to speak of 'probability' in the continuous case. However, using this heuristic interpretation it seems reasonable to choose as an estimator for θ the value that maximizes the likelihood.

Maximum Likelihood Estimator (MLE): The m.l.e. $\hat{\theta}_{ML}$ for θ is given as the value maximizing $L(\theta)$.

Note: We may maximize $L(\theta)$ to write down our estimator, even though we have not recorded the values $x_1, \ldots, x_n$. If the observations have been recorded too, we may compute the estimate and use it accordingly.

It is often more convenient to **maximize the *log–likelihood*** (natural logarithm) $l(\theta) = \ln L(\theta)$, since then we do not need to apply the product rule when differentiating:

$$l(\theta) = \ln L(\theta) = \begin{cases} \sum_{i=1}^n \ln p(x_i;\theta), & \text{if discrete} \\ \sum_{i=1}^n \ln f(x_i;\theta), & \text{if continuous} \end{cases}$$

The (log)likelihood function is the basis of a variety of statistical inference methods, not only for statistical estimation!

Example 6.5 (Bernoulli): It holds that the marginal probability mass function is

$$P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}, \ x_i = 0, 1, \text{ for } i = 1, ..., n.$$

The joint probability mass function is then

$$\begin{aligned} p(x_1, x_2, ..., x_n; p) &= P(X_1 = x_1) \times ... \times P(X_n = x_n) \\ &= p^{x_1}(1-p)^{(1-x_1)} \times ... \times p^{x_n}(1-p)^{(1-x_n)} \\ &= p^{x_1+...+x_n}(1-p)^{n-(x_1+...+x_n)} \\ &= p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i}. \end{aligned}$$

Thus,

$$L(p) = p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i}, \qquad p \in (0,1).$$

Example 6.6 (Binomial): $X_i \sim Bin(m_i, p), \ i = 1, \ldots, n$, and the marginal probability mass functions are equal to

$$P(X_i = x) = \binom{m_i}{x} p^x (1-p)^{m_i - x}, \ x = 0, \ldots, m_i, \ i = 1, \ldots, n.$$

The joint probability mass function is equal to

$$\begin{aligned} p(x_1, ..., x_n) &= \binom{m_1}{x_1} p^{x_1}(1-p)^{m_1-x_1} \times ... \times \binom{m_n}{x_n} p^{x_n}(1-p)^{m_n-x_n} \\ &= \binom{m_1}{x_1} \cdots \binom{m_n}{x_n} p^{(x_1+...+x_n)}(1-p)^{(m_1+...+m_n)-(x_1+...+x_n)}. \end{aligned}$$

Thus, we may write

$$L(p) \propto p^{\sum_{i=1}^n x_i}(1-p)^{\sum_{i=1}^n m_i - \sum_{i=1}^n x_i}$$

(*Note*: the symbol $\propto$ means 'proportional to', i.e. $y \propto x$ when $y = kx$ for k : constant)

and

$$l(p) = (\sum_{i=1}^n x_i) \ln p + (\sum_{i=1}^n m_i - \sum_{i=1}^n x_i) \ln(1-p) + \text{additive constant}.$$

Question: Which value of p maximizes the likelihood function?

Example 6.7 (Exponential): The p.d.f of an Exp(λ) random variable X_i is given by:

$$f_{X_i}(x;\lambda) = \lambda e^{-\lambda x}, \ x > 0.$$

For the random sample $X_1, \ldots, X_n$ from the Exp(λ) distribution, the joint probability density function is

$$f(x_1, \ldots, x_n) = \lambda e^{-\lambda x_1} \times \ldots \times \lambda e^{-\lambda x_n} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}, \ x_i > 0, \ i = 1, \ldots, n, \ \lambda > 0.$$

Thus, we may write

$$L(\lambda) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}, \ \lambda > 0$$

and

$$l(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i, \ \lambda > 0.$$

Chapter 7

Interval estimation

7.1 Confidence intervals

We now move from POINT ESTIMATION to INTERVAL ESTIMATION. Instead of trying to 'guess' the value for the unknown parameter from the estimate, we take advantage of the fact that the ESTIMATOR behind the estimate is a RANDOM VARIABLE and it has a distribution. We have seen that a 'good' estimator is such that its variance/st.error reduces as the sample size increases. For example, if we estimate $\hat{\mu} = 5$ from a sample of size $n = 4$ we will usually have much less confidence in this estimated value than if the sample had been of size $n = 100$. In order to include information on the PRECISION of the estimator, we should therefore consider *confidence intervals.*

Confidence Interval (CI): Based on a sample $X_1, \ldots, X_n$ a $(1-\alpha)$–confidence interval $[L, U]$ for θ is given by a lower bound $L(X_1, \ldots, X_n)$ and an upper bound $U(X_1, \ldots, X_n)$ satisfying

$$P(L(X_1, \ldots, X_n) \leq \theta \leq U(X_1, \ldots, X_n)) = 1 - \alpha.$$

Interpretation: With a $(1-\alpha)$ confidence (probability) the RANDOM interval $[L, U]$ will contain the true parameter value θ. Note that the bounds of the interval are random since they are functions of the sample variables $X_1, \ldots, X_n$.

Construction of confidence intervals: This is a bit different than the construction of estimators as seen in the previous chapter. Typically, we have to find a sample statistic which depends on the unknown parameter θ, but such that the distribution of this statistic is independent of θ. In fact, the sample statistic must depend on θ and everything else that we have available (numbers/data/known parameters), while its distribution may only depend on what we have available (numbers/data/known parameters). We can then use this distribution to get the required quantiles and transform the statistic into lower and upper bounds for θ. This sounds complicated and it is indeed complicated, but we will see several examples below yielding confidence intervals for most standard situations.

7.2 Sampling distributions of $\overline{X}$ and S^2 for Normal Samples

In this section we explicitly **assume that the sample comes from a normal distribution**. Thus, suppose $X_1, X_2, ..., X_n$ are independent $N(\mu, \sigma^2)$ random variables.

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \text{and} \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

are both random variables as well, since they are transformations of the (random) sample variables.

Results:

1. $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$ or $\sqrt{n}(\overline{X} - \mu)/\sigma \sim N(0,1)$;
2. $\overline{X}$ and S^2 are independent;
3. $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ or $\sum_{i=1}^{n}(X_i - \overline{X})^2/\sigma^2 \sim \chi^2_{n-1}$;
4. $\sqrt{n}(\overline{X} - \mu)/S \sim t_{n-1}$.

<u>Proofs</u>: 1. may be shown using m.g.f.
3. uses 1. and 2.
Also, 4. follows on from 1., 2., and 3. due to the next reasoning: By 1.

$$Z = \sqrt{n}\frac{\overline{X} - \mu}{\sigma} \sim N(0,1).$$

By 3.

$$Y = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Moreover,

$$\sqrt{\frac{Y}{n-1}} = \frac{S}{\sigma}.$$

Thus, by definition of the t-distribution

$$T = \frac{Z}{\sqrt{\frac{Y}{(n-1)}}} = \sqrt{n}\frac{\overline{X} - \mu}{\sigma}\left(\frac{S}{\sigma}\right)^{-1} = \sqrt{n}\frac{\overline{X} - \mu}{S} \sim t_{n-1}.$$

Question: When should we use Results 1. to 4.?
Answer. Result 1 will be used when we are interested in making inference about the unknown mean μ from its estimator $\overline{X}$ and we DO KNOW the population variance σ^2. This will be replaced by result 4, when we are interested in making inference about the unknown mean μ from its estimator $\overline{X}$ and we DO NOT KNOW the population variance σ^2. Usually, when we do not know the variance σ^2, we definitely do not know the mean μ either. Thus, result 3 will be used to make inference about the unknown variance σ^2 from its estimator S^2, when we DO NOT KNOW the population mean μ. In the unlikely event that we DO KNOW the population mean μ and we want to make inference about the unknown variance σ^2 from its estimator S^2, we use the following result:

$$X_i \sim N(\mu, \sigma^2) \quad \text{or} \quad \frac{X_i - \mu}{\sigma} \sim N(0,1) \quad \text{or} \quad \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2_1 \quad \text{or} \quad \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2_n.$$

or

$$\frac{n\sigma^{*2}}{\sigma^2} \sim \chi^2_n, \text{ where } \qquad \sigma^{*2} = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n}.$$

7.3 Confidence intervals for the unknown mean of a Normal population

We have a random sample $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$:

(i) **I AM INTERESTED IN THE MEAN AND I KNOW THE VARIANCE**

Suppose that we want to construct a $(1-\alpha)$ confidence interval for the unknown mean μ and we know the population variance σ^2.

Test statistic:

$$T = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \sim N(0,1).$$

Indeed, T depends on the unknown parameter of interest (μ), it depends on the data and everything else is known (eg. σ). The result is PARAMETER-FREE (standard normal distribution always has $(0,1)$ parameters).

We may write

$$P\left(z_{\alpha/2} \leq \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \leq z_{1-\alpha/2} \right) = 1 - \alpha,$$

where we consider $F(z_\alpha) = P(Z \leq z_\alpha) = \alpha,\ Z \sim N(0,1)$. In other words, $z_{\alpha/2}$ is the lower $\alpha/2$-quantile, while $z_{1-\alpha/2}$ is the upper $\alpha/2$-quantile. Thus, thanks to the symmetry of the normal distribution

$$z_{\alpha/2} = -z_{1-\alpha/2}.$$

The value $z_{1-\alpha/2}$ can be found in proper tables (or from statistical packages), given the probability α. If we re-write the probability, with respect to the unknown parameter μ, then it holds that

$$P\left(\overline{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

The C.I. $(\overline{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \overline{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}})$ has a probability $(1-\alpha)$ of including the true (but unknown) value of μ.

How can you use the classical definition of probability to INTERPRET a C.I.?

(ii) **I AM INTERESTED IN THE MEAN AND I DO NOT KNOW THE VARIANCE**

Suppose that we want to construct a $(1-\alpha)$ confidence interval for the unknown mean μ and we do not know the population variance σ^2.

Test statistic:

$$T = \frac{\sqrt{n}(\overline{X} - \mu)}{S} \sim t_{n-1}.$$

Indeed, T depends on the unknown parameter of interest (μ), it depends on the data and everything else is known (eg. I have 'replaced' σ by the function of the data S). We may consider this result to be PARAMETER-FREE too (form of t distribution only depends on the sample size).

We may write

$$P\left(t_{n-1;\alpha/2} \leq \frac{\sqrt{n}(\overline{X} - \mu)}{S} \leq t_{n-1;1-\alpha/2}\right) = 1 - \alpha,$$

where we consider $P(T \leq t_{\nu,\alpha}) = \alpha$, $T \sim t_\nu$. In other words, $t_{n-1;\alpha/2}$ is the lower $\alpha/2$-quantile, while $t_{n-1;1-\alpha/2}$ is the upper $\alpha/2$-quantile from the t-distribution with $(n-1)$ degrees of freedom (d.f.). Thus, thanks to the symmetry of the t-distribution, we may write

$$t_{n-1;\alpha/2} = -t_{n-1;1-\alpha/2}.$$

If we re-write the probability, with respect to the unknown parameter μ, then it holds that

$$P\left(\overline{X} - t_{n-1;1-\alpha/2}\frac{S}{\sqrt{n}} \leq \mu \leq \overline{X} + t_{n-1;1-\alpha/2}\frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

The C.I. $(\overline{X} - t_{n-1;1-\alpha/2}\frac{S}{\sqrt{n}}, \overline{X} + t_{n-1;1-\alpha/2}\frac{S}{\sqrt{n}})$ has a probability $(1-\alpha)$ of including the true (but unknown) value of μ.

Questions: 1. What is the effect of the fact that we do not know the variance σ^2, on our interval for the mean μ? (Remember the properties of t-distribution compared to $N(0,1)$.) 2. From the same sample, if we construct a $(1-\alpha)$ C.I. using the $N(0,1)$ distribution (and the true value σ) and another $(1-\alpha)$ C.I. using the t-student distribution, which one is going to be wider? 3. What happens for large sample sizes? Why?

Example 7.1: An enthusiastic cook was interested in the accuracy of his kitchen timer. So, on ten different occasions he set it to ring after a 5 minute (300 second) delay. The 10 different times (secs) recorded on a stopwatch are given by:

293.7	296.2	296.4	294.0	297.3
293.7	294.3	291.3	295.1	296.1

Is it reasonable to assume $N(\mu, \sigma^2)$ for these data? We have $\overline{x} = 294.81$, $s^2 = 3.1232$ and $s = 1.77$. What is a 90% confidence interval for μ (= true mean delay)?

Interpretation of Confidence Intervals (*Important*!)

1. Note that the above confidence intervals are based on the random variable $\overline{X}$ (and maybe on the random variable S^2 too). Therefore, different people who collect different samples will get different confidence intervals.

2. The correct interpretation of the $100(1-\alpha)\%$ confidence interval is that with probability $1-\alpha$, the (random) interval $[L, U]$ contains the unknown true value of μ; it is the bounds $[L, U]$ that are random!

3. Once the sample is realized and you have found specific values $[l, u]$ for the confidence limits, no probability statement is possible anymore because there is no randomness left! We then say "we have $(1-\alpha)100\%$ confidence that $[l, u]$ contains the true parameter value".

4. If the statistical experiment (giving $\overline{X}$) was repeated many times and the confidence interval was computed each time, then approximately 95% (or $100 \times (1-\alpha)$%) of these intervals would contain the unknown μ (from which distribution does this 'expected number of successes' come from?).

Width of Confidence Interval: It is obviously desirable to have short confidence intervals because this narrows the range of possible parameter values.

The width of an <u>approximate</u> confidence interval for the population mean is given by

$$\text{width } = 2z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

where, as an approximation, σ can be replaced by the sample standard deviation S if σ is unknown.

Question: What does the width of the interval depend on and how can we change it?

1. We can choose $z_{1-\alpha/2}$ small. This corresponds to choosing α large implying in turn that the confidence level (1-α) will be small. This might be too high a price to pay!

2. It would also help if σ^2 or its estimator S^2 is small. However, this can often not be influenced since it is the natural variability in the population. We may, however, make sure that we consider a homogenous population. Or, the population should be subdivided into subpopulations that are homogenous and different confidence intervals computed for the different subpopulations.

3. Our last chance is to increase n in order to decrease the interval width. It is plausible that high precision can be obtained by increasing the sample size since larger samples contain more information.

Example 7.2: If a given width ω_0 of a confidence interval is to be obtained with a given level of confidence $(1-\alpha)$, find a formula for the required sample size.

7.4 Approximate Confidence Intervals for the Population Mean

Suppose $X_1, X_2, ..., X_n$ are independent random variables from a given distribution (that is not necessarily normal), and a confidence interval for the unknown population mean, $\mu = E(X_i)$, is desired. ***Trick*:** use the Central Limit Theorem!

Variance known and 'one parameter distributions'

Assume for the moment that $Var(X_i) = \sigma^2$ is known. We have for large sample size n:

$$Z = \sqrt{n}\,\frac{\overline{X} - \mu}{\sigma} \overset{approx}{\sim} N(0,1).$$

Thus, Z is a statistic (i.e. a transformation of the sample) with a known distribution that is independent of the unknown μ. We therefore know:

$$P\left(-z_{1-\alpha/2} \leq \sqrt{n}\,\frac{\overline{X} - \mu}{\sigma} \leq z_{1-\alpha/2}\right) \approx 1 - \alpha.$$

Solving with respect to the unknown value μ, we can write again the lower bound

$$L(X_1, X_2, ..., X_n) = \overline{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

and the upper bound

$$U(X_1, X_2, ..., X_n) = \overline{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Obviously, this is only useful if we know σ. For some distributions, however, $Var(X)$ is a function of $E(X)$ — so–called 'one parameter distributions'. Then we can modify the above procedure as is demonstrated in the following example for the case of an $Exp(\lambda)$ distribution.

Example 7.3: Find a $(1-\alpha)$ confidence interval (CI) for the parameter λ of an $\text{Exp}(\lambda)$ density. Remember that for $\text{Exp}(\lambda)$ it holds that $\mu = E(X) = \frac{1}{\lambda}$ and $\sigma^2 = \text{Var}(X) = \frac{1}{\lambda^2} = \mu^2$.

Confidence Intervals for the mean of other one 'parameter distributions' can be obtained similarly as for the exponential example and the results are summarized in the table below.

Model	approx. $100(1-\alpha)\%$ Confidence Interval
Exponential (for $\lambda = 1/E(X)$)	$\left[\frac{\hat{\mu}\sqrt{n}}{(\sqrt{n}+q_{1-\alpha/2})}, \frac{\hat{\mu}\sqrt{n}}{(\sqrt{n}-q_{1-\alpha/2})}\right],\ \hat{\mu} = \overline{X}$
Poisson (for $\mu = E(X)$)	$\left[\hat{\mu} \pm q_{1-\alpha/2}\sqrt{\frac{\hat{\mu}}{n}}\right],\ \hat{\mu} = \overline{X}$
Bernoulli (for $p = E(X)$)	$\left[\hat{p} \pm q_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right],\ \hat{p} = \overline{X}$

NOTE — *Rule of Thumb*: Since the previous confidence intervals are based on the CLT we need to make sure that the sample size is 'large enough' for the approximation to hold. A rule of thumb is that $n \geq 30$, but for very skewed distributions this might not be enough.

Variance unknown

In cases where we cannot express the variance of a distribution as a function of its expected value, we may still use the CLT and the above ideas to get an approximate $(1-\alpha)$ confidence interval. The following procedure is particularly useful when *no distributional assumption* is made. It is valid whenever the CLT is valid (i.e. independent observations with the same population mean and variance).

Consider $X_1, X_2, ..., X_n$ (with n large) from a population where no distributional assumptions are made. All you care about is the population mean μ. If the variance σ^2 is unknown, then use the estimator $S^2 \Rightarrow$ ***approximate*** $100(1-\alpha)\%$ confidence intervals for μ is:

$$\left[\overline{X} \pm z_{1-\alpha/2}\frac{S}{\sqrt{n}}\right]$$

with $z_{1-\alpha/2} = 1 - \frac{\alpha}{2}$ quantile of $N(0,1)$.

Note: For $\alpha = 5\%$, $z_{0.975} = 1.96 \approx 2$, so often people tend to say $\overline{X} \pm 2 \times \text{stdev}(\overline{X})$ is a 95% confidence interval for μ. Here, $\text{stdev}(\overline{X}) \approx s/\sqrt{n}$.

The approximation for the above CI is *very* crude! As we make no assumptions we must make sure that the sample is very large for the CI to be valid.

We have seen how this is an approximation. Only when we know the distribution is Gaussian, we come up with t-distribution (which tends to $N(0,1)$ for large sample size). If the population is not normal, things become worse. Thus, we need a very LARGE SAMPLE SIZE to use this result.

7.5 Confidence intervals for the unknown variance of a Normal population

We have a random sample $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$:

(i) **I AM INTERESTED IN THE VARIANCE AND I KNOW THE MEAN**

Suppose that we want to construct a $(1-\alpha)$ confidence interval for the unknown variance σ^2 and we know the population mean μ. This is rare, because when we reach that level that we do not know the second-order properties of the random variables ($E(X^2)$), it is highly unlikely that we know the first-order properties ($E(X)$).

Test statistic:

$$T = \frac{n\sigma^{*2}}{\sigma^2} \sim \chi^2_n.$$

Indeed, T depends on the data, it depends on quantities that we know (eg. μ in σ^{*2}) and on the unknown parameter of interest σ^2. The distribution of the statistic, only depends on the sample size.

We may write

$$P\left(\chi^2_{n;\alpha/2} \leq \frac{n\sigma^{*2}}{\sigma^2} \leq \chi^2_{n;1-\alpha/2}\right) = 1-\alpha,$$

where we consider $P(Y \leq \chi^2_{n;\alpha}) = \alpha$, $Y \sim \chi^2_n$. In other words, $\chi_{n,\alpha/2}$ is the lower $\alpha/2$-quantile, while $\chi_{n,1-\alpha/2}$ is the upper $\alpha/2$-quantile. Unfortunately, the distribution is not symmetric (and $Y > 0$ with probability 1).

The values $\chi^2_{n;\alpha/2}, \chi^2_{n;1-\alpha/2}$ can be found in proper tables (or from statistical packages), given the probability α. If we re-write the probability, with respect to the unknown parameter σ^2, then it holds that

$$P\left(\frac{n\sigma^{*2}}{\chi^2_{n;1-\alpha/2}} \leq \sigma^2 \leq \frac{n\sigma^{*2}}{\chi^2_{n;\alpha/2}}\right) = 1-\alpha.$$

Thus, the lower bound of the C.I. is $L(X_1,\dots,X_n) = \frac{n\sigma^{*2}}{\chi^2_{n;1-\alpha/2}}$ and the upper bound is $U(X_1,\dots,X_n) = \frac{n\sigma^{*2}}{\chi^2_{n;\alpha/2}}$. The C.I. has a probability $(1-\alpha)$ of including the true (but unknown) value of σ^2.

(ii) **I AM INTERESTED IN THE VARIANCE AND I DO NOT KNOW THE MEAN**

Suppose that we want to construct a $(1-\alpha)$ confidence interval for the unknown variance σ^2 and we do not know the population mean μ. This is usually the case when we are interested in σ^2.

Test statistic:

$$T = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Indeed, T depends on the data and on the unknown parameter of interest σ^2. The distribution of the statistic, only depends on the sample size. Similarly to before, the lower bound of the C.I. is $L(X_1,\ldots,X_n) = \frac{(n-1)S^2}{\chi^2_{n-1;1-\alpha/2}}$ and the upper bound is $U(X_1,\ldots,X_n) = \frac{(n-1)S^2}{\chi^2_{n-1;\alpha/2}}$. The C.I. has a probability $(1-\alpha)$ of including the true (but unknown) value of σ^2.

Questions: What is the effect of the fact that we do not know the mean μ, on our interval for the variance σ^2? What happens to the χ^2-distribution when we increase the degrees of freedom?

Example 7.4.: Give a 90%–confidence interval for the variance of the cook's kitchen timer (Example 7.1).

7.6 Sampling distributions of some statistics for samples from two normal populations

(i)

For the **first case**, we consider that we have two normal populations, say $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$, and we have selected TWO random samples, one from each one of them:

$$X_1,\ldots,X_n \sim N(\mu_X,\sigma_X^2), \qquad Y_1,\ldots,Y_m \sim N(\mu_Y,\sigma_Y^2).$$

We write

$$\overline{X} = \frac{\sum_{i=1}^n X_i}{n}, \qquad S_X^2 = \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{n-1}$$

and

$$\overline{Y} = \frac{\sum_{j=1}^m Y_j}{m}, \qquad S_Y^2 = \frac{\sum_{j=1}^m (Y_j - \overline{Y})^2}{m-1}.$$

We also define

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} = \frac{\sum_{i=1}^n (X_i - \overline{X})^2 + \sum_{j=1}^m (Y_j - \overline{Y})^2}{n+m-2}.$$

Results:

1. $\overline{X}-\overline{Y} \sim N(\mu_X-\mu_Y, \frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m})$ or $\frac{\overline{X}-\overline{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$.
2. $\overline{X}-\overline{Y}$ and S_p^2 are independent.
3. $\frac{S_X^2}{S_Y^2}\frac{\sigma_Y^2}{\sigma_X^2} \sim F_{n-1,m-1}$.
4. If $\sigma_X^2=\sigma_Y^2=\sigma^2$, then it holds that $\frac{\overline{X}-\overline{Y}-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t_{n+m-2}$.

Can you imagine the proofs for these results, similarly to the proofs in 7.2?

(ii)

For the **second case**, we consider that we have two normal populations, say $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$, and we have selected ONE random sample of PAIRS $(X_1, Y_1), \ldots, (X_n, Y_n)$ (for each pair, there is something that relates X_i and Y_i, $i = 1, \ldots, n$, and we cannot change the pairs, for example from (X_1, Y_1) and (X_2, Y_2) to (X_1, Y_2) and (X_2, Y_1)).

We can still write

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2), \qquad Y_1, \ldots, Y_n \sim N(\mu_Y, \sigma_Y^2).$$

If $\overline{X}, \overline{Y}$ are defined as in (i), can we write down $E(\overline{X}-\overline{Y})$ in terms of μ_X and μ_Y? How about $\mathrm{Var}(\overline{X}-\overline{Y})$ in terms of σ_X^2 and σ_Y^2? Are the random variables $\overline{X}$ and $\overline{Y}$ uncorrelated (or independent)?

We create the random sample of differences

$$D_1, \ldots, D_n \sim N(\mu_D, \sigma_D^2), \qquad D_i = X_i - Y_i,\ i = 1, \ldots, n.$$

Then, all the results we have demonstrated in 7.2 also hold for

$$\overline{D} = \frac{\sum_{i=1}^n D_i}{n}, \qquad S_D^2 = \frac{\sum_{i=1}^n (D_i-\overline{D})^2}{n-1}.$$

7.7 Confidence intervals for the difference of means from two normal populations

(i) **WE HAVE TWO INDEPENDENT SAMPLES, WE ARE INTERESTED IN COMPARING THE MEANS AND WE KNOW THE TWO DIFFERENT VARIANCES**

Two samples: $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$.

Use $T = \frac{(\overline{X}-\overline{Y})-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$.

The lower bound of the $(1-\alpha)$ C.I. for the difference of the means is

$$L(X_1, \ldots, X_n, Y_1, \ldots, Y_m) = (\overline{X}-\overline{Y}) - z_{1-\alpha/2}\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}$$

and the upper bound is

$$U(X_1, \ldots, X_n, Y_1, \ldots, Y_m) = (\overline{X}-\overline{Y}) + z_{1-\alpha/2}\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}.$$

(ii) **WE HAVE TWO INDEPENDENT SAMPLES, WE ARE INTERESTED IN COMPARING THE MEANS AND WE DO NOT KNOW THE TWO DIFFERENT VARIANCES; We can assume that they are equal**

Two samples: $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$ and we can assume $\sigma_X^2 = \sigma_Y^2 = \sigma^2$.

Use $T = \frac{(\overline{X}-\overline{Y})-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t_{n+m-2}$.

The lower bound of the $(1-\alpha)$ C.I. for the difference of the means is

$$L(X_1, \ldots, X_n, Y_1, \ldots, Y_m) = (\overline{X}-\overline{Y}) - t_{n+m-2;1-\alpha/2}\ S_p\sqrt{\frac{1}{n}+\frac{1}{m}}$$

and the upper bound is

$$U(X_1, \ldots, X_n, Y_1, \ldots, Y_m) = (\overline{X}-\overline{Y}) + t_{n+m-2;1-\alpha/2}\ S_p\sqrt{\frac{1}{n}+\frac{1}{m}}.$$

(iii) **WE HAVE ONE SAMPLE OF PAIRS, WE ARE INTERESTED IN COMPARING THE MEANS**

One sample $(X_1, Y_1), \ldots, (X_n, Y_n)$, which we turn to $D_1, \ldots, D_n \sim N(\mu_X - \mu_Y, \sigma_D^2)$, where $D_i = X_i - Y_i,\ i = 1, \ldots, n$.

Use $T = \sqrt{n}\ \frac{(\overline{X}-\overline{Y})-(\mu_X-\mu_Y)}{S_D} \sim t_{n-1}$.

The lower bound of the $(1-\alpha)$ C.I. for the difference of the means is

$$L(X_1, \ldots, X_n, Y_1, \ldots, Y_n) = (\overline{X} - \overline{Y}) - t_{n-1;1-\alpha/2}\ \frac{S_D}{\sqrt{n}}$$

and the upper bound is

$$U(X_1, \ldots, X_n, Y_1, \ldots, Y_n) = (\overline{X} - \overline{Y}) + t_{n-1;1-\alpha/2}\ \frac{S_D}{\sqrt{n}}.$$

7.8 Confidence intervals for the ratio of variances from two normal populations

(iv) **WE HAVE TWO INDEPENDENT SAMPLES, WE ARE INTERESTED IN COMPARING THE VARIANCES AND WE KNOW THE TWO DIFFERENT MEANS**

In this very unlikely event, we remember that

$$\frac{n{\sigma^*}_X^2}{\sigma_X^2} \sim \chi_n^2, \qquad {\sigma_X^*}^2 = \frac{\sum_{i=1}^n (X_i - \mu_X)^2}{n}$$

and

$$\frac{m{\sigma^*}_Y^2}{\sigma_Y^2} \sim \chi_m^2, \qquad {\sigma_Y^*}^2 = \frac{\sum_{j=1}^m (Y_j - \mu_Y)^2}{m}.$$

Thus, we use the statistic $T = \frac{{\sigma_X^*}^2}{{\sigma_Y^*}^2}\ \frac{\sigma_Y^2}{\sigma_X^2} \sim F_{n,m}$.

For the $(1-\alpha)$ C.I for the ratio of the true variances, we may write

$$P\left(F_{n,m;\alpha/2} \leq \frac{{\sigma_X^*}^2}{{\sigma_Y^*}^2}\ \frac{\sigma_Y^2}{\sigma_X^2} \leq F_{n,m;1-\alpha/2}\right) = 1 - \alpha$$

or

$$P\left(\frac{{\sigma_X^*}^2}{{\sigma_Y^*}^2}\frac{1}{F_{n,m;1-\alpha/2}} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq \frac{{\sigma_X^*}^2}{{\sigma_Y^*}^2}\frac{1}{F_{n,m;\alpha/2}}\right) = 1 - \alpha.$$

For one of the two quantiles, we will need to 'change' degrees of freedom and use the property that

$$F_{n,m;\alpha} = \frac{1}{F_{m,n;1-\alpha}}.$$

(v) **WE HAVE TWO INDEPENDENT SAMPLES, WE ARE INTERESTED IN COMPARING THE VARIANCES AND WE DO NOT KNOW THE TWO DIFFERENT MEANS**

Use the statistic $T = \frac{S_X^2}{S_Y^2} \frac{\sigma_Y^2}{\sigma_X^2} \sim F_{n-1,m-1}$.

Similarly to before,

we derive the lower bound

$$L(X_1, \ldots, X_n, Y_1, \ldots, Y_m) = \frac{S_X^2}{S_Y^2} \frac{1}{F_{n-1,m-1;1-\alpha/2}}$$

and the upper bound

$$U(X_1, \ldots, X_n, Y_1, \ldots, Y_m) = \frac{S_X^2}{S_Y^2} \frac{1}{F_{n-1,m-1;\alpha/2}}.$$

7.9 Hypotheses tests

So far we have used data (random samples) to explore distributional properties of populations from which random samples were drawn.

- Population (e.g. $N(\mu, \sigma^2)$): random sample $X_1, ..., X_n$ to learn about μ and σ^2.
- Point estimation
- Confidence intervals give a range of plausible values.

Now we want to investigate "claims" about population parameters; we want to assess formally whether the data support a "claim" or not.

Example 7.5: Advertiser claims: 8 out of 10 dogs prefer Pedigree Plum to any other dog food.

It looks like the company (Plum) is saying $p = 0.8$ where $p \equiv$ "probability of a dog preferring Pedigree Plum." A statistician wants to check Plum's advert and collects data. She finds 100 dogs and records X = total number of dogs who prefer Pedigree Plum. Do the data support this claim?

7.10 Principles of Statistical Tests

Aim: Investigate claims about properties of the distribution in a population.

These claims have to be 'translated' into *hypotheses* about the population parameter θ. If θ is the population mean then, typically, we have the following pairs of hypotheses:

Null–hypothesis		Alternative hypothesis	
$H_0: \ \theta = \theta_0$	vs.	$H_1: \ \theta \neq \theta_0$	(two–sided alternative)
$H_0: \ \theta \leq \theta_0$	vs.	$H_1: \ \theta > \theta_0$	(one–sided alternative)
$H_0: \ \theta = \theta_0$	vs.	$H_1: \ \theta > \theta_0$	(one–sided alternative)
$H_0: \ \theta \geq \theta_0$	vs.	$H_1: \ \theta < \theta_0$	(one–sided alternative)
$H_0: \ \theta = \theta_0$	vs.	$H_1: \ \theta < \theta_0$	(one–sided alternative)

A hypothesis is called *simple* if it completely determines the distribution of the observed data; an example of a simple hypothesis is $\theta = \theta_0$ if θ is the only parameter in the model. If the converse is true (for example if a hypothesis contains an unknown parameter), then the hypothesis is called *composite*; an example is a hypothesis of the form $\theta > \theta_0$, since it does not exactly determine all the parameters. A null hypothesis is called 'one-sided' or 'one-tailed' if only values above (or below) a threshold are included, and two-sided otherwise.

Note: Usually, it is not possible to make an unambiguous decision between H_0 and H_1. Statisticians have therefore agreed on formulating the null–hypothesis such that it is *falsifiable.* But it may not necessarily be possible to prove that it is true. The data may therefore provide strong evidence *against* H_0, but they cannot provide strong evidence *for* H_0 (in other words, we will be able to reject H_0, but we will not be able to accept it).

Decisions: Given the above formulation of the hypotheses we can make the following decisions based on some collected DATA:

- Either, reject H_0 — it has been falsified.
- Or, H_0 cannot be rejected, which does no necessarily mean that H_0 is true — we just do not have enough evidence against it.

We still need a formal decision rule to decide between these two possibilities! This has to take the following into account:

Errors: Since the decision is based on a random sample (as opposed to observing the whole population) it is possible to make erroneous decisions due to random variations in the samples. E.g. even if more than 80% of the dogs prefer Pedigree Plum it might happen that in the sample only 50% of the dogs prefer Pedigree Plum. This is less likely in a large sample than in a small, but it can still happen! The types of errors are classified as follows.

Type I Error: H_0 is rejected even though it is true.

Type II Error: H_0 is not rejected even though it is false.

Level of Significance: It is not possible, in general, to prevent erroneous decisions. But we can try to limit the *probability for erroneous decisions.* But, we can not at the same time minimize the probabilities for both types of errors. Statisticians have agreed that a statistical test should limit the probability for a Type I error.

If we can find a *decision rule* such that

$$P(\text{Type I error}) \leq \alpha$$

we call this decision rule a level–α–test; α is the *level of significance* of this test.

Common choices for α are 5% or 1%. This procedure is also known as *fixed level testing.* Note that the maximum probability of Type I error for one-sided tests, where the null hypothesis is of the form

$$H_0 : \theta \leq \theta_0 \qquad \text{or} \qquad H_0 : \theta \geq \theta_0$$

is usually taking place when $\theta = \theta_0$.

Construction: There are several methods of finding decision rules that lead to a level–α–test. Some examples will be presented in the following sections.

7.11 Two-sided tests via confidence intervals

Suppose we are testing $H_0 : \mu = \mu_0$ (μ the population mean). You have collected data $X_1, ..., X_n$. One easy thing to do would be to calculate a 90% confidence interval for μ based on your data and see if it contains μ_0.

Example 7.5 (continued): Do dogs prefer Pedigree Plum? Assume the number of dogs (out of our sample of 100 dogs) preferring Pedigree Plum is $Bin(100, p)$, then the testing problem is

$$H_0 : \; p = 0.8 \text{ (here } p_0 \text{ is 0.8)} \quad vs. \quad H_1 : \; p \neq 0.8$$

and has to be answered using the available data. We will do this by constructing a $(1-\alpha)$ confidence interval for p based on the single observation $X = 70$ (actual number of dogs found to prefer Pedigree Plum). What needs to be changed if the null hypothesis is one-sided?

General Procedure: Given a (1-α) confidence interval $[L, U]$ for θ the decision rules

- reject $H_0 : \theta = \theta_0$ (in favor of $H_1 : \theta \neq \theta_0$) if $\theta_0 \notin [L, U]$
- cannot reject H_0 if $\theta_0 \in [L, U]$

result in a level–α test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Notes:
1. Only two–sided hypotheses can be tested in this way! If a one-sided test is to be performed at level-α, then the confidence interval must be constructed with $1 - 2\alpha$ probability.

2. For a large sample you can calculate approximate confidence intervals using the Central Limit Theorem and perform the test in the same way.

7.12 Using Test Statistics

The idea is very similar to constructing a confidence interval. That is why in 7.11, we saw the relationship between the two. We will need a statistic that is a function of the parameter of interest and everything else is known - this is the same as for C.I. However, the distribution of the statistic may depend on quantities that we know and THE PARAMETER OF INTEREST. A Hypothesis test (H.T.) will give us the opportunity to 'assume' that the parameter is taking specific values, and, thus, under specific assumptions, it will not be unknown, but it will be equal to a certain value.

Test Statistic: This is a transformation (function) of the sample into a single value containing all relevant information to make the decision.

Assuming that H_0 is true, the distribution of the test statistic *(null–distribution)* should not depend on the unknown parameter.

Rejection Region: All values of the test statistic that result in a decision against H_0.

- Given the hypotheses H_0, H_1 and the test statistic T, we see which of its values favor the alternative hypothesis (small or big or both?).
- Given a fixed significance level (probability of type I error), we determine our critical region.
- For the specific set of data we compute the realization of the statistic T.
- If it falls into the critical region, we reject the null hypothesis. Otherwise, we do not manage to reject it.

A graphical illustration of a one-sided and two-sided toy example are shown in Figure 7.1.

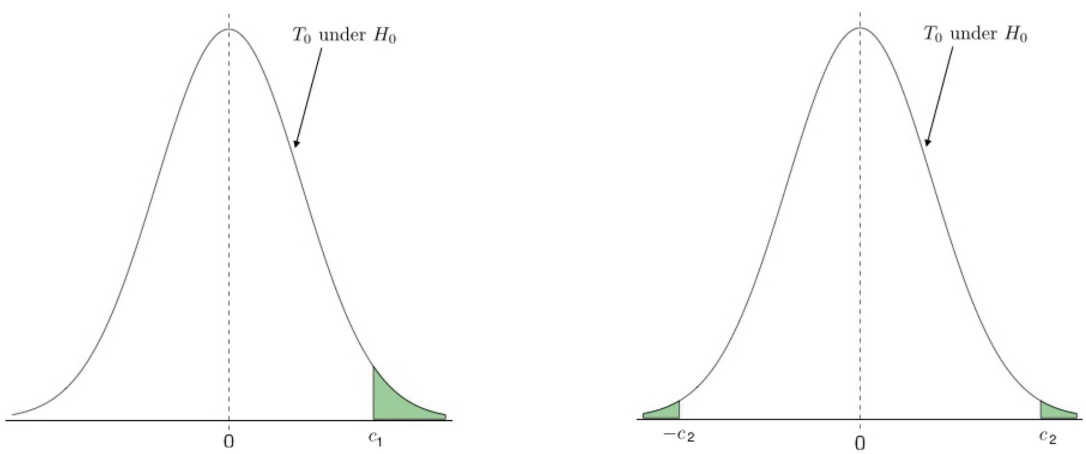

Figure 7.1: The distribution of the test statistic T_0 is shown as the solid black line, symmetrically around some null parameter value μ_0. In the case of a one-sided test, the critical region falls on one extreme of the null distribution (left-hand panel), whereas in the case of a two-sided test it is evenly split on either side (right-hand panel).

7.13 Testing for the mean of a population

We have a random sample $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ and we want to test whether

$$H_0 : \mu = \mu_0 \qquad \text{against} \qquad H_1 : \mu \neq \mu_0$$

and μ_0 is the specific value that we are testing. We know that the statistic

$$T = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma}$$

will always have distribution $T \sim N(0,1)$. Thus, both very big (very positive) and very small (very negative) values of

$$T_0 = \frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma}$$

favor the alternative hypothesis. On the one hand, IF H_0 IS TRUE, then $T_0 \sim N(0,1)$. On the other hand, if we have observed a very big absolute value of T_0, it is very unlikely that it has been generated by a μ_0 mean.

We reject the null hypothesis $H_0 : \mu = \mu_0$ in favor of the alternative hypothesis $H_1 : \mu \neq \mu_0$ at α significance level, if

$$|T_0| = \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma} \right| \geq z_{1-\alpha/2}$$

or if

$$\overline{X} \geq \mu_0 + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \qquad \text{or} \qquad \overline{X} \leq \mu_0 - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

However, if we were testing

$$H_0 : \mu \leq \mu_0 \qquad \text{against} \qquad H_1 : \mu > \mu_0$$

or

$$H_0 : \mu = \mu_0 \qquad \text{against} \qquad H_1 : \mu > \mu_0,$$

then we would reject H_0 in favor of H_1 at α significance level, if

$$\overline{X} \geq \mu_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}}.$$

Finally, if we were testing

$$H_0 : \mu \geq \mu_0 \qquad \text{against} \qquad H_1 : \mu < \mu_0$$

(and similarly if the null hypothesis were simple) then we would reject H_0 in favor of H_1 at α significance level, if

$$\overline{X} \leq \mu_0 - z_{1-\alpha} \frac{\sigma}{\sqrt{n}}.$$

As we have seen before, this result also holds, for the cases that we can use the central limit theorem and T is distributed approximately as a standard normal random variable.

On the other hand, if we do have a sample from a normal population and the population variance σ^2 is not known, then we may follow a similar path. We reject the null hypothesis H_0 in favor of the alternative hypothesis H_1, at α significance level as follows:

$$
\begin{array}{lll}
H_0 : \mu = \mu_0 & \text{against} & H_1 : \mu \neq \mu_0 \\
\text{R}\textit{eject} \text{ if} & \overline{X} \geq \mu_0 + t_{n-1;1-\alpha/2} \dfrac{S}{\sqrt{n}} & \text{or} \quad \overline{X} \leq \mu_0 - t_{n-1;1-\alpha/2} \dfrac{S}{\sqrt{n}} \\
H_0 : \mu \leq \mu_0 & \text{against} & H_1 : \mu > \mu_0 \\
\text{R}\textit{eject} \text{ if} & \overline{X} \geq \mu_0 + t_{n-1;1-\alpha} \dfrac{S}{\sqrt{n}} & \\
H_0 : \mu \geq \mu_0 & \text{against} & H_1 : \mu < \mu_0 \\
\text{R}\textit{eject} \text{ if} & \overline{X} \leq \mu_0 - t_{n-1;1-\alpha} \dfrac{S}{\sqrt{n}}. &
\end{array}
$$

Testing for the mean of a population in one-parameter distributions

We will demonstrate that with the example of Binomial, or better, with the random sample

$X_1, \ldots, X_n$ from the distribution $\text{Bin}(1, p)$. If we want to perform, for example, the two-sided test

$$H_0 : p = p_0 \qquad \text{against} \qquad H_1 : p \neq p_0,$$

we may use

$$T = \frac{\sqrt{n}(\overline{X} - p)}{\sqrt{p(1-p)}},$$

and subsequently the test statistic

$$T_0 = \frac{\sqrt{n}(\overline{X} - p_0)}{\sqrt{p_0(1-p_0)}}.$$

Indeed, under the null hypothesis, we know everything about the parameter p and, thus, the expected value and the variance of $\overline{X}$.

For large sample sizes n, we know that, under H_0, it holds that T can be considered as a standard normal random variable. When the null hypothesis $p = p_0$ is true, T_0 is also $N(0, 1)$ distributed. When the central limit theorem cannot be used, we may consider the exact distribution of $\overline{X}$, since $\sum_{i=1}^{n} X_i \sim \text{Bin}(n, p)$.

Thus, we reject $H_0 : p = p_0$ in favor of $H_1 : p \neq p_0$ at α significance level, if

$$\overline{X} \leq p_0 + q_{\alpha/2}\frac{\sqrt{p_0(1-p_0)}}{\sqrt{n}} \qquad \text{or} \qquad \overline{X} \geq p_0 + q_{1-\alpha/2}\frac{\sqrt{p_0(1-p_0)}}{\sqrt{n}}.$$

Of course, if $T_0 \sim N(0,1)$ under H_0, then $q_{1-\alpha/2} = z_{1-\alpha/2}$ and $q_{\alpha/2} = -z_{1-\alpha/2}$.

Otherwise, if we perform the test

$$H_0 : p \leq p_0 \qquad \text{against} \qquad H_1 : p > p_0,$$

then we reject H_0 in favor of H_1 at α significance level, if

$$\overline{X} \geq p_0 + q_{1-\alpha}\frac{\sqrt{p_0(1-p_0)}}{\sqrt{n}}.$$

If we perform the test

$$H_0 : p \geq p_0 \qquad \text{against} \qquad H_1 : p < p_0,$$

then we reject H_0 in favor of H_1 at α significance level, if

$$\overline{X} \leq p_0 + q_{\alpha}\frac{\sqrt{p_0(1-p_0)}}{\sqrt{n}}.$$

Note: In order to find the numbers that relate to fixed significance levels for tests in a Binomial data situation you will need to find x, such that

$$P(X \leq x) = \sum_{k=0}^{x}\binom{n}{k}p_0^k(1-p_0)^{n-k} = 1 - \alpha/2.$$

However, these probabilities might be cumbersome to compute if n is large. Tables don't necessarily cover all combinations of (n, p). Therefore, it is advisable to use approximations if they are justified. In cases where we do decide to approximate a Binomial random variable by a Gaussian random variable, we should not forget the CONTINUITY CORRECTION either.

Suppose $X \sim Bin(n, p)$ where np is large (***rule of thumb*:** $np > 5$ and $n(1-p) > 5$), then:

$$\begin{aligned} P(X \leq x) &\approx P\left(X \leq x + \tfrac{1}{2}\right) = P\left(\frac{X - np}{\sqrt{np(1-p)}} \leq \frac{x + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right) \\ &\approx P\left(Z \leq \frac{x + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right), \end{aligned}$$

where $Z \sim N(0,1)$.

7.14 Testing for the variance of a normal population

Suppose now that we have a random sample $X_1, \ldots, X_n$ from a population $N(\mu, \sigma^2)$ and we are interested in the true value of the variance. Thus, we want to test

$$H_0 : \sigma^2 = \sigma_0^2, \qquad \text{against} \qquad H_1 : \sigma^2 \neq \sigma_0^2.$$

Since μ is unknown, we may use that the statistic

$$T = \frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\sigma^2},$$

is distributed as a χ^2 random variable with $(n-1)$ degrees of freedom, so that

$$T_0 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\sigma_0^2},$$

is also χ^2_{n-1} distributed under H_0. Then,

we will reject H_0 in favor of H_1 at α significance level, if

$$\sum_{i=1}^n (X_i - \overline{X})^2 \geq \sigma_0^2 \ \chi^2_{n-1;1-\alpha/2} \qquad \text{or} \qquad \sum_{i=1}^n (X_i - \overline{X})^2 \leq \sigma_0^2 \ \chi^2_{n-1;\alpha/2}.$$

Similarly, if we are testing

$$H_0 : \sigma^2 \leq \sigma_0^2, \qquad \text{against} \qquad H_1 : \sigma^2 > \sigma_0^2.$$

we reject H_0 at α significance level, if

$$\sum_{i=1}^n (X_i - \overline{X})^2 \geq \sigma_0^2 \ \chi^2_{n-1;1-\alpha}.$$

If we are testing

$$H_0 : \sigma^2 \geq \sigma_0^2, \qquad \text{against} \qquad H_1 : \sigma^2 < \sigma_0^2.$$

we reject H_0 at α significance level, if

$$\sum_{i=1}^n (X_i - \overline{X})^2 \leq \sigma_0^2 \ \chi^2_{n-1;\alpha}.$$

What would we do, if we knew the population mean μ? Would anything change then?

7.15 Testing for the difference of the means from two normal populations

2 independent samples t-test

For the first case, we assume that we have two independent samples $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ from the two populations $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$, respectively.

For example, we want to test

$$H_0 : \mu_X = \mu_Y \qquad \text{against} \qquad H_1 : \mu_X \neq \mu_Y.$$

If we **know the variances** σ_X^2, σ_Y^2, we may use the test statistic

$$T = \frac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}},$$

which is distributed as a $N(0, 1)$ random variable, so that

$$T_0 = \frac{(\overline{X} - \overline{Y}) - 0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}},$$

is also distributed as $N(0, 1)$ if H_0 holds. Thus, we reject H_0 in favor of H_1 at α significance level, if

$$\overline{X} - \overline{Y} \leq -z_{1-\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \qquad \text{or} \qquad \overline{X} - \overline{Y} \geq z_{1-\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}.$$

Now, if the two **variances** σ_X^2 and σ_Y^2 are **not known**, but we can still assume that they are equal to the same value, say σ^2, in the two populations, then we may use the test statistic

$$T_0 = \frac{(\overline{X} - \overline{Y}) - 0}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}}$$

which, under H_0, is distributed according to the t-student distribution with $(n+m-2)$ degrees of freedom. In this case, we reject H_0, if

$$\overline{X} - \overline{Y} \leq -t_{n+m-2;1-\alpha/2}\ S_p\sqrt{\frac{1}{n}+\frac{1}{m}} \qquad \text{or} \qquad \overline{X} - \overline{Y} \geq t_{n+m-2;1-\alpha/2}\ S_p\sqrt{\frac{1}{n}+\frac{1}{m}}.$$

The one-sided tests will flow similarly, like for the previous cases.

1 sample matched pairs t-test

However, if instead of two independent samples, we come up with one sample of pairs, say $(X_1, Y_1), \ldots, (X_n, Y_n)$, from the two populations, we will need the test statistic

$$T_0 = \frac{\sqrt{n}((\overline{X} - \overline{Y}) - 0)}{S_D}$$

since we cannot approximate $\mathrm{Var}(\overline{D}) = \mathrm{Var}(\overline{X} - \overline{Y})$ by σ_X^2 and σ_Y^2 (or their estimates) only. Thus, under the null hypothesis that assumes that $\mu_X = \mu_Y$ or that $\mu_D = 0$, it holds that $T \sim t_{n-1}$. We reject the null hypothesis in favor of the alternative hypothesis (which assumes that the means are not equal), if

$$\overline{D} \leq -t_{n-1;1-\alpha/2}\ S_D\sqrt{\frac{1}{n}} \qquad \text{or} \qquad \overline{D} \geq t_{n-1;1-\alpha/2}\ S_D\sqrt{\frac{1}{n}}.$$

Again, one-sided tests are possible to perform too.

7.16 Testing for the ratio of the variances from two normal populations

For two independent samples $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$, we want to test

$$H_0 : \sigma_X^2 = \sigma_Y^2 \qquad \text{against} \qquad \sigma_X^2 \neq \sigma_Y^2.$$

Assuming that both the mean values μ_X and μ_Y are unknown, the test statistic to be used is

$$T_0 = \frac{S_X^2}{S_Y^2},$$

which is distributed according to the $F_{n-1,m-1}$ distribution when H_0 holds. Thus, we reject the null hypothesis in favor of the alternative hypothesis, if

$$T_0 = \frac{S_X^2}{S_Y^2} \leq F_{n-1,m-1;\alpha/2} \qquad \text{or} \qquad T_0 = \frac{S_X^2}{S_Y^2} \geq F_{n-1,m-1;1-\alpha/2}.$$

What would be the test statistic, if we knew μ_X and μ_Y? Would the test statistic T that we used above be wrong then? What would be the best statistic to use, if we knew μ_X, but we did not know μ_Y?

7.17 The Power of a test

Definition: The Power of a statistical test it the probability with which H_0 is rejected in case that H_0 is indeed false.

The power of a test therefore, gives the probability for a correct decision in case that H_1 is true. A graphical representation of a toy example of the power versus size of a test is given in Figure 7.2.

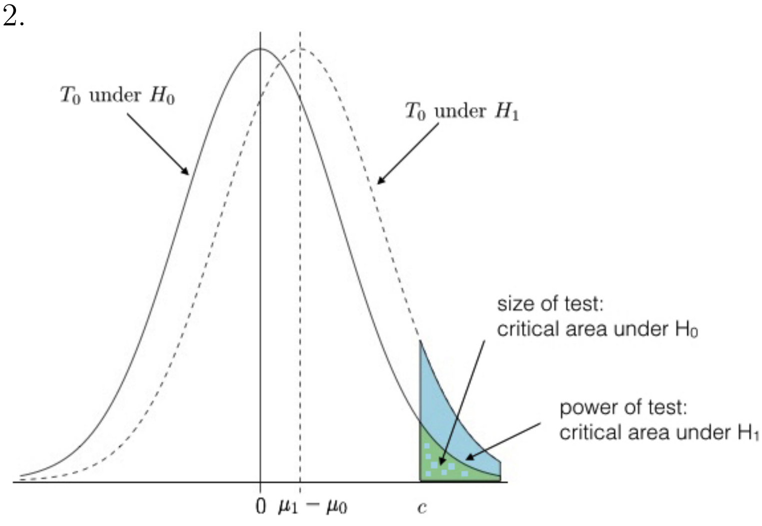

Figure 7.2: The distribution of the test statistic T_0 is shown as the solid black line, whereas its distribution if $\mu = \mu_1$ is shown as the dashed line. Here we show a one-sided test such that the critical region is above μ_0, with area shown in green equal to the size of the test. The corresponding power of the test is then the area under the μ_1 curve over the same critical region, shown in blue.

Note: The power is actually a function of the parameters in H_1, i.e. it is typically not only a single number because the probability for rejection might vary depending on which value of θ in H_1 is considered.

Example 7.6: A machine is producing, in general, chocolates of mean weight $\mu = 454$g. However, due to a machine malfunction the true average weight is $\mu = 440$g. So clearly $H_0: \ \mu = 454$ is false and ought to be rejected. Further, assume for simplicity that $\sigma = 20$ is *known.* Based on $\overline{X}$, what is the probability that the null hypothesis will be rejected indeed, when it is tested against the alternative $\mu \neq 454$? We need to find

$$P(H_0 \text{ is rejected}|\mu = 440)$$

where $P(|\mu = 440)$ means that the probability has to be calculated assuming that the hypothesis $\mu = 440$ is true.

A graphical representation of the quantities of interest is shown in Figure 7.3.

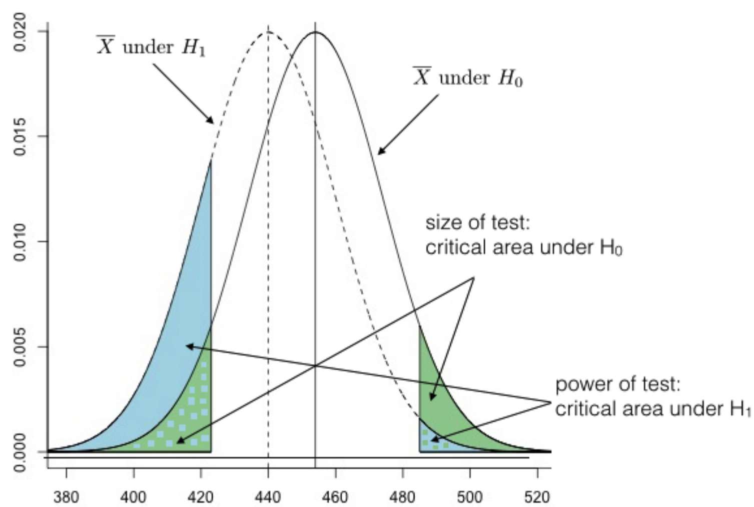

Figure 7.3: The distribution of $\overline{X}$ is shown as the solid black line, whereas its distribution if $\mu = 440$ is shown as the dashed line. Since this is a two-sided test and the distribution of $\overline{X}$ is symmetric, the critical region is symmetrically above and below $\mu_0 = 454$, with total area shown in green equal to the size of the test. The corresponding power of the test is then the area under the $\mu = 440$ curve over the same critical region, shown in blue.

Answer: In cases where the power (or the probability of type II error) has to be computed, we follow these steps.

1) Remember the case where you have assumed that H_0 is true. Under this assumption, you wrote down your critical region. Thus, write down your REJECTION RULE again in

symbols (nothing in numbers yet). When doing that, remember to KEEP THE RANDOM VARIABLES APART in the left side of your equalities, and anything else in the right side.

Since the test was two-sided

$$H_0 : \mu = 454 \qquad \text{against} \qquad H_1 : \mu \neq 454$$

we would reject the null hypothesis in favor of the alternative hypothesis (at, say α, significance level), if

$$\frac{\sqrt{n}(\overline{X} - 454)}{\sigma} < -z_{1-\alpha/2} \qquad \text{or} \qquad \frac{\sqrt{n}(\overline{X} - 454)}{\sigma} > z_{1-\alpha/2}.$$

Now remember to write this in a convenient form keeping the random variables in the left side. There is only one function of the data and, thus, one random variable only, the $\overline{X}$. REMEMBER THAT WE DID NOT HAVE TO ESTIMATE σ^2. We may write, that we reject, if

$$\overline{X} < 454 - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \qquad \text{or} \qquad \overline{X} > 454 + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

2) Now, we may assume that A SPECIFIC VALUE OF THE ALTERNATIVE HYPOTHESIS IS TRUE. Then, we may compute the probability that we are looking for, UNDER THE NEW ASSUMPTION.

Back to our example, we now assume that it holds that $\mu = 440$ (because the 'power' will be to compute the probability of rejecting the null, given that this value of the alternative holds and $\mu = 440$). Thus, under that new assumption, it holds that

$$T_1 = \frac{\sqrt{n}(\overline{X} - 440)}{\sigma} \sim N(0, 1).$$

3) We put these two (outcomes from 1) and 2)) together; on the one hand we are looking for the probability of rejecting, and we know what is the rejection rule from (1). On the other hand, we need the new distribution of the random variable (under the assumed value of the alternative hypothesis), which comes from (2). We find the probability.

$$\begin{aligned} P(\text{Reject } H_0|\mu = 440) &= P(\overline{X} < 454 - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}|\mu = 440) \\ &+ P\left(\overline{X} > 454 + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}|\mu = 440\right) \end{aligned}$$

$$
\begin{aligned}
&= P\left(\frac{\sqrt{n}(\overline{X}-440)}{\sigma} < \frac{\sqrt{n}(454-440)}{\sigma} - z_{1-\alpha/2} | \mu = 440\right) \\
&+ P\left(\frac{\sqrt{n}(\overline{X}-440)}{\sigma} > \frac{\sqrt{n}(454-440)}{\sigma} + z_{1-\alpha/2} | \mu = 440\right) \\
&= P\left(Z < \frac{\sqrt{n}(454-440)}{\sigma} - z_{1-\alpha/2}\right) \\
&+ P\left(Z > \frac{\sqrt{n}(454-440)}{\sigma} + z_{1-\alpha/2}\right),
\end{aligned}
$$

where $Z \sim N(0,1)$ and we may proceed as we already know, from the tables with the cumulative probabilities of the standard normal distribution. Indeed, Z is the only random variable, which is in the left-hand side, and everything in the right-hand side for the two probabilities are just numbers.

We usually write β for the probability of type II error. For example, we could have written the probability we were looking for in our example as $1-\beta(440)$. In the same way, we know that we write α for the probability of type I error. When the null hypothesis is assuming more than one values (one-sided tests) then α is a function of the assumed parameter too.

How easy would it be to find the power, had we not known the variance σ^2? What would be the test statistic then? How many random variables does it involve? Would it be possible to compute the power at all? What is the problem, according to the three rules we have used to compute the power/ probability of type II error?

However, we have plenty of examples, with other statistics, which we can use to compute the power.

Example 7.7: A researcher has two independent samples of sizes n, m available from two normal populations $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$. She wants to test whether the two means μ_X and μ_Y are equal, but she cannot proceed without the assumption that the variances σ_X^2 and σ_Y^2 are equal. She has decided to perform the test

$$
H_0 : \sigma_X^2 \leq \sigma_Y^2 \qquad \text{against} \qquad H_1 : \sigma_X^2 > \sigma_Y^2,
$$

and she will proceed only if she manages not to reject H_0 at α significance level. She suspects that the ratio of the variances is actually

$$
\frac{\sigma_X^2}{\sigma_Y^2} = 2,
$$

rather than equal to 1. If this is the case, what is the probability that the researcher will be able to perform the test for the equality of the population means? Is this a probability of type II error or power?

7.18 Observed significance level: p -value

We have already seen two ways that we can perform a statistical test. One way is when there is a confidence interval already available. By looking at an interval of confidence $1-\alpha$, we may know the decision of two-sided tests at α significance level, or one-sided tests at $\alpha/2$ significance level.

The second way is the standard one, via the use of a test statistic that we know its distribution, under the null hypothesis. The test statistic is a random variable and a rejection rule can be formed. The rejection region comes according to a fixed probability, the significance level α. Thus, the observed value of the test statistic is compared to a fixed value; a decision is made.

An alternative way of performing the same test is comparing probabilities, instead of comparing the values of the statistic. Instead of finding the fixed value(s) of the test statistic that determine(s) the critical region according to α, and then compare them to the observed value of the test statistic, we leave the fixed probability α as it is, and relate the test statistic to a probability (the realization of a random variable). The comparison between this observed value of the random variable and the fixed significance level α, will give the answer whether we should reject or not reject H_0.

Definition: Formally the p–value is given as follows.

1. Determine H_0 and H_1;
2. Determine a suitable test statistic and null distribution of test statistic;
3. Collect data $X_1, ..., X_n$;
4. Calculate observed value of test statistic using data;
5. Identify all other values of test statistic that, under H_0, are *as extreme or more extreme* than the observed value of test statistic (these will depend on H_1!);
6. Probability <u>under H_0</u> of all these extreme values is the p-value.

The p-value is a random variable, but we only see its realization for the specific sample (which is also called p-value) and compare it to α.

The decision is that 'we reject H_0 for any significance level GREATER THAN the p-value'. Thus, small values for the p-value (usually ≤ 0.05) result in the rejection of H_0.

A graphical illustration for a toy example of a two-sided test for μ of a normally-distributed population with known σ^2 is shown in Figure 7.4.

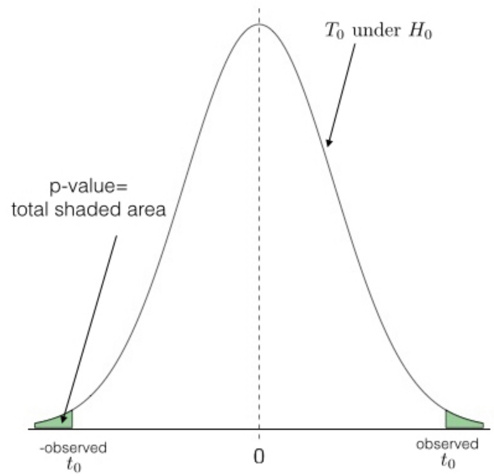

Figure 7.4: The solid line shows the distribution of the test statistic T_0 under the null hypothesis. Assume that the observed value of the test statistic is t_0, then the p-value is the total area under the T_0 curve (i.e. the total probability) of values more extreme than t_0 and $-t_0$.

For a better understanding, we will demonstrate the case of making a decision using the p-values with the following examples:

Example 7.8: For a random sample of size $n = 10$ from a normal population with variance $\sigma^2 = 9$, we have computed the sample mean $\overline{x} = 2.5$. We are interested in testing whether the true population mean μ can be considered as to be greater than 2. Make a decision from the relevant test using the p-value.

First of all, we may write the hypotheses

$$H_0 : \mu \leq 2 \qquad \text{against} \qquad H_1 : \mu > 2.$$

If the alternative hypothesis is true, we would expect that the statistic

$$T = \frac{\sqrt{n}(\overline{X} - 2)}{\sigma}$$

will have the tendency to generate positive values, or that $\overline{X}$ will generate big values (probably bigger than 2). Thus, extreme and unlikely values of $\overline{X}$, under H_0, are these that are very big only. The p-value is the (maximum) probability of observing something as extreme or more extreme, than what I actually observed, given that H_0 holds. Thus,

$$\begin{aligned} p - \text{value} &= \max_\mu P(\overline{X} \geq 2.5 | \mu \leq 2) = P(\overline{X} \geq 2.5 | \mu = 2) \\ &= P\left(\frac{\sqrt{n}(\overline{X} - 2)}{\sigma} \geq \frac{\sqrt{n}\ 0.5}{\sigma} | \mu = 2\right) = P\left(Z \geq \frac{\sqrt{10}\ 0.5}{3}\right), \end{aligned}$$

where $Z \sim N(0, 1)$ and the probability can be computed.

What would you really change if we were testing

$$H_0 : \mu = 2 \qquad \text{against} \qquad H_1 : \mu \neq 2?$$

What would be the p-value if we were testing

$$H_0 : \mu \geq 2 \qquad \text{against} \qquad H_1 : \mu < 2?$$

If we did not know the variance σ^2, but we had estimated $s^2 = 9$ from the sample, the right statistic for the tests would be

$$T = \frac{\sqrt{n}(\overline{X} - 2)}{S}.$$

Would we be able to find the p-value then (approximately)? Or, would that be a problem like it was for the case of power/probability of type II error? Why?

Chapter 8

Statistical analysis of associated variables

8.1 Simple Linear Regression

We consider data $\{x_i, y_i, i = 1, 2, ..., n\}$ which are n pairs of observations on a response variable Y and an explanatory variable X.

The $y_1, y_2, ..., y_n$ are realizations of independent random variables $Y_1, Y_2, ..., Y_n$, such that the distribution of Y typically depends on the realization x of X. How does the distribution of Y depend on X? In particular, how do $E(Y)$ and $\text{Var}(Y)$ depend on X?

Simple Linear Regression Model

Model: For each Y_i independently we assume $Y_i \sim$ Normal with

$$\begin{aligned} E(Y_i|X_i = x_i) &= \alpha + \beta x_i \quad \leftarrow \text{ straight line} \\ Var(Y_i|X_i = x_i) &= \sigma^2 \qquad\quad \leftarrow \text{ constant variance} \end{aligned}$$

i.e. $Y_i|X_i = x_i \sim N(\alpha + \beta x_i, \sigma^2)$ independent (but not identically distributed).

Assumptions: The above model essentially implies that we assume:

1. The relation between X and Y is basically *linear* with random deviations;
2. The size of the variation in Y does not depend on the corresponding X–value;
3. For fixed X–value, Y has a normal distribution;
4. Finally, the Y-s are assumed to be independent (given the values for the X-s).

Parameters: α = intercept; β = slope; σ^2 = variance (= residual variance)

Question: How do we estimate α, β, σ^2?

1. Least Squares

"Minimize distance between y_i and $E(Y_i|X_i = x_i) = \alpha + \beta x_i$." Minimize the residual sum of squares (RSS)

$$RSS(\alpha, \beta) = \sum_{i=1}^{n}(y_i - \alpha - \beta x_i)^2,$$

i.e. $\hat{\alpha}$ and $\hat{\beta}$ are the arguments which minimize RSS:

$$(\hat{\alpha}, \hat{\beta}) = \text{argmin}_{\alpha,\beta} \sum_{i=1}^{n}(y_i - \alpha - \beta x_i)^2.$$

In order to minimize this, it is helpful to re-write it as

$$\begin{aligned}
RSS(\alpha, \beta) &= \sum_{i=1}^{n}(y_i - \alpha - \beta x_i)^2 \\
&= \sum_{i=1}^{n}(((y_i - \overline{y}) - \beta(x_i - \overline{x})) + (\overline{y} - \beta\overline{x} - \alpha))^2 \\
&= \sum_{i=1}^{n}((y_i - \overline{y}) - \beta(x_i - \overline{x}))^2 + \sum_{i=1}^{n}(\overline{y} - \beta\overline{x} - \alpha)^2 \\
&+ 2(\overline{y} - \beta\overline{x} - \alpha)\sum_{i=1}^{n}((y_i - \overline{y}) - \beta(x_i - \overline{x})) \\
&= \sum_{i=1}^{n}((y_i - \overline{y}) - \beta(x_i - \overline{x}))^2 + \sum_{i=1}^{n}(\overline{y} - \beta\overline{x} - \alpha)^2 \\
&= \sum_{i=1}^{n}((y_i - \overline{y}) - \beta(x_i - \overline{x}))^2 + n(\overline{y} - \beta\overline{x} - \alpha)^2.
\end{aligned}$$

This is a sum of two nonnegative values and, α only appears in the second term . Thus, its minimum value 0 can be achieved for the second term when

$$\overline{y} - \hat{\beta}\overline{x} - \hat{\alpha} = 0 \qquad \text{or} \qquad \hat{\alpha} = \overline{y} - \hat{\beta}\overline{x}.$$

We may write for the first term that

$$\begin{aligned} Q(\alpha, \beta) &= \sum_{i=1}^{n}((y_i - \overline{y}) - \beta(x_i - \overline{x}))^2 = \sum_{i=1}^{n}((y_i - \overline{y})^2 + \beta^2(x_i - \overline{x})^2 \\ &- 2\ \beta(y_i - \overline{y})(x_i - \overline{x})) = \sum_{i=1}^{n}(y_i - \overline{y})^2 + \beta^2 \sum_{i=1}^{n}(x_i - \overline{x})^2 \\ &- 2\ \beta \sum_{i=1}^{n}(y_i - \overline{y})(x_i - \overline{x}) = C_{yy} + \beta^2 C_{xx} - 2\ \beta C_{xy}, \end{aligned}$$

where $C_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $C_{xx} = \sum(x_i - \bar{x})^2$ and $C_{yy} = \sum_{i=1}^{n}(y_i - \overline{y})^2$. To minimize $Q(\alpha, \beta)$, we will need to find its derivative

$$\frac{\partial}{\partial \beta}\ Q(\alpha, \beta) = 2\ \beta\ C_{xx} - 2\ C_{xy}$$

and this should be equal to 0, when

$$\left.\frac{\partial}{\partial \beta} Q(\alpha, \beta)\right|_{\alpha = \hat{\alpha}, \beta = \hat{\beta}} = 0.$$

Thus, we may find our estimator for the slope, from

$$2\ \hat{\beta}\ C_{xx} - 2\ C_{xy} = 0 \qquad \text{or} \qquad \hat{\beta} = \frac{C_{xy}}{C_{xx}}.$$

Indeed, it holds that

$$\frac{\partial}{\partial \beta^2}\ Q^2(\alpha, \beta) = 2\ C_{xx}$$

and we have found a minimum of $Q(\alpha, \beta)$ or $RSS(\alpha, \beta)$ when $\alpha = \hat{\alpha}$ and $\beta = \hat{\beta}$. We will often denote our estimators by $\hat{\beta}_{LS}$ and $\hat{\alpha}_{LS}$, in order to remember that these are least squares estimators (LSE).

2. Maximum Likelihood

The above method of least squares gives no estimate for the residual variance σ^2. This can instead be derived using the Maximum Likelihood method (where the normal distribution is exploited). The resulting estimator is

$$\hat{\sigma}^2_{ML} = \frac{RSS(\hat{\alpha}, \hat{\beta})}{n} = \frac{n-2}{n} \frac{RSS(\hat{\alpha}, \hat{\beta})}{n-2}.$$

and it is biased (but asymptotically unbiased!).
Actually, it also holds that

$$\hat{\alpha}_{ML} = \hat{\alpha}_{LS} = \hat{\alpha}, \qquad \hat{\beta}_{ML} = \hat{\beta}_{LS} = \hat{\beta}.$$

In fact for Gaussian random variables, it holds that

$$\frac{(n-2)}{\sigma^2} \frac{\sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}\, x_i)^2}{n-2} = \frac{\sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}\, x_i)^2}{\sigma^2} \sim \chi^2_{n-2}.$$

Does this remind you of anything else? What if we knew there is no slope and all the variables had the same mean α (or μ)? What is the expected value of a χ^2 random variable with $(n-2)$ degrees of freedom?

A simple correction gives

$$\hat{\sigma}^2 = \frac{RSS(\hat{\alpha}, \hat{\beta})}{n-2}$$

which is unbiased (called *sample residual variance*).

Sampling Distributions for the Normal Linear Model with two parameters

From now on, we will remember that the whole statistical analysis is taking place conditionally on the realizations $x_1, \ldots, x_n$ of $X_1, \ldots, X_n$. Thus, the only random variables involved now are $Y_1, \ldots, Y_n$.

Model: $Y_i \sim N(\alpha + \beta x_i, \sigma^2)$ independently for $i = 1, 2, ..., n$.

Theoretical Results:

1. $$\boxed{\overline{Y} \sim N\left(\alpha + \beta\bar{x}, \frac{\sigma^2}{n}\right)}$$

   As $\overline{Y} = \frac{1}{n}\sum Y_i$ is a linear combination of normal random variables, it is a normal random variable too (use the moment generating function to see that for independent random variables)! Thus

   $$\begin{array}{rclclcl} E(\overline{Y}) & = & \frac{1}{n}\sum E(Y_i) & = & \frac{1}{n}\sum(\alpha + \beta x_i) & = & \alpha + \beta\bar{x} \\ Var(\overline{Y}) & = & \frac{1}{n^2}\sum Var(Y_i) & = & \frac{1}{n^2}n\sigma^2 & = & \frac{\sigma^2}{n} \end{array}$$

2. $$\boxed{\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{C_{xx}}\right)}$$

   Recall

   $$\begin{aligned} \hat{\beta} = \frac{C_{xY}}{C_{xx}} &= \frac{1}{C_{xx}}\sum(x_i - \overline{x})(Y_i - \overline{Y}) \\ = \frac{1}{C_{xx}} & \left[\sum(x_i - \overline{x})Y_i - \overline{Y}\underbrace{\sum(x_i - \overline{x})}_{=\,0}\right] \end{aligned}$$

   Thus, $\hat{\beta}$ is a linear combination of normal variables, so is also normal.

3. $RSS(\hat{\alpha}, \hat{\beta})$, $\overline{Y}$ and $\hat{\beta}$ are mutually independent (no proof).

4. Estimate value $E(Y|X = x_0) = \alpha + \beta x_0$ of regression line at x_0:

   $$\boxed{\hat{\alpha} + \hat{\beta}x_0 \sim N\left(\alpha + \beta x_0, \sigma^2\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{C_{xx}}\right)\right)}$$

   $\hat{\alpha} + \hat{\beta}x_0 = \overline{Y} + \hat{\beta}(x_0 - \bar{x})$ is a linear combination of normals so it is a normal random variable too.

5. For the residual sum of squares we have

   $$\boxed{\frac{RSS(\hat{\alpha}, \hat{\beta})}{\sigma^2} \sim \chi^2_{n-2}}$$

Implications: From (2):

$$\frac{\hat{\beta}-\beta}{\sigma/\sqrt{C_{xx}}} \sim N(0,1).$$

Let $\hat{\sigma}^2 = \frac{RSS(\hat{\alpha},\hat{\beta})}{n-2}$ then from (5)

$$\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$$

Also:

$$\begin{aligned} T &= \frac{\frac{\hat{\beta}-\beta}{\sigma/\sqrt{C_{xx}}}}{\sqrt{[(n-2)\hat{\sigma}^2]/[\sigma^2(n-2)]}} \sim t_{n-2} \\ \Rightarrow T &= \frac{\hat{\beta}-\beta}{\hat{\sigma}/\sqrt{C_{xx}}} \sim t_{n-2}. \end{aligned}$$

Use this to test values for β or construct confidence intervals for β.

Confidence Intervals for the Regression Line

We have seen that

$$\hat{Y}_{x_0} = \hat{E}(Y|X=x_0) = \hat{\alpha} + \hat{\beta}x_0 \sim N\left(\alpha + \beta x_0, \sigma^2\left(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{C_{xx}}\right)\right)$$

Therefore:

$$\frac{\hat{\alpha}+\hat{\beta}x_0 - (\alpha+\beta x_0)}{\hat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{C_{xx}}}} \sim t_{n-2}$$

Hence:

$$(\hat{\alpha}+\hat{\beta}x_0) \pm \hat{\sigma}\, t_{n-2;1-\frac{\alpha}{2}} \sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{C_{xx}}}$$

is called a $100(1-\alpha)\%$ *confidence interval for the regression line* at x_0 (or for the mean $E(Y|X=x_0) = \alpha + \beta x_0$).

Prediction Intervals

So far, we have been very well acquainted with the construction of confidence intervals. We have also seen their correct interpretation. Even in the previous subsection, we were still dealing with the construction of a confidence interval. Since all the statistical analysis in regression, is taking place given the fixed values of $X_1, \ldots, X_n$, we may consider $\alpha + \beta x_0$ as a parameter (a conditional expectation). In fact, conditional expectations are RANDOM VARIABLES (once the X-s will be set free to vary randomly).

For a confidence interval on a parameter, we have seen how its bounds are random and they have a probability of including the fixed value of the parameter. We are now going to see the essence of a PREDICTION INTERVAL instead. Not only are we going to deal with RANDOM BOUNDS, but also the value we will be trying to 'guess' is going to be a REALIZATION OF A RANDOM VARIABLE. This is going to be reflected in the bigger variances of the random variables (based on which we will construct the interval).

Now suppose that we want to predict a new response Y_{n+1} at a given new point x_{n+1}. We need a random variable involving Y_{n+1} and anything else that we know (observed, estimated or fixed values); this random variable will be normally distributed with known mean and variance.

Indeed, we consider the random variable $\hat{Y}_{n+1} - Y_{n+1}$, where

$$Y_{n+1} = \alpha + \beta x_{n+1} + \varepsilon_{n+1}$$

is what we are interested in predicting, and

$$\hat{Y}_{n+1} = \hat{\alpha} + \hat{\beta} x_{n+1}$$

is the estimator for the conditional expectation of Y_{n+1} given x_{n+1}. It holds that

$$E(\hat{Y}_{n+1} - Y_{n+1}) = E(\hat{\alpha}) + E(\hat{\beta}) x_{n+1} - \alpha - \beta x_{n+1} - E(\varepsilon_{n+1}) = 0$$

and

$$\mathrm{Var}(\hat{Y}_{n+1} - Y_{n+1}) = \mathrm{Var}(\hat{Y}_{n+1}) + \mathrm{Var}(\varepsilon_{n+1}) = \sigma^2 \left[\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{C_{xx}} + 1 \right].$$

Using the usual arguments:

$$\frac{\hat{Y}_{n+1} - Y_{n+1}}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{C_{xx}} + 1}} \sim t_{n-2}$$

Solving with respect to the unknown Y_{n+1},

> a $100(1-\alpha)\%$ prediction interval for Y_{n+1} is:
>
> $$(\hat{\alpha} + \hat{\beta} x_{n+1}) \pm t_{n-2;1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{X})^2}{C_{xx}} + 1}.$$

So now the interval is wider, because we are not concluding about the mean value $E(Y|X = x_{n+1})$ but we are going further; trying to say something about the value, which will have an extra variation around this mean.

Model Checking

We define the residuals

$$\hat{\varepsilon}_i = Y_i - \hat{\alpha} - \hat{\beta}\, x_i,\ i = 1, \ldots, n,$$

and we check whether they look random, homoscedastic (equal variance σ^2 remains) and normally distributed (actually and I.I.D. assumption of the random errors is often enough to proceed with the tests). To test the normality, we might use a QQ-plot (or histogram). Once the assumptions are satisfied, we may use R^2 or R^2-adjusted to see how well a line fits the data.

F-Tests for nested models

We will demonstrate that with three very simple models. In more advanced courses in regression, more general versions of this result will be used.

For pairs of observations $(X_1, Y_1), \ldots, (X_n, Y_n)$, we consider three different ways to model the Y-s.
<u>Model 1</u>:

$$Y_i = \varepsilon_i,\ \{\varepsilon_i\} \sim NID(0, \sigma^2).$$

<u>Model 2</u>:

$$Y_i = \alpha + \varepsilon_i,\ \{\varepsilon_i\} \sim NID(0, \sigma^2).$$

<u>Model 3</u>:

$$Y_i = \alpha + \beta\, x_i + \varepsilon_i,\ \{\varepsilon_i\} \sim NID(0, \sigma^2).$$

The first model is just assuming that the Y-s are a random sample from a normal population with zero mean. Knowledge of the X-s does not affect the Y-s. As a first minimal assumption, we usually consider that the set of observations $Y_i,\ i = 1, \ldots, n$, comes from an unknown mean α (or μ), rather than 0. Thus, we move to the second model, which is also called the 'Null Model' (often considered as the starting point with almost no assumptions at all). Again, knowledge of the X-s does not have any effect on the Y-s. The third model, is taking into account a linear relationship between the X-s and the Y-s.

Model 1 is nested in Model 2, since it is its special case when $\alpha = 0$. Similarly, Model 2 is nested in Model 3, as a special case when $\beta = 0$.

For the null model, we will estimate $\hat{\alpha} = \overline{Y}$. The null model uses n pairs of observations to estimate 1 parameter only. Thus, it leaves $(n-1)$ degrees of freedom for the 'error' (any variability that the use of this parameter did not manage to explain). The residual sum of squares might be written as

$$RSS_2 = \sum_{i=1}^{n} (Y_i - \overline{Y})^2$$

and it holds that $\frac{\mathrm{RSS}_2}{\sigma^2} \sim \chi^2_{n-1}$.

For Model 3, we will estimate $\hat{\beta} = \frac{C_{xY}}{C_{xx}}$ and $\hat{\alpha} = \overline{Y} - \hat{\beta}\overline{x}$. This model uses n pairs, in order to estimate 2 parameters. Thus, it leaves $(n-2)$ degrees of freedom for the 'error'. The residual sum of squares might be written as

$$RSS_3 = \sum_{i=1}^{n} (Y_i - \hat{\alpha} - \hat{\beta}x_i)^2$$

and it holds that $\frac{\mathrm{RSS}_3}{\sigma^2} \sim \chi^2_{n-2}$.

If Model 3 is taking place and we want to test whether $\beta = 0$ (and, thus, under H_0 the null model is correct), then we use the statistic

$$F = \frac{\frac{RSS_2 - RSS_3}{1}}{\frac{RSS_3}{n-2}},$$

which is distributed as $F_{1,n-2}$, under H_0. Indeed, under the null hypothesis, we may write both

$$\frac{\mathrm{RSS}_2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \frac{\mathrm{RSS}_3}{\sigma^2} \sim \chi^2_{n-2}$$

and a χ^2_1 random variable can be derived from their difference. In a similar way, we may always perform tests for such nested models.

In the previous test, what do you think F will be equal to? For which values of F (always taking positive values) will we reject H_0? Will we be able to test one-sided tests for β? Which test that we have seen is more appropriate in such cases?

8.2 Bivariate Continuous Data

Here we consider a random sample of the type:

$$(X_1, Y_1), (X_2, Y_2), ..., (X_n, Y_n)$$

Both X's and Y's are random variables and the interest lies in their relationship. Not explanatory–response relationship is assumed or even sensible (e.g. height and weight).

Measures of Association

We need a formal method of measuring strengths of (linear) association between two random variables.

Data:

$$(X_1, Y_1), (X_2, Y_2), ..., (X_n, Y_n)$$

Pearson correlation coefficient:

$$r = \frac{\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2 \sum_{i=1}^{n}(Y_i - \overline{Y})^2}} = \frac{C_{XY}}{\sqrt{C_{XX}C_{YY}}} \qquad -1 \leq r \leq 1$$

Interpretation: $0 < r \leq 1$ positive correlation (as X increases so does Y)

-1 $\leq r < 0$ negative correlation (as X increases, Y decreases)

$r = 0$ zero correlation (no *linear* relationship — there might be another kind of relation).

To say there is correlation or association between two variables X and Y is merely to say that the values of the two variables vary together in some way for any of the following reasons:

1. Changes in X cause changes in Y.
2. Changes in Y cause changes in X.
3. changes in some 3rd variable Z causes similar or opposite changes in X and Y.
4. Observed relationship is just a coincidence.

Correlation is not <u>causation</u>.

Relation to Linear Regression: If we regress Y on X (despite knowing that X is not the cause for Y) we find that the measure of determination R^2 is the same as the squared Pearson correlation coefficient r^2 (in numbers). Further (and in numbers only - forget about random variables for this)

$$\hat{\beta} = \frac{C_{xy}}{C_{xx}} = r\sqrt{\frac{C_{yy}}{C_{xx}}}.$$

Since the square root is always positive, we find that the sign of $\hat{\beta}$ is the same as the one of the correlation coefficient r.

The remainder of this chapter should be considered as 'Part 1: Statistics for populations'.

Bivariate Normal Distribution

Vector (X, Y) is <u>bivariate Normal</u> $(X, Y) \sim N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$ if:

1. marginal distribution of $X \sim N(\mu_X, \sigma_X^2)$ and marginal distribution of $Y \sim N(\mu_Y, \sigma_Y^2)$.
2. Conditional distribution of $X|Y = y$:

$$(X|Y = y) \sim N(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma_X^2(1 - \rho^2))$$

(similar definition for $Y|X = x$)

The joint density of (X, Y) is

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]\right\}$$

for all real numbers x, y.

Role of ρ: Define the new random variables

$$Z_X = \frac{X - \mu_X}{\sigma_X}, \qquad Z_Y = \frac{Y - \mu_Y}{\sigma_Y}$$

or

$$X = \mu_X + \sigma_X Z_X, \qquad Y = \mu_Y + \sigma_Y Z_Y.$$

Then

$$E(Z_X Z_Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{z_1 z_2}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}[z_1^2 + z_2^2 - 2\rho z_1 z_2]\right\} dz_1 dz_2$$

Can you compute these two integrals consecutively? What is then the correlation between X and Y equal to?

Example 8.1: We have available the heights (cm) and weights (kg) of $n = 100$ 11-years old boys in Bradford School.

Sample statistics:

$$\overline{x} = 145,\ \overline{y} = 36,\ s_X^2 = 58,\ s_Y^2 = 60,\ r = 0.75$$

Note that r is a sample estimate of ρ. We want to model variation in heights and weights of 11-yr old boys by a Bivariate Normal Distribution with parameters:

$$\mu_X = 145,\ \mu_Y = 36,\ \sigma_X^2 = 58,\ \sigma_Y^2 = 60,\ \rho = 0.75$$

i.e:

$$(X, Y) \sim N(145, 36, 58, 60, 0.75)$$

Then the proportion of the 11-yr old school boys in the population who should be more than 140cm tall is:

$$P(X > 140) = 1 - P\left(Z \leq \frac{140 - 145}{\sqrt{58}}\right) = 0.744$$

Considering only those schoolboys 150cm tall (taller than the average $\mu_X = 145$) the distribution of the weights has mean:

$$\begin{aligned} E(Y|X = 150) &= \mu_Y + \rho\tfrac{\sigma_Y}{\sigma_X}(x - \mu_X) \\ &= 36 + 75\sqrt{\tfrac{60}{58}}(150 - 145) \\ &= 36 + 3.8 \\ &= 39.8 \text{ kg} \end{aligned}$$

which is heavier than the average $\mu_Y = 36$kg.

Note, $P(X > 150, Y > 40)$ is not possible to compute analytically. Why? Can we write a formula for this probability?

8.3 Other multivariate distributions (discrete data)

Multinomial distribution

Let a trial with k possible different outcomes. Let $p_i,\ i = 1, \ldots, k$ be the probability of the i-th outcome ($\sum_{i=1}^{k} p_i = 1$). We repeat n independent trials. We write the r.v. $X_i,\ i = 1, \ldots, k$, which is 'the number of times the i-th outcome came up in the n independent trials'. Then, it holds that

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \ldots x_k!} p_1^{x_1} \ldots p_k^{x_k},\ x_i = 0, \ldots, n,\ i = 1, \ldots, k,\ \sum_{i=1}^{k} x_i = n.$$

Multivariate Hypergeometric distribution

Let a population of N units, out of which M_1 have the characteristic '1', M_2 have the characteristic '2',..., M_k have the characteristic 'k' ($\sum_{i=1}^{k} M_i = N$). We select n units without replacement. We write the r.v. $X_i,\ i = 1, \ldots, k$, to be 'the number of units with the characteristic 'i' in the sample'. Then, it holds that

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{\binom{M_1}{x_1} \cdots \binom{M_k}{x_k}}{\binom{N}{n}},\ x_i = 0, \ldots, \min(M_i, n),\ i = 1, \ldots, k,$$

under the restriction $\sum_{i=1}^{k} x_i = n$ again (otherwise the probability is equal to 0).

Example 8.2: In the neighborhood of Helen, there are overall ten single ladies (including her), five married ladies and two ladies that have been divorced. We select three ladies at random. What is the probability that two are divorced and one is single if (i) we select without replacement (ii) we select with replacement?